ETH zürich

ETH Library

Quantum Nonlocality in Bilocal Networks: An Operator Algebraic Perspective

Master Thesis

Author(s):
Xu, Xiangling

Publication date:
2023-05-25

Permanent link:
https://doi.org/https://doi.org/10.3929/ethz-b-000613670

Rights / license:
In Copyright - Non-Commercial Use Permitted

This page was generated automatically upon download from the ETH Zurich Research Collection.
For more information, please consult the Terms of use.

ETH
Eidgenössische Technische Hochschule Zürich
Swiss Federal Institute of Technology Zurich

Quantum Nonlocality in Bilocal Networks: An Operator Algebraic Perspective

Thesis for MSc ETH Mathematics

Xiangling Xu (许湘灵)

25th May, 2023

Supervisor: Prof. Dr. Marc-Olivier Renou
Official Supervisor: Prof. Dr. Renato Renner

Institute for Theoretical Physics, ETH Zürich

Abstract

In this thesis, we explore the applications of the theory of operator algebras, such as C^*-algebras and von Neumann algebras, in quantum physics with a focus on the quantum bilocal network scenario. Originating from Heisenberg's picture, the theory of operator algebras has become a fruitful field and a powerful mathematical tool in the study of quantum systems. Quantum nonlocality is essential for understanding the foundations of quantum mechanics, and quantum correlations in network scenarios provide a robust framework for its investigation.

We combine the algebraic perspective with the study of quantum nonlocality to analyse the quantum bilocal network scenario, involving three parties, Alice, Bob, and Charlie, arranged linearly. In this scenario, two independent quantum sources are present, one shared between Alice and Bob, and the other shared between Bob and Charlie. Recent developments from [57, Renou & Xu, arXiv:2210.09065] and [35, Ligthart & Gross, arXiv:2212.11299] demonstrate the applicability of operator algebras in this context, offering a deeper understanding of quantum nonlocality in network settings.

This thesis serves as a reference for results in the theory of operator algebras relevant to the study of quantum nonlocality, introducing notable results and attempting to draw connections between them. By building upon the findings of the two papers mentioned above, the thesis highlights the potential of these abstract mathematical tools in uncovering new insights into quantum mechanics and motivates further research in the field using an operator algebraic perspective.

Acknowledgements

First and foremost, I wish to express my heartfelt gratitude to Prof. Marc-Olivier Renou. Marco has been an outstanding supervisor, guiding me into the captivating world of quantum information theory, an insightful mentor on my academic path, a stimulating colleague for intellectual discussions, and a cherished friend in life. I count myself truly fortunate to have such an encouraging future PhD supervisor, and I eagerly anticipate the next three years at Inria, working closely with Marco as his PhD student.

My appreciation extends to Prof. Renato Renner and the Quantum Information Theory Group at ETH Zürich for graciously agreeing to supervise this thesis formally. In addition, I am indebted to all the professors and lecturers I have encountered at ETH Zürich and the University of Toronto. Their guidance has not only broadened my knowledge but also, more importantly, fostered my curiosity for lifelong learning.

I wish to acknowledge my dear friends from the physics program during my bachelor's degree at the University of Toronto. Together, we shared conversations spanning the vastness of the cosmos to the simplicity of everyday life in McLennan. As we all continue to walk the daring path of academia, their camaraderie and mutual encouragement are truly inspiring.

A gentle mention goes to the delightful Janine Alina Gugger, whose infinite kindness and warmth have made this foreign land of Switzerland feel more like home, providing me with the strength and courage to persevere and forge ahead.

Lastly, I offer my sincerest thanks to my family for their steadfast support. Their unconditional encouragement of my endeavours has emboldened me to pursue my aspirations and soar without constraints. I wish to share a poem by Nalan Xingde that echoes my sentiments in our native language, which I have translated to the best of my ability:

> 山一程，水一程，身向榆关那畔行，夜深千帐灯。
> 风一更，雪一更，聒碎乡心梦不成，故园无此声。
>
> *Mountains stretch a journey, waters span a journey,*
> *Toward Yu Pass I tread, a myriad of lamps in deepest night.*
> *Wind whispers through a watch, snow flurries through a watch,*
> *Shattering dreams, in my hometown such sounds are scarce.*

Contents

Contents **iii**

1 Introduction **1**
1.1 Thesis outline 3
1.2 Contribution and discussion 4

2 Algebraic structures in quantum mechanics **6**
2.1 C^*-algebras prelimnary 7
2.1.1 Selected C^*-algebra objects 9
2.1.2 Commutative Gelfand–Naimark Theorem 10
2.2 States and representations of C^*-algebras 11
2.2.1 States 11
2.2.2 Representations 13
2.2.3 Gelfand–Naimark–Segal (GNS) construction 15
2.2.4 Structure of C^*-algebras 18
2.3 von Neumann algebras 19
2.3.1 Definitions and basic properties 19
2.3.2 Affiliated unbounded self-adjoint operators 21
2.3.3 Factors and structures of von Neumann algebras . . . 22
2.3.4 Superselection rules 23
2.4 Composite systems and independent subsystems 24
2.4.1 Sptial and maximal tensor products of C^*-algebras . . 25
2.4.2 States on tensor products of C^*-algebras 27
2.4.3 Tensor products of von Neumann algebras 27
2.4.4 Independent subsystems 28
2.5 Algebraic formulation of quantum theory 29

3 Basics of quantum nonlocal correlations **31**
3.1 Nonlocal correlations 31
3.1.1 Tsirelson's problem 34

3.1.2 Geometry of the correlation sets 34
3.1.3 The CHSH game 35
3.2 Navascués-Pironio-Acín (NPA) hierarchy 36
3.2.1 Moment matrix and necessity of the hierarchy 37
3.2.2 Sufficiency of the hierarchy 39
3.2.3 Stopping criteria 41
3.2.4 Semi-definite Programs 42

4 Algebraic approaches to quantum bilocal scenarios 43
4.1 Bilocal quantum correlations with generalised NPA hierarchies 45
4.1.1 Generalising NPA hierarchy for tripartite case 45
4.1.2 Key concepts in factorisation bilocal NPA hierarchy . . 46
4.1.3 Convergence of factoristaion bilocal NPA hierarchy . . 47
4.1.4 Remarks on factorisation bilocal NPA hierarchy 49
4.2 Algebraic insights into quantum bilocal network 51
4.2.1 Models for quantum bilocal scenario 51
4.2.2 Equivalence of bilocal models 53
4.2.3 Bilocal Tsirelson's Problem 56
4.3 Quantum inflation technique in bilocal scenario 56

Bibliography 59

A Selected functional analysis concepts for quantum mechanics 65
A.1 Banach spaces and Banach algebras 65
A.1.1 Metrics, norms, inner products, and completeness . . 65
A.1.2 Basic notions of Banach spaces 67
A.1.3 Banach algebras and specturm 69
A.2 Spectral Theorem for bounded self-adjoint operators 71
A.2.1 Spectrum of Hilbert space operators 71
A.2.2 Projector-valued measures (PVMs) and integration . . 72
A.2.3 Spectral Theorem 74
A.3 Hilbert space operator topologies 75
A.4 Important theorems in functional analysis 76
A.4.1 Three cornerstones in functional analysis 77
A.4.2 Banach–Alaoglu and Krein–Milman Theorems 78

Chapter 1

Introduction

The study of quantum mechanics has led to the development of two alternative formulations, namely Schrödinger's picture and Heisenberg's picture. Schrödinger's picture, introduced by Schrödinger in 1926, is based on wave functions representing quantum states, which evolve with time according to the Schrödinger equation [59]. In contrast, Heisenberg's picture, introduced by Heisenberg in 1925, employs the concept of matrices representing observables, with the quantum states seemingly absent [26].

Von Neumann initially demonstrated the equivalence of these two seemingly different formulations in 1927 [43, 48]. Building upon the operator algebra ideas inherent in Heisenberg's matrix mechanics, von Neumann, in collaboration with Murray in the 1930s, deepened the understanding through a series of works [39, 40, 41, 44, 45, 46, 47, 49], which laid the groundwork for the theory of operator algebras. This theory was later developed further by Gelfand, Naimark, and Segal [23, 60].

In 1947, Segal's paper [61] marked another turning point in the relationship between operator algebras and mathematical quantum physics, introducing the concept of C^*-algebras to the field. Since then, the theory of operator algebras has become a powerful tool, finding applications in quantum statistical mechanics [11, 24], quantum field theory [3, 24], and even giving rise to the field of noncommutative geometry [18] with potential applications in quantum field theory, quantum gravity, and string theory [5, 19, 37, 62, 65]. The algebraic perspective has proven invaluable in describing and analysing quantum systems, showcasing the profound interplay between operator algebras and quantum mechanics.

In 1935, the EPR paradox, proposed by Einstein, Podolsky, and Rosen, questioned the completeness of quantum mechanics and introduced the concept of entanglement [20]. Building on the idea of entanglement, in 1964, Bell introduced the concept of nonlocality by proving Bell's theorem [7], which

showed that quantum mechanics inherently exhibits nonlocal correlations that cannot be explained by local hidden variable theories. This result revealed that nonlocality is a fundamental feature of quantum mechanics and has led to a plethora of applications, such as quantum cryptography [1, 2, 6, 71], randomness [17, 51], and quantum device certification [4, 36, 66].

In particular, understanding the set of probability distributions feasible by quantum theory in any given setup is crucial in the study of quantum nonlocality. In the standard Bell scenario (see, e.g., Figure 3.2 and 4.1), where multiple parties share one quantum source, one can approximate the set of feasible quantum correlations from within by considering all possible Hilbert spaces and operators. On the other hand, the seminal paper by Navascués, Pironio, and Acín [42] provides an outer approximation of the quantum set, introducing the NPA hierarchy—a convergent hierarchy of semi-definite programs (SDP) that can be solved by computers [70, 75].

More recently, there has been growing interest in studying network scenarios involving multiple independent quantum sources [9, 22, 54, 55, 68]. These scenarios provide new insights into the foundations of quantum mechanics and can help to understand quantum correlations in more complex settings [15, 16, 27, 67]. The bilocal scenario [10], which is the simplest network scenario involving three parties sharing two independent quantum sources (see Figure 4.2), is a prime candidate for investigation [53, 56, 73].

In order to characterise the set of quantum correlations feasible in these more complex quantum network scenarios, various methods were considered. For example, the quantum inflation technique [74], which involves duplicating quantum sources in an inflated network so that the additional symmetries can be exploited to characterise network correlations. In the bilocal scenario, the scalar extension hierarchy is introduced to linearise factorisation constraints and achieve a tighter approximation than the standard NPA hierarchy [52]. A nonlinear generalisation of the NPA hierarchy is investigated in [57] to study the bilocal scenario, further formalising the scalar extension hierarchy. Subsequently, the Positivstellensätze of state polynomials are proven, generalising the scalar extension polynomials in [57] and leading to convergent SDP hierarchies capable of probing network scenarios [32]. In addition to addressing the open questions posed in [57], the authors of [35] demonstrate the convergence of the inflation-NPA hierarchy in the bilocal scenario. Throughout these advancements, the algebraic aspects of quantum theory naturally surface and prove to be essential.

Motivated by these developments, this thesis aims to integrate the algebraic perspective of quantum mechanics with the study of quantum nonlocality in quantum network scenarios, establishing a foundation for future research and uncovering potential connections and insights.

1.1 Thesis outline

This thesis is divided into four main chapters and an appendix, providing a comprehensive treatment of the relevant concepts and techniques in operator algebras and their application to quantum nonlocality.

In Chapter 2, surveying across [8, 11, 30, 33, 38], we delve into the algebraic formulation of quantum mechanics, examining the underlying algebraic structures and their applications. This chapter begins with an introduction to the theory of C^*-algebras, which models physical observables and provides a powerful abstraction for quantum mechanics. We then explore (algebraic) states and representations of C^*-algebras, allowing us to associate abstract algebras with concrete Hilbert spaces and their bounded operators through the Gelfand-Naimark-Segal (GNS) construction. Following this, we investigate von Neumann algebras, a more specialised subclass of C^*-algebras with unique properties that are particularly relevant to quantum theory. Superselection rules, which are important physical phenomena restricting feasible observables, are briefly introduced after the von Neumann–Murray decomposition theorem. Next, composite systems and independent subsystems are discussed, where the notion of tensor product of C^*-algebras and von Neumann algebras are examined. Finally, the chapter concludes by presenting the algebraic quantum postulates A'–C' based on Heisenberg's picture to contrast the standard quantum postulates A–C.

Chapter 3 provides an overview of quantum correlations, focusing on nonlocal correlations in the context of two-player nonlocal games. We discuss the geometry of correlation sets [12], the CHSH game [14], and Tsirelson's problem [21, 58, 69]. Moreover, we introduce the Navascués-Pironio-Acín (NPA) hierarchy [42], a powerful technique for analysing quantum correlations and characterising the set of feasible quantum correlations. This chapter serves as a foundation for understanding the significance and implications of quantum correlations for the subsequent chapter.

In Chapter 4, we apply the mathematical formalism introduced in previous chapters to study quantum nonlocality, specifically focusing on the quantum bilocal network scenario, which involves a line arrangement of three parties, namely Alice, Bob, and Charlie, who share two independent quantum sources as in Figure 4.2. The chapter discusses the factorisation bilocal NPA hierarchy and its convergence to a relaxation of tensor bilocal quantum correlations [57]. Furthermore, the chapter explores algebraic insights into the quantum bilocal network, discussing alternative models and their equivalence, as well as addressing the bilocal Tsirelson's problem [35]. Although relevant, the primary focus of the chapter is not on scalar extension [52, 57], inflation technique [34, 35, 74], and Positivstellensätze of state polynomials [32], thus we only briefly mention these topics in remarks.

In Appendix A, we provide an overview of the basics of functional analysis based on [8, 30, 33, 38], including Banach spaces and Banach algebras, the Spectral Theorem for bounded self-adjoint operators, Hilbert space operator topologies, and important theorems in functional analysis. This appendix serves as a preparation for Chapter 2 and includes some key functional analysis results that are ubiquitous in quantum mechanics.

1.2 Contribution and discussion

This thesis serves as a reference for results in the theory of operator algebras relevant to the study of quantum nonlocality, introducing notable results and attempting to draw connections between them. While the main contribution is not the original application of operator algebras to quantum bilocal network scenarios, it does summarise two recent works [35, 57] in the same notation, highlighting the significance of these works in solving problems in the bilocal scenario with new insights. The thesis emphasises the value of algebraic insights in understanding quantum nonlocality and showcases the usefulness of operator algebras in providing a deeper understanding of quantum mechanics and its underlying structure. Therefore, the thesis' contribution is mainly in surveying various publications and making the connections between these works more explicit.

Potential future research directions include incorporating the existing methods in further investigation of bilocal quantum scenarios, and the development of algebraic models for more general quantum networks and their implications in quantum foundations. Among many possibilities, there are:

- One of the open problems is the convergence of the scalar extension technique [52] in the bilocal scenario, as attempted in [57, Appendix C] without a full resolution. The recent work on Positivstellensätze of state polynomials [32] seems to provide insights from the primal point of view, and understanding it in the context of the dual perspective [57] might be the key.
- The generalisation of the algebraic formulation to more general quantum network scenarios is a promising direction. In particular, one can consider the quantum inflation technique [74], which is a powerful method for characterising correlations in general network scenarios. Therefore, it would be interesting to explore the convergence of the quantum inflation-NPA hierarchy to alternative quantum models in generalised networks.

In summary, this thesis presents a comprehensive exploration of the algebraic perspective in quantum mechanics and its applications in understanding quantum correlations and the bilocal network scenario. Serving as a reference for quantum nonlocality studies, the thesis connects recent works and

highlights the significance of algebraic insights. Furthermore, it lays the groundwork for future research directions, such as investigating bilocal quantum scenarios and developing algebraic models for more general quantum networks. In conclusion, this thesis offers insights from theory of operator algebras into the current understanding of quantum correlations and may serve as a basis for further exploration in this field.

Chapter 2

Algebraic structures in quantum mechanics

In this chapter, we delve into the algebraic formulation of quantum mechanics, a complementary perspective to the traditional Schrödinger's picture. By examining the underlying algebraic structures of quantum mechanics, we gain insights and discover numerous applications.

Primarily based on [8, 11, 30, 33, 38], we begin with an introduction to C^*-algebras, which model physical observables. Following this, we explore (algebraic) states, representations, and the Gelfand-Naimark-Segal (GNS) construction, which associates abstract algebras with concrete Hilbert spaces and their bounded operators. Subsequently, we discuss von Neumann algebras, a specialized subclass with properties particularly relevant to quantum theory. The chapter also investigates superselection rules and their implications for quantum theory, as well as the notion of tensor products in the algebraic framework, essential for multipartite quantum systems. Finally, we conclude the chapter by contrasting algebraic quantum postulates with standard quantum postulates, highlighting their differences and similarities.

To provide context for the forthcoming algebraic definitions, we present the standard formulation of postulations in quantum theory, namely the Schrödinger's Hilbert space–vector state approach. For the purposes of this thesis, we will not consider the evolution of quantum systems, thus focusing on a framework that omits this aspect. The following discussion assumes finite-dimensionality for simplicity and draws on the classical reference for quantum information theory [50].

Definition 2.1 *A* density operator *(or density matrix) ρ on a Hilbert space $\mathcal{H}$ is a positive operator that is normalised. That is, $\langle\varphi|\rho|\psi\rangle \geq 0$ for any $|\varphi\rangle, |\psi\rangle \in \mathcal{H}$ and $\mathrm{Tr}(\rho) = 1$. The set of all density operators $\mathcal{S}(\mathcal{H})$ is called the* state space.

We say that ρ is a pure state *if it is an extreme point of the state space, i.e. an element of $\mathcal{P}(\mathcal{H}) = \mathrm{ex}(\mathcal{S}(\mathcal{H}))$, otherwise it is a* mixed state.

From the definition, it is clear that the state space $\mathcal{S}(\mathcal{H})$ is convex, and with the (finite-dimensional) spectral theorem, we conclude that any pure state assumes the form $|\varphi\rangle\langle\varphi|$, where $|\varphi\rangle \in \mathcal{H}$ is a normalised vector.

The postulates, modulo the evolutionary aspect, are as follows:

A An isolated physical system S is described by a density operator ρ on some Hilbert space $\mathcal{H}$.

B An *observable* of the system S is represented by a self-adjoint operator $O = \sum_x P_x$ on $\mathcal{H}$, where each x is an eigenvalue and P_x is the (orthogonal) projector onto the respective eigenspace. Each eigenvalue x of O corresponds to a possible measurement outcome, with the probability of observing x given by the *Born's rule*:

$$p(x) = \mathrm{Tr}(\rho P_x) = \mathrm{Tr}_\rho(P_x). \tag{2.1}$$

Finally, the post-measurement state is given by

$$\rho' = \frac{P_x \rho P_x}{\mathrm{Tr}_\rho(P_x)}. \tag{2.2}$$

C The composite system is described by the tensor product of the state space of its component system. That is, if we have two systems represented by $\mathcal{H}_A, \mathcal{H}_B$ with states ρ_A, ρ_B, then the joint state space is $\mathcal{H}_A \otimes \mathcal{H}_B$ with the joint state $\rho_A \otimes \rho_B$.

With respect to the standard quantum postulates A–C, goal of this chapter is to develop the necessary notions for the algebraic quantum postulates A′–C′ in Section 2.5.

2.1 C^*-algebras prelimnary

Theory of C^*-algebra was developed with the aim of modelling physical observables, and is a powerful abstraction that happens to be a speacial case of Banach algebras; see A.1 for more detailed definition of Banach spaces and algebras. We refer to [11, 33] for this section.

Definition 2.2 *1. An* involution *on an algebra $\mathfrak{A}$ is a real linear map $\dagger : \mathfrak{A} \to \mathfrak{A}$, $\alpha \mapsto \alpha^\dagger$, such that for $\alpha, \beta \in \mathfrak{A}$ and $a, b \in \mathbb{C}$,*

(a) $\alpha^{\dagger\dagger} = \alpha$

(b) $(\alpha\beta)^\dagger = \beta^\dagger \alpha^\dagger$

(c) $(a\alpha + b\beta)^\dagger = a^\dagger\alpha^\dagger + b^\dagger\beta^\dagger$.

2. *A* *-algebra $\mathfrak{A}$ *is an algebra equipped with an involution. A subset* $\mathfrak{B} \subset \mathfrak{A}$ *is* self-adjoint *if it is closed under involution*

3. *A* Banach algebra *is a Banach space* $\mathfrak{A}$ *that is also an algebra such that*

$$\|\alpha\beta\| \leq \|\alpha\|\|\beta\| \tag{2.3}$$

for any $\alpha, \beta \in \mathfrak{A}$.

4. *A* C^*-algebra $\mathfrak{A}$ *is a Banach* **-algebra satisfying*

$$\|\alpha^\dagger\alpha\| = \|\alpha\|^2 \tag{2.4}$$

for any $\alpha \in \mathfrak{A}$.

We note that

$$\|\alpha^\dagger\| = \|\alpha\|. \tag{2.5}$$

Since $\|\alpha\|^2 = \|\alpha^\dagger\alpha\| \leq \|\alpha^\dagger\|\|\alpha\|$ implies $\|\alpha\| \leq \|\alpha^\dagger\|$, and the same argument shows another side of the inequality.

We motivate the definition of C^*-algebras with the following examples.

Example 2.3 *1. Consider the set of all bounded operators* $\mathcal{B}(\mathcal{H})$ *of some Hilbert space* $\mathcal{H}$*. It is an* **-algebra with the usual composition and the usual adjoint operator* $\dagger$*, equipped with the operator norm*

$$\|\alpha\| = \sup_{|\varphi\rangle \in \mathcal{H}, \||\varphi\rangle\| = 1} \{\|\alpha\,|\varphi\rangle\|\}. \tag{2.6}$$

We check the C^**-algebra property that*

$$\begin{aligned} \|\alpha\|^2 &= \sup_{\||\varphi\rangle\|=1} \{\|\alpha\,|\varphi\rangle\|^2\} \\ &= \sup_{\||\varphi\rangle\|=1} \{\langle\varphi|\alpha^\dagger\alpha|\varphi\rangle\} \\ &\leq \sup_{\||\varphi\rangle\|=1} \{\|\alpha^\dagger\alpha\,|\varphi\rangle\|\} \\ &= \|\alpha^\dagger\alpha\| \leq \|\alpha^\dagger\|\|\alpha\| = \|\alpha\|^2. \end{aligned} \tag{2.7}$$

2. *The space* $\mathcal{B}_0(\mathcal{H})$ *of all compact operators with all operations inherited from* $\mathcal{B}(\mathcal{H})$ *is a* C^**-algebra. It is uniformly closed, as the uniform limit of a set of compact operators is again compact, and is a two-sided ideal in* $\mathcal{B}(\mathcal{H})$ *that is not unital if* $\mathcal{H}$ *is infinite-dimensional.*

3. *The space $C_0(X)$ of all continuous functions $f : X \to \mathbb{C}$ vanishing at infinity (i.e. for any $\varepsilon > 0$ there exists a compact set $K \subset X$ such that $|f(x)| < \varepsilon$ for all $x \in K$), where X is a locally compact space, is a commutative C^*-algebra. It is a commutative *-algebra with the usual pointwise addition, scalar multiplication, and pointwise function multiplication, and the *-structure comes from the usual complex conjugation $\dagger$. We introduce the* sup *norm*

$$\|f\| = \sup_{x \in X}\{|f(x)|\} \tag{2.8}$$

and one can check $\|ff^\dagger\| = \sup_{x\in X}\{|f(x)|^2\} = \|f\|^2$.

Example 2.3.(2) above shows that the C^*-algebra $\mathfrak{A}$ does not need to admit a unit in general. However, there is a procedure (cf. [11, Proposition 2.1.5]) to embed them into a larger unital algebra $\tilde{\mathfrak{A}} = \{(a, \alpha) \mid a \in \mathbb{C}, \alpha \in \mathfrak{A}$, in which the algebraic rules are $(a, \alpha) + (b, \beta) = (a + b, \alpha + \beta)$, $(a, \alpha)^\dagger = (a^\dagger, \alpha^\dagger)$, and $(a, \alpha)(b, \beta) = (ab, a\beta + b\alpha + \alpha\beta)$. The norm of the enlarged algebra is $\|(a, \alpha)\| = \sup_{\beta\in\mathfrak{A},\|\beta\|=1}\{\|a\beta + \alpha\beta\|\}$. For the scope of this thesis, we consider only unital C^*-algebras.

2.1.1 Selected C^*-algebra objects

Recall that the residue $\rho(\alpha)$ of $\alpha \in \mathfrak{A}$ is the subset of $\mathbb{C}$ such that $\mathbb{1} - \lambda\alpha$ is invertible, and the spectrum $\sigma(\alpha) = \mathbb{C} \setminus \rho(\alpha)$; see Appendix A.1.3 and Appendix A.2.1. We introduce some special classes of elements.

Definition 2.4 *An element of some unital C^*-algebra, $\alpha \in \mathfrak{A}$, is*

1. normal *if $\alpha\alpha^\dagger = \alpha^\dagger\alpha$;*
2. self-adjoint *if $\alpha^\dagger = \alpha$;*
3. isometry *if $\alpha^\dagger\alpha = \mathbb{1}$;*
4. unitary *if $\alpha\alpha^\dagger = \mathbb{1} = \alpha^\dagger\alpha$;*
5. positive *if α is self-adjoint and $\sigma(\alpha) \subset [0, \infty)$, denoted as $\alpha \geq 0$.*

The significance of these operators is that their spectrum has special properties; see Theorem A.13.

We observe that $\alpha \in \mathfrak{A}$ is positive iff $\alpha = \beta^\dagger\beta$ for some $\beta \in \mathfrak{A}$. Positive operators are significant in the theory of C^*-algebras, as they provide the notion of comparing different elements, underpin the notion of square root of operators, and give unique decompositions of self-adjoint operators (cf. [8, Chapter II.3.1] and [11, Chapter 2.2.2].

Finally, we include the definitions of homomorphisms and isomorphisms.

Definition 2.5 *1. A* *-morphisms (homomorphism) *between the C^*-algebra $\mathfrak{A}, \mathfrak{B}$ is a linear map $\Phi : \mathfrak{A} \to \mathfrak{B}$ such that*

$$\Phi(\alpha\beta) = \Phi(\alpha)\Phi(\beta),\ \Phi(\alpha^\dagger) = \Phi(\alpha)^\dagger. \tag{2.9}$$

2. An isomorphism $\Phi : \mathfrak{A} \to \mathfrak{B}$ *is an invertible homomorphism.*

Any * -morphisms preserve positivity (i.e. $\alpha \geq 0$ then $\Phi(\alpha) \geq 0$) and are continuous with $\|\Phi(\alpha)\| \leq \|\alpha\|$. A subtler fact is that any C^*-algebra isomorphism is also an isometry (injectivity is already enough). Analogously to other branches of algebra, the image of $\Phi(\mathfrak{A})$ is a closed C^*-subalgebra of $\mathfrak{B}$, and the kernal $\ker\Phi$ is an ideal of $\mathfrak{A}$; see [33, Theorem C.62].

Definition 2.6 *Let $\mathfrak{A}$ be a Banach algebra (in particular a C^*-algebra). A* left (right) ideal *in $\mathfrak{A}$ is a closed linear subspace $\mathfrak{J}$ such that for any $\alpha \in \mathfrak{A}$, $\beta \in \mathfrak{J}$, the element $\alpha\beta \in \mathfrak{J}$ ($\beta\alpha \in \mathfrak{J}$). An* ideal *is a two-sided ideal (both left and right).*

2.1.2 Commutative Gelfand–Naimark Theorem

In fact, Example 2.3.(3) has more significance as it captures the structure of commutative C^*-algebras.

Theorem 2.7 (Gelfand–Naimark theorem, commutative case) *Every commutative C^*-algebra $\mathfrak{A}$ is isomorphic to the space $C_0(X)$ of complex-valued continuous functions on some locally compact space X. The isomorphism is unique up to homeomorphisms.*

The proof is technical and less relevant to the main theme of this thesis; we present only the main ideas of this theorem. Let $\mathfrak{A}$ be a commutative C^*-algebra.

1. The space X can be taken to be the *Gelfand spectrum* $\Sigma(\mathfrak{A})$, which is the set of all *characters* on $\mathfrak{A}$. A *character*[1] $\omega : \mathfrak{A} \to \mathbb{C}$ is a nonzero linear map satisfying

   $$\omega(\alpha\beta) = \omega(\alpha)\omega(\beta) \tag{2.10}$$

   for all α, β.

2. The isomorphism is given by the *Gelfand transform* $\mathfrak{A} \to C_0(\Sigma(\mathfrak{A}))$, which maps each $\alpha \in \mathfrak{A}$ to a function $\hat{\alpha} : \Sigma(\mathfrak{A}) \to \mathbb{C}$ defined by

   $$\hat{\alpha}(\omega) = \omega(\alpha). \tag{2.11}$$

3. We endow $X = \Sigma(\mathfrak{A})$ with *Gelfand topology*, which is the weakest topology on $\Sigma(\mathfrak{A})$ making all functions $\hat{\alpha}$ continuous. With Gelfand topology, $\Sigma(\mathfrak{A})$ is compact iff it is unital, otherwise it is locally compact.

[1] For commutative C^*-algebra, the notions of characters and pure states (discussed in later section) are equivalent.

Example 2.8 *We provide a simple example with a finite space* $X = \{1, \cdots, N\}$ *and the space of complex-valued continuous functions* $C(X) = C_0(X)$*. Obviously* $C(X) \simeq \mathbb{C}^N$ *with the identification* $f \sim \mathrm{diag}\,(f(1), \cdots, f(n))$*.*

For each $x \in X$*, it can be checked that the evaluation map* $\mathrm{ev}_x \in C_0(X), \mathrm{ev}_x(f) = f(x)$*, is a character, while a more involved proof of the maximality of the kernel of* $\mathrm{ev} : x \mapsto \mathrm{ev}_x$ *shows that every character must be an evaluation map. Hence* $\Sigma(C_0(X)) = \{\mathrm{ev}_x \mid x \in X\} \simeq X$*.*

Finally, it is clear that the Gelfand transform reads $f \mapsto \hat{f}$*, where* $\hat{f}(\mathrm{ev}_x) = \mathrm{ev}_x(f) = f(x)$*. We observe an isomorphism between* $f : x \mapsto f(x)$ *and* $\hat{f} : \mathrm{ev}_x \mapsto f(x)$*.*

This correspondence, often referred to as Gelfand duality, is crucial for connecting abstract commutative C^*-algebras with more concrete spaces of continuous functions, where geometrical structures can be recovered by studying spectral objects. The generalisation of this idea to noncommutative cases has led to a fruitful field called noncommutative geometry [65].

2.2 States and representations of C^*-algebras

In this section, we delve into one of the more important definitions, namely the (algebraic) states on C^*-algebras, which are abstract generalisations of density operators in the Schrödinger picture. Furthermore, states allow us to associate abstract C^*-algebras with concrete Hilbert spaces and their bounded operators through the renowned Gelfand–Naimark–Segal (GNS) construction. Using representation theory and GNS construction, we can investigate the structure of C^*-algebras beyond the scope of the Gelfand–Naimark theorem for commutative C^*-algebras.

From now on, we focus on unital C^*-algebras for the sake of simplicity. The general cases can be dealt with unitalisation and use of approximate units. This section is jointly referenced from [8, 11, 30, 33, 38].

2.2.1 States

Let us begin with the abstraction of the usual quantum states.

Definition 2.9 (Algebraic states) *A* (algebraic) state *on a unital* C^**-algebra* $\mathfrak{A}$ *is a linear functional* $\omega : \mathfrak{A} \to \mathbb{C}$ *that is*

- positive*:* $\omega(a^\dagger a) \geq 0$ *for all* $a \in \mathfrak{A}$*;*
- normalised[2]*:* $\omega(\mathbb{1}) = 1$*.*

[2]This definition relies on the existence of an identity. In general, the normalisation condition reads $\|\omega\| = 1$, where the norm of the space $\mathfrak{A}^*$ of linear functionals is defined by$\|\omega\| = \sup_{\|a\|=1}\{|\omega(a)|\}$.

The state space $\mathcal{S}(\mathfrak{A})$ *is the set of all states on* $\mathfrak{A}$. *The* pure state space $\mathcal{P}(\mathfrak{A})$ *is the extreme boundary of the state space* $\mathrm{ex}(\mathcal{S}(\mathfrak{A}))$ *with elements called* pure states. *More explicitly, suppose that a state* $\omega = t\omega_1 + (1-t)\omega_2$, *for some* $t \in [0,1]$ *and states* ω_1, ω_2. *Then* ω *is pure iff* $\omega = \omega_1 = \omega_2$.

From the definition it is immediate that the state space $\mathcal{S}(\mathfrak{A})$ is convex. Moreover, by the Banach–Alaoglue Theorem A.24 we further conclude that $\mathcal{S}(\mathfrak{A})$ is a compact convex subset of $\mathfrak{A}^*$ relative to the weak-* topology. Then the application of Krein–Milman Theorem A.26 leads to the following result; observe the analogy of this result with the usual notion pure-state/mixed-state in Definition 2.1.

Theorem 2.10 *Consider a unital* C^**-algebra* $\mathfrak{A}$ *with the state space* $\mathcal{S}(\mathfrak{A})$ *and the pure state space* $\mathcal{P}(\mathfrak{A})$. *Then* $\mathcal{P}(\mathfrak{A}) \neq \emptyset$ *and* $\mathcal{S}(\mathfrak{A}) = \overline{\mathrm{co}(\mathcal{P}(\mathfrak{A}))}$.

That is, pure states on $\mathfrak{A}$ *always exist and all states on* $\mathfrak{A}$ *can be approximated by a sequence of probability mixtures (i.e., convex combinations) of pure states.*

Physically, given a self-adjoint $\alpha \in \mathfrak{A}$, the value of $\omega(\alpha)$ should be interpreted as the expectation value of the observable α in the state ω. Similarly, for a projector P_E given by a projector-valued measure (PVM) $X \to \mathfrak{A}$ as in Definition A.14, $\omega(P_E)$ is to be understood as the probability of observing the event E, where X is some measure space and E is a Borel set [38, Chapter 8.2.3]. For a more extensive discussion of PVMs, see Appendix A.2.2.

Example 2.11 *Consider a Hilbert space* $\mathcal{H}$ *with the* C^**-algebra* $\mathcal{B}(\mathcal{H})$.

1. *The formula* $\omega(\alpha) = \langle\varphi|\alpha|\varphi\rangle$ *defines a state* ω *on* $\mathcal{B}(\mathcal{H})$, *where* $|\varphi\rangle \in \mathcal{H}$ *is a normalised vector. The state* ω *is, in fact, pure and any pure state also assumes this form. Similarly, any density operator gives rise to an algebraic state using the formula* $\omega(\alpha) = \mathrm{Tr}(\rho\alpha)$.

   That is, any density operator defines an algebraic state ω.

2. *Let us assume that* $\mathcal{H}$ *is finite-dimensional. Given any state* ω, *we define the matrix* ρ *through* $\rho_{ij} = \omega(E_{ji})$, *where* E_{ji} *is the matrix with* 0 *everywhere but* 1 *at the* (j,i)*-th entry.*

   We show that ρ *is indeed a density operator. Indeed, for any matrix* α *we check*

$$\mathrm{Tr}(\rho\alpha) = \sum_{i,j} \rho_{ij}\alpha_j i = \sum_{i,j} \omega(E_{ji})\alpha_{ji} = \omega(\sum_{i,j} E_{ji}\alpha_{ji}) = \omega(\alpha), \qquad (2.12)$$

   recovering the formula in the previous example. Then ρ *is normalised considering* $\alpha = \mathbb{1}_{\mathcal{H}}$, *and it is positive since* $\langle\varphi|\rho|\varphi\rangle = \omega(|\varphi\rangle\langle\varphi|) \geq 0$.

The example above shows that in a finite-dimensional Hilbert space, algebraic states and the usual quantum states are equivalent to each other. This realisation is sufficient for many applications in quantum information theory. However, this is in general not the case.

Remark 2.12 *The notion of algebraic state is weaker than the usual quantum states. We define a state ω on $\mathcal{B}(\mathcal{H})$ as a* normal state *if it is induced by a density operator ρ through $\omega(\alpha) = \mathrm{Tr}(\rho\alpha)$. Then there are algebraic states that are not normal.*

Nonnormal states are typically difficult to express explicitly, similar to pathological examples of continuity and differentiability in analysis. Nevertheless, one can attempt to gain further insight with measure theory. If we fix a PVM $P : X \to \mathcal{B}(\mathcal{H})$ as in Definition A.14, then a state ω defines a measure $\mu : X \to \mathbb{R}$ through the formula $\mu(E) = \omega(P_E)$ for any set $E \subset X$.

Recall that a measure μ is additive if for any finite family of disjoint sets $E_1, \cdots, E_N$, we have $\mu(\cup_i E_i) = \sum_i \mu(E_i)$, and is σ-additive (resp. completely additive) if we can generalise this to a countable (resp. uncountable) disjoint union. It should then be noted that, in general, algebraic states induce additive measures that are not necessarily σ-additive or completely additive [38]. On the other hand, the measures induced by normal states are completely additive [30].

We present technical results found in [11, Lemma 2.3.10] and [8, II.6.3.3] that are useful for the construction of GNS, providing further insight into the relationship between states and observable quantities in quantum systems.

Lemma 2.13 (Cauchy–Schwarz inequality) *If $\omega : \mathfrak{A} \to \mathbb{C}$ is a positive linear functional (in particular a state) on a unital *-algebra $\mathfrak{A}$ (in particular a C^*-algebra state), then*

(a) $\omega(\alpha^\dagger\beta) = \omega(\alpha^\dagger\beta)^\dagger$,

(b) $|\omega(\alpha^\dagger\beta)| \leq \omega(\alpha^\dagger\alpha)\omega(\beta^\dagger\beta)$,

(c) $|\omega(\beta^\dagger\alpha\beta)| \leq \|\alpha\|\omega(\beta^\dagger\beta)$,

(d) and $\omega(\alpha^\dagger) = \omega(\alpha)^\dagger$.

Lemma 2.14 *Let $\alpha \in \mathfrak{A}$ be a self-adjoint element. Then there exists a pure state ω such that $|\omega(\alpha)| = \|\alpha\|$*

2.2.2 Representations

In the discussion of normality in Remark 2.12, we have to work with a concrete Hilbert space as opposed to abstract C^*- algebras. To overcome this, we introduce the notion of representation.

Definition 2.15 *A* representation *of a C^*-algebra $\mathfrak{A}$ is a pair $(\mathcal{H}, \pi)$, where $\mathcal{H}$ is a Hilbert space, and $\pi : \mathfrak{A} \to \mathcal{B}(\mathcal{H})$ is a *-morphism. Moreover, we have the following:*

1. *A representation $(\mathcal{H}, \pi)$ is* faithful *if π is injective, whence $\mathfrak{A} \simeq \pi(\mathfrak{A})$.*
2. *A nonzero representation $(\mathcal{H}, \pi)$ is* nondegenerate *if for $|\psi\rangle \in \mathcal{H}$ and all $\alpha \in \mathfrak{A} \setminus \{0\}$, $\pi(\alpha)|\psi\rangle = 0$ implies $|\psi\rangle = 0$.*

3. *A vector* $|\varphi_1\rangle \in \mathcal{H}$ *is* cyclic *for* π *if the closure of* $\pi(\mathfrak{A})|\varphi_1\rangle$ *equals to* $\mathcal{H}$. *A* cyclic representation *is a triple* $(\mathcal{H}, \pi, |\varphi_1\rangle)$, *where* $(\mathcal{H}, \pi)$ *is a representation, and* $|\varphi_1\rangle$ *is cyclic for* π.

4. *Let* $(\mathcal{H}_x, \pi_x)_{x\in I}$ *be a family of representations of* $\mathfrak{A}$, *indexed by a possibly uncountable set* I. *The* direct sum representation[3] $(\mathcal{H} = \oplus_{x\in I}\mathcal{H}_x, \pi = \oplus_{x\in I}\pi_x)$ *is defined by*

$$(\pi(\alpha)|\psi\rangle)_x = \pi_x(\alpha)|\psi_x\rangle\,,$$

for all $\alpha \in \mathfrak{A}$ *and* $|\psi\rangle \in \mathcal{H}$.

5. *Two representations* $(\mathcal{H}_1, \pi_1)$ *and* $(\mathcal{H}_2, \pi_2)$ *of* $\mathfrak{A}$ *are* unitarily equivalent *if there exists a unitary* $u : \mathcal{H}_1 \to \mathcal{H}_2$ *such that* $u\pi_1(\alpha)u^\dagger = \pi_2(\alpha)$ *for each* $\alpha \in \mathfrak{A}$.

We observe that cyclic representations are automatically nondegenerate, via a proof by contradiction with the orthogonality argument.

Remark 2.16 *From now on, we use the "hat" notation to distinguish elements* α *in the abstract* C^**-algebra and concrete operators* $\hat{\alpha}$ *in* $\mathcal{B}(\mathcal{H})$. *Furthermore, if there is no ambiguity in the represented Hilbert space* $\mathcal{H}$, *we write* π *in place of* $(\mathcal{H}, \pi)$ *and denote* $\pi(\alpha) = \hat{\alpha}$.

The faithfulness is important since it is equivalent to π being norm-preserving ($\|\pi(\alpha)\| = \|\hat{\alpha}\| = \|\alpha\|$) and to positive definiteness preserving ($\pi(\alpha) > 0$ for all $\alpha > 0$) [11, Proposition 2.3.3].

Any cyclic representation, as mentioned, is nondegenerate. We observe a "converse statement" relating these notions to direct sum representations.

Proposition 2.17 *Any nondegenerate representation* $(\mathcal{H}, \pi)$ *of a* C^**-algebra* $\mathfrak{A}$ *is a direct sum of cyclic representations of* $\mathfrak{A}$.

Proof We provide a sketch as the construction will be useful later. Using Zorn's lemma, we define a maximal family of nonzero vectors $(\varphi_1^x)_{x\in I}$ in $\mathcal{H}$ such that $(\langle\varphi_1^y|\,\pi(\alpha)^\dagger)(\pi(\beta)|\varphi_1^x\rangle) = 0$, for all $\alpha, \beta \in \mathfrak{A}$ whenever $x \neq y$. Then, for each $x \in I$, define $\mathcal{H}_x$ as the closure of $\pi(\mathfrak{A})|\varphi_1^x\rangle$ in $\mathcal{H}$ and π_x as the restriction to $\mathcal{H}_x$. This leads to the cyclic representation $(\mathcal{H}_x, \pi_x, |\varphi_1^x\rangle)$ for each $x \in I$. □

We end the section by introducing the notion of irreducibility.

Definition 2.18 *Let* $(\mathcal{H}, \pi)$ *be a representation of* $\mathfrak{A}$.

[3] A finite direct sum of Hilbert space is defined the usual way, and infinite direct sum is defined as a direct limit. Thus, each $\psi \in \mathcal{H}$ takes the form $|\psi\rangle = (|\psi_x\rangle)_{x\in I}$ where $|\psi_x\rangle \in \mathcal{H}_x$. It can be checked that the map π defined this way is indeed a representation, see [33, p. 694-695] and [11, 45]

1. *A closed subspace $\mathcal{H}'$ is* invariant *under π of $\pi(\mathfrak{a})\mathcal{H}' \subset \mathcal{H}'$ for all $\mathfrak{a} \in \mathfrak{A}$. We then say that the restriction π' on $\mathcal{H}'$ is a* subrepresentation *$(\mathcal{H}', \pi')$ of $(\mathcal{H}, \pi)$.*
2. *A representation $(\mathcal{H}, \pi)$ of $\mathfrak{A}$ is* irreducible *if the only closed invariant subspaces are $\{0\}$ and $\mathcal{H}$, or equivalently, $(\mathcal{H}, \pi)$ admits no proper non-trivial subrepresentation.*

Denote the commutant of $S \subset \mathcal{B}(\mathcal{H})$ by $S' = \{\hat{\mathfrak{a}} \mid [\hat{\mathfrak{a}}, \hat{\beta}] = 0,\ \forall \hat{\beta} \in S\}$. We have the following characterisations of irreducibility [33, Theorem C.90].

Theorem 2.19 (Schur's Lemma) *For a representation $(\mathcal{H}, \pi)$ of $\mathfrak{A}$, the following conditions are equivalent:*

1. *$(\mathcal{H}, \pi)$ is irreducible;*
2. *$\pi(\mathfrak{A})' = \mathbb{C}\mathbb{1}_{\mathcal{H}}$;*
3. *$\pi(\mathfrak{A})'' = \mathcal{B}(\mathcal{H})$*
4. *Every vector in $\mathcal{H}$ is cyclic for π.*

2.2.3 Gelfand–Naimark–Segal (GNS) construction

We have discussed the properties of representations of C^*-algebra, now we show explictly the existence with *Gelfand–Naimark–Segal (GNS) construction* since they are important to the discussion in the latter chapters.

Theorem 2.20 (Gelfand–Naimark–Segal construction) *Let ω be a state on the C^*-algebra $\mathfrak{A}$. Then there exists a unique (up to unitary equivalence) cyclic representation $(\mathcal{H}_\omega, \pi_\omega)$ of $\mathfrak{A}$ and a cyclic vector $|\varphi_1^\omega\rangle$ such that*

$$\omega(\mathfrak{a}) = \langle \varphi_1^\omega | \pi(\mathfrak{a}) | \varphi_1^\omega \rangle = \langle \varphi_1^\omega | \hat{\mathfrak{a}} | \varphi_1^\omega \rangle \tag{2.13}$$

for all $\mathfrak{a} \in \mathfrak{A}$. Consequently, $|\varphi_1^\omega\rangle$ is a unit vector.

The triple $(\mathcal{H}_\omega, \pi_\omega, |\varphi_1^\omega\rangle)$ is often referred to as *GNS representation*, and the subscripts/superscripts ω are often omitted if the context is clear.

Proof The proof is consisted of four steps.

Step 1. We first construct the Hilbert space $\mathcal{H}_\omega$ as the completion of the quotient space $\mathfrak{A}/N_\omega$ with respect to the state-induced inner product.

- Defining $N_\omega = \{\mathfrak{a} \in \mathfrak{A} \mid \omega(\mathfrak{a}^*\mathfrak{a}) = 0\}$, we claim that N_ω is a left ideal. Indeed, letting $\mathfrak{a} \in N_\omega$ and $\beta \in \mathfrak{A}$, we use the Cauchy-Schwarz lemma 2.13 to compute

$$0 \leq \omega((\beta\mathfrak{a})^\dagger(\beta\mathfrak{a})) \leq \omega(\mathfrak{a}^*\mathfrak{a})\omega(\beta^*\beta) = 0 \tag{2.14}$$

  which implies that $\beta\mathfrak{a} \in N_\omega$.

- Define a pre-inner product on $\mathfrak{A} \times \mathfrak{A}$ by mapping (α, β) to $\omega(\beta^\dagger \alpha)$. The hermitian property and positivity of this pre-inner product can be attributed to the Cauchy-Schwarz lemma 2.13 once more. However, it may not be positive definite, since there may be some $\alpha \in \mathfrak{A}$ with $\alpha \neq 0$ but $\omega(\alpha^\dagger \alpha) = 0$. But then this α belongs to N_ω by definition.
- Therefore, for the quotient space $\mathfrak{A}/N_\omega$ where the equivalence class of each $\alpha \in \mathfrak{A}$ is denoted by $[\alpha] = |\varphi_\alpha^\omega\rangle = |\varphi_{\alpha+\nu}^\omega\rangle \, \forall \nu \in N_\omega$, we have constructed a well-defined inner product by

$$\langle \varphi_\alpha^\omega | \varphi_\beta^\omega \rangle = \omega(\alpha^\dagger \beta). \tag{2.15}$$

  The completion of $\mathfrak{A}/N_\omega$ relative to this inner product is the Hilbert space $\mathcal{H}_\omega$ that we wish.

Step 2. Next, we construct the representation $\pi_\omega : \mathfrak{A} \to \mathcal{B}(\mathcal{H})$ as the left-multiplication

$$\pi_\omega(\alpha) \left| \varphi_\beta^\omega \right\rangle = \left| \varphi_{\alpha\beta}^\omega \right\rangle. \tag{2.16}$$

- We begin by considering the map $\pi_\omega(\alpha) : \mathfrak{A}/N_\omega \to \mathfrak{A}/N_\omega$ defined by the same formula. Then for any $|\varphi_\alpha^\omega\rangle \in \mathfrak{A}/N_\omega$, it is well-defined since N_ω is a left-ideal and its *-morphism properties can be checked directly.
- It is also bounded (hence continuous), since

$$\begin{aligned} \left\| \pi_\omega(\alpha) \left| \varphi_\beta^\omega \right\rangle \right\|^2 &= \langle \varphi_\beta^\omega | \varphi_\beta^\omega \rangle \\ &= \omega(\beta^\dagger \alpha^\dagger \alpha \beta) \leq \|\alpha\|^2 \omega(\beta^\dagger \beta) \\ &= \|\alpha\|^2 \langle \varphi_\beta^\omega | \varphi_\beta^\omega \rangle = \|\alpha\|^2 \left\| \left| \varphi_\beta^\omega \right\rangle \right\|^2 \end{aligned} \tag{2.17}$$

  by Lemma 2.13.
- By continuity, we extend $\pi_\omega(\alpha) : \mathfrak{A}/N_\omega \to \mathfrak{A}/N_\omega$ to an operator in $\mathcal{B}(\mathcal{H}_\omega)$. Thus, we check $\pi_\omega : \mathfrak{A} \to \mathcal{B}(\mathcal{H})$ is indeed a representation.

Step 3. We set the cyclic vector to be the equivalence class of the identity $[\mathbb{1}] = |\varphi_1^\omega\rangle$. With this we recover the desired equation $\langle \varphi_1^\omega | \pi(\alpha) | \varphi_1^\omega \rangle = \langle \varphi_1^\omega | \varphi_\alpha^\omega \rangle = \omega(\alpha)$. To see the cyclicity, note that $\mathfrak{A}/N_\omega = \{ |\varphi_\alpha^\omega\rangle \mid \alpha \in \mathfrak{A} \} = \{ \pi(\alpha) \, |\varphi_1^\omega\rangle \mid \alpha \in \mathfrak{A} \}$.

Step 4. It remains to show the uniqueness, that is, suppose that there is another cyclic representation $(\mathcal{H}, \pi, |\psi\rangle)$ satisfying Equation 2.13, then there exists a unitary operator $u : \mathcal{H}_\omega \to \mathcal{H}$ such that

$$u \pi_\omega(\alpha) u^\dagger = \pi(\alpha) \text{ and } u \, |\varphi_1^\omega\rangle = |\psi\rangle \tag{2.18}$$

for all $\alpha \in \mathfrak{A}$.

To this end, we simply define $u : \mathcal{H}_\omega \to \mathcal{H}$ by

$$u\pi_\omega(\alpha)\,|\varphi_1^\omega\rangle = \pi(\alpha)\,|\psi\rangle\,, \tag{2.19}$$

which is well-defined and isometric as it preserves the scalar product,

$$\langle\varphi_1^\omega|(u\pi_\omega(\alpha))^\dagger(u\pi_\omega(\beta))|\varphi_1^\omega\rangle_{\mathcal{H}_\omega} = \langle\psi|\pi(\alpha)^\dagger\pi(\beta)|\psi\rangle_{\mathcal{H}} \tag{2.20}$$

$$= \omega(\alpha^\dagger\beta) = \langle\varphi_1^\omega|\pi_\omega(\alpha)^\dagger\pi_\omega(\beta)|\varphi_1^\omega\rangle_{\mathcal{H}_\omega}. \tag{2.21}$$

As an isometry, it can be extended by continuity from $\mathfrak{A}/N_\omega$ to $\mathcal{H}_\omega$, then a direct density argument shows that it is the desired unitary. □

One nice feature of GNS representations is that they relate irreducibility to the purity of states on $\mathfrak{A}$.

Theorem 2.21 (Irreducibility and purity of GNS representations) *Let ω be a state on $\mathfrak{A}$ and let $(\mathcal{H}, \pi, |\varphi_1^\omega\rangle)$ be its GNS representation. Then the representation $(\mathcal{H}, \pi, |\varphi_1^\omega\rangle)$ is irreducible iff the state ω is pure.*

Proof For the "⇒" direction we refer to the proofs of [11, Theorem 2.3.19] or [8, II.6.4.], which relies on the Cauchy–Schwarz inequality for positive functionals and the property of a densely defined bounded sesquilinear form.

The converse direction is a nice exercise of Schur's lemma and the concept of commutant. Suppose, for contradiction, that there exists a non-trivial orthogonal projector P onto the commutant $\pi_\omega(\mathfrak{A})'$, i.e. $P \neq 0, \mathbb{1}$. Therefore, the commutativity implies $P\,|\varphi_1^\omega\rangle \neq 0 \neq (\mathbb{1} - P)\,|\varphi_1^\omega\rangle$.

With commutativity, we define

$$\begin{aligned} \lambda &= \|P\,|\varphi_1^\omega\rangle\| = \langle\varphi_1^\omega|P|\varphi_1^\omega\rangle \\ \omega_1(\alpha) &= \frac{\omega(P\alpha)}{\lambda} = \frac{\langle\varphi_1^\omega|P\pi(\alpha)P|\varphi_1^\omega\rangle}{\lambda} \\ \omega_1(\alpha) &= \frac{\omega((\mathbb{1}-P)\alpha)}{1-\lambda} = \frac{\langle\varphi_1^\omega|(\mathbb{1}-P)\pi(\alpha)(\mathbb{1}-P)|\varphi_1^\omega\rangle}{1-\lambda}, \end{aligned} \tag{2.22}$$

which we can check ω_1, ω_2 are states and $\omega = \lambda\omega_1 + (1-\lambda)\omega_2$. It follows from the purity assumption that $\omega = \omega_1 = \omega_2$.

In particular, we have

$$\langle\varphi_1^\omega|\pi(\alpha)\lambda|\varphi_1^\omega\rangle = \langle\varphi_1^\omega|P\pi(\alpha)P|\varphi_1^\omega\rangle = \langle\varphi_1^\omega|\pi(\alpha)P|\varphi_1^\omega\rangle. \tag{2.23}$$

By cyclicity we conclude that $P\,|\varphi_1^\omega\rangle = \lambda\,|\varphi_1^\omega\rangle$, which implies either $\lambda = 0$ or $\lambda = 1$ as P is projector. But then either $P\,|\varphi_1^\omega\rangle = 0$ or $(1-P)\,|\varphi_1^\omega\rangle = 0$, which is a contradiction. □

Remark 2.22 *Suppose that we have a density operator ρ on some Hilbert space $\mathcal{H}$ that is not pure, in the sense of standard quantum theory. Define the corresponding algebraic state ω on $\mathcal{B}(\mathcal{H})$ by $\omega(\hat{\alpha}) = \mathrm{Tr}(\rho\hat{\alpha})$ and construct a GNS representation $(\mathcal{H}_\omega, \pi_\omega, |\varphi_1^\omega\rangle)$. Then we obtain a pure state $|\varphi_1^\omega\rangle$ such that $\mathrm{Tr}(\rho\hat{\alpha}) = \langle\varphi_1^\omega|\hat{\alpha}|\varphi_1^\omega\rangle$. It is tempted to conclude that "a mixed state is transformed into a pure state".*

However, such naive deduction is incorrect, for there is no such transformation. First, $|\varphi_1^\omega\rangle$ lives in a different Hilbert space $\mathcal{H}_\omega$. Moreover, there is no spatial isomorphism $u : \mathcal{H} \to \mathcal{H}_\omega$ with $u\hat{\alpha}u^{-1} = \pi_\omega(\hat{\alpha})$ such that $u^{-1}|\varphi_1^\omega\rangle \in \mathcal{H}$. To see this, we note that ω is not pure, as its corresponding ρ is already not extremal in a smaller set (than in the space of algebraic states). But then the representation of $\mathcal{B}(\mathcal{H})$ in itself by inclusion is irreducible, while the representation π_ω must be reducible by Theorem 2.21, and therefore no such isomorphism can exist.

GNS construction associates each abstract C^*-algebra with a very concrete Hilbert space. However, we observe that the representation π_ω is not necessarily faithful, yet physically meaningful representations are generally expected to be faithful [38, Remark 8.8].

2.2.4 Structure of C^*-algebras

Any Hilbert space operator space $\mathcal{B}(\mathcal{H})$ is trivially isomorphic to a C^*-algebra (i.e. itself). We now demonstrate that the converse is also true, that is, any C^*-algebra is isomorphic to a closed *-algebra in $\mathcal{B}(\mathcal{H})$ thanks to the GNS construction. The following theorem can be seen as an noncommutative analogy to Gelfand–Naimark Theorem 2.7.

Theorem 2.23 (Gelfand–Naimark theorem, general case) *Each C^* algebra $\mathfrak{A}$ is isomorphic to a norm-closed *-algebra in $\mathcal{B}(\mathcal{H})$ for some Hilbert space $\mathcal{H}$. Equivalently, there exists a faithful representation $\pi : \mathfrak{A} \to \mathcal{B}(\mathcal{H})$.*

Proof By the GNS construction theorem 2.20, for any state ω on $\mathfrak{A}$, there exists a cyclic representation $(\mathcal{H}_\omega, \pi_\omega)$ of $\mathfrak{A}$ such that $\omega(a) = \langle\varphi_1^\omega|\pi_\omega(a)|\varphi_1^\omega\rangle$ for all $a \in \mathfrak{A}$. But π_ω is not faithful.

Consider the direct sum representation

$$(\mathcal{H} = \oplus_{\omega\in\mathcal{P}(\mathfrak{A})}, \pi = \oplus_{\omega\in\mathcal{P}(\mathfrak{A})}\pi_\omega) \tag{2.24}$$

summing over the GNS representations of all pure states $\omega \in \mathcal{P}(\mathfrak{A})$ on $\mathfrak{A}$. We claim that the direct sum representation $(\mathcal{H}, \pi)$ is faithful, which is what we need.

We now show that this representation is faithful. Suppose, for contradction, that there exists a nonzero $a \in \mathfrak{A}$ such that $\pi(a) = 0$. Then, for all states ω on $\mathfrak{A}$, we have $\pi_\omega(a) = 0$. This implies that

$$\omega(a^\dagger a) = \langle\varphi_1^\omega|\pi(a^\dagger a)|\varphi_1^\omega\rangle = \langle\varphi_1^\omega|\pi(a^\dagger)\pi(a)|\varphi_1^\omega\rangle = 0. \tag{2.25}$$

However, since $a \neq 0$, we have $a^\dagger a \neq 0$, which contradicts Lemma 2.14 that there should exist at least one pure state $|\omega(a)| = \|a\| \neq 0$. □

2.3 von Neumann algebras

C^*-algebras offer a powerful abstraction in the study of quantum mechanics, yielding numerous results. However, their limitations call for a more specialised subclass: von Neumann algebras. While C^*-algebras are closed in the uniform topology, it is often too restrictive; weak topology arises more naturally in quantum theory, and von Neumann algebras are both strongly and weakly closed. Furthermore, unbounded operators, abundant in quantum theory, are challenging to handle within C^*-algebras due to domain changes induced by adjoint operations. In contrast, von Neumann algebras can model unbounded operators through associated PVMs. Also, the von Neumann–Murray factor classification provides valuable structural insights. Finally, the special structure of von Neumann algebras enables concepts such as superselection rules to be easily modelled.

This section is mainly based on the discussion in [38, Chapter 6].

2.3.1 Definitions and basic properties

Recall that for a set of operator $\mathfrak{M} \subset \mathcal{B}(\mathcal{H})$ of some Hilbert space, its commutant is denoted by $\mathfrak{M}' = \{\hat{\alpha} \in \mathcal{B}(\mathcal{H}) \mid [\hat{\alpha}, \hat{\beta}] = 0,\ \forall \hat{\beta} \in \mathfrak{M}\}$. We begin by including some basic facts about commutants from [38, Proposition 6.3].

Proposition 2.24 *The commutant $\mathfrak{M}'$ of $\mathfrak{M} \subset \mathcal{B}(\mathcal{H})$ satisfies the following properties:*

(a) If $\mathfrak{M}$ is closed under the involution $\dagger$, then $\mathfrak{M}'$ is a C^*-subalgebra;

(b) $\mathfrak{M}'$ is closed in both weak and strong topologies;

(c) $\mathfrak{M}' = ((\mathfrak{M}')')'$, hence no more new information above the second commutant.

As mentioned before, the weak operator topology arises much more naturally from quantum theory, rooted from the convergence of matrix elements of quantum operators. We shall model such a feature with the following definition.

Definition 2.25 *A* von Neumann algebra $\mathfrak{A}$ *is a unital *-subalgebra of $\mathcal{B}(\mathcal{H})$ that is weakly closed. The* center *of a von Neumann algebra $\mathfrak{M}$ is the set $Z(\mathfrak{M}) = \mathfrak{M} \cap \mathfrak{M}'$, that is, it contains all operators in $\mathfrak{M}$ that commute with every element of $\mathfrak{M}$.*

Now we introduce the bicommutant theorem, which is central in the study of von Neumann algebras.

Theorem 2.26 (von Neumann bicommutant theorem) *Let $\mathfrak{M}$ be a unital *-subalgebra of $\mathcal{B}(\mathcal{H})$, then the following conditions are equivalent:*

(a) $\mathfrak{M} = \mathfrak{M}''$;

(b) $\mathfrak{M}$ *is weakly closed, i.e. a von Neumann algebra;*

(c) $\mathfrak{M}$ *is strongly closed.*

Therefore, one other common definition for von Neumann algebras is that they are the algebras that equal their double commutants.

Remark 2.27 *We discuss some examples of von Neumann algebras.*

1. *We note that the space of all bounded operators $\mathcal{B}(\mathcal{H})$ is a von Neumann algebra, as it is clearly weakly closed. Alternatively, we can think of $\mathcal{B}(\mathcal{H})$ as having a trivial representation on $\mathcal{H}$, which is irreducible. Then Schur's lemma 2.19 indicates that $\mathcal{B}(\mathcal{H}) = \mathcal{B}(\mathcal{H})''$ and hence it is a von Neumann algebra.*
2. *If $\mathfrak{M}$ is a subset of $\mathcal{B}(\mathcal{H})$ closed under involution, then its commutant $\mathfrak{M}'$ is a von Neumann algebra. Indeed, by Proposition 2.24.(c), we have $(\mathfrak{M}')'' = \mathfrak{M}'$ which proves the claim.*
3. *As claimed in the introductory text of this section, von Neumann algebras are a special subclass of C^*-algebras. They are even unital. To see this, given any von Neumann algebra we know that $\mathfrak{M} = (\mathfrak{M}')'$, then Proposition 2.24.(a) implies that $\mathfrak{M}$ is a C^*-subalgebra. It must contain the unit since $\mathbb{1} \in \mathfrak{M}'$ and thus $\mathbb{1} \in \mathfrak{M}'' = \mathfrak{M}$.*
4. *An arbitrary intersection of von Neumann algebras is again a von Neumann algebra, see [38, Remark 6.6].*
5. *Given any $\mathfrak{M} \subset \mathcal{B}(\mathcal{H})$ closed under involution, the smallest von Neumann algebra containing $\mathfrak{M}$ is precisely the bicommutant $\mathfrak{M}''$. Indeed, suppose that we have a von Neumann algebra $\mathfrak{A} \supset \mathfrak{M}$, then $\mathfrak{A}' \subset \mathfrak{M}'$ and so $\mathfrak{A} = \mathfrak{A}'' \supset \mathfrak{M}''$. Consequently, $\mathfrak{M}''$ is the intersection of all von Neumann algebras containing $\mathfrak{M}$.*

The last remark above inspires the following definition.

Definition 2.28 *The* von Neumann algebra generated *by $\mathfrak{M} \subset \mathcal{B}(\mathcal{H})$ is the double commutant $\mathfrak{M}''$.*

Next, we discuss the notion of isomorphism of von Neumann algebras.

Definition 2.29 *Two von Neumann algebras $\mathfrak{M}_1 \subset \mathcal{B}(\mathcal{H}_1)$ and $\mathfrak{M}_2 \subset \mathcal{B}(\mathcal{H}_2)$ for some Hilbert spaces $\mathcal{H}_1, \mathcal{H}_2$ are* isomorphic *(or* quasi equivalent*) if there exists a unital *-isomorphism between $\Phi : \mathfrak{M}_1 \to \mathfrak{M}_2$.*

They are completely isomorphic *if the isomorphism Φ is a homeomorphism for both weak and strong topolgies.*

They are spatially isomorphic *if there exists a surjective linear isometry $V : \mathcal{H}_1 \to \mathcal{H}_2$ such that the map $\hat{\alpha} \mapsto V\hat{\alpha}V^{-1}$ is surjective, for $\hat{\alpha} \in \mathfrak{M}_1$. The map is also a complete isomorphism.*

We remark that any unital *-isomorphisms between von Neumann algebras are norm-preserving [38, Proposition 6.9].

2.3.2 Affiliated unbounded self-adjoint operators

We now discuss how von Neumann algebras are adapted to unbounded self-adjoint operators, which are quite standard in quantum theory. For this, let us further extend our notion of commutant.

Recall that, for a general self-adjoint operator $\hat{\alpha}$, the (unbounded version of) Theorem A.16 indicates the existence of a spectral PVM $P^{(\hat{\alpha})}$ such that $\hat{\alpha} = \int_{\sigma(\hat{\alpha})} \lambda dP^{(\hat{\alpha})}(\lambda)$.

Definition 2.30 *Let $\mathfrak{R}$ be a set of (possibly unbounded) self-adjoint operators on $\mathcal{H}$.*

(a) The commutant *$\mathfrak{R}'$ of $\mathfrak{R}$ is defined as the commutant (as in the previous subsection) of the set of all spectral measures $P^{(\hat{\alpha})}$ of every $\hat{\alpha} \in \mathfrak{R}$.*

(b) The von Neumann algebra generated by $\mathfrak{R}$ is $\mathfrak{R}'' = (\mathfrak{R}')'$, where the inner $'$ is as in (a) and the outer $'$ is as in the previous subsection.

(c) Let $\mathfrak{M}$ be a von Neumann algebra, a self-adjoint operator $\hat{\alpha} : D(\hat{\alpha}) \to \mathcal{H}$, where the domain $D(A) \subset \mathcal{H}$, is said to be affiliated *to $\mathfrak{M}$ if its PVM $P^{(\hat{\alpha})}$ also lie in $\mathfrak{M}$.*

Note that each self-adjoint element $\hat{\alpha}$ in the set $\mathfrak{R}$ is affilated to the genreated von Neumann algebra $\mathfrak{R}''$, since $P^{(\hat{\alpha})}$ commutes with all self-adjoint operators in $\mathcal{B}(\mathcal{H})$ that also commute with $\hat{\alpha}$, it specifically commutes with any operator in $\mathcal{B}(\mathcal{H})$ that commutes with $\hat{\alpha}$, as they admit decomposition into self-adjoint operators.

Finally, we observe that for any (unbounded) self-adjoint operator $\hat{\alpha} : D(\hat{\alpha}) \to \mathcal{H}$ can be written as

$$\hat{\alpha} |v\rangle = \lim_{n\to\infty} \hat{\alpha}_n |v\rangle = \lim_{n\to\infty} \left(\int_{[-n,n]\cap\sigma(\hat{\alpha})} \lambda dP^{(\hat{\alpha})}(\lambda) |v\rangle \right) \tag{2.26}$$

by the spectral theorem. Note that the operators $\hat{\alpha}_n$ are in the space $\mathcal{B}(\mathcal{H})$, and so $\hat{\alpha}$ is a strong limit of bounded operators with respect to its domain. With a bit more work, it can be shown that if $\hat{\alpha}$ is affiliated to a von Neumann algebra, then all $\hat{\alpha}_n$ belongs to the algebra [38, Proposition 6.12].

Proposition 2.31 *A self-adjoint operator $\hat{\alpha} : D(\hat{\alpha}) \to \mathcal{H}$ affilated to a von Neumann algebra $\mathfrak{M}$ is a strong limit (with respect to $D(\hat{\alpha})$) of a sequence of self-adjoint operator in $\mathfrak{M}$. Moreover, $\hat{\alpha} \in \mathfrak{M}$ if $D(\hat{\alpha}) = \mathcal{H}$.*

Proposition 2.31 enables us to study unbounded self-adjoint operators affiliated to a given von Neumann algebra more easily, as a limit of the elements of the von Neumann algebra itself.

2.3.3 Factors and structures of von Neumann algebras

One feature of von Neumann algebras is that they have many projectors (in fact, they are generated by them) [33, Appendix C.21]. The study of these projectors leads to the concept of factors which are the "irreducible" components in von Neumann algebras. Factors are the building blocks of the von Neumann-Murray decomposition theorem.

Definition 2.32 *A von Neumann algebra $\mathfrak{M}$ is called a* factor *if it has a trivial center, that is, $Z(\mathfrak{M}) = \mathfrak{M} \cap \mathfrak{M}' = \mathbb{C}$.*

Von Neumann and Murry classified factors based on the properties of their minimal projectors: type I, type II, and type III. While type I factors are easier to describe, it would be hard to define the other two rigorously without a long setup beyond the scope of this thesis, as they are less relevant to the elementary quantum theory.

- *Type I factors:* A von Neumann algebra is a Type I factor if it contains a minimal projection. These factors can be further divided into Type I_n factors, where n is the dimension of the central projections, while Type $\mathrm{I}\infty$ is infinite-dimensional.

  Examples of Type I factors include the algebra of all bounded operators on a Hilbert space, $\mathcal{B}(\mathcal{H})$. Additionally, a factor is of type I if and only if it is isomorphic to some $\mathcal{B}(\mathcal{H})$ [38, Chapter 6.1.4].

- *Type II factors:* A von Neumann algebra is a Type II factor if it has no minimal projections but has projections equivalent to the identity. Type II factors are also divided into Type II_1 and Type II_∞. Type II_1 factors have a unique, normalised, finite trace, while Type II_∞ factors have traces that are only semifinite. Type II factors often arise in the context of infinite tensor products of Hilbert spaces and crossed product constructions.

- *Type III factors:* A von Neumann algebra is a Type III factor if it has no minimal projections and no projections equivalent to the identity. Type III factors can be further divided into subtypes based on their modular theory properties. Although very abstract, type III factors play a crucial role in extended quantum thermodynamical systems and in algebraic quantum field theory [11, 25].

The von Neumann-Murray decomposition theorem serves as a crucial tool for breaking down a general von Neumann algebra into a direct integral of

factors. This decomposition can include a combination of various types of factors, with their distribution determined by a measure on an index set Γ. This influential theorem ensures us to comprehend the fundamental structure of von Neumann algebras more efficiently [11, 30].

2.3.4 Superselection rules

The entire framework of von Neumann algebras (and C^*-algebras) is built on the concept of modeling quantum observables. However, in reality, not all self-adjoint operators can be considered as observables.

A well-known example is the electrical charge observable, which commutes with all other observables. As a result, since all quantum systems possess electrical charges, any physically viable observable must be able to commute with the electrical charge observable. Self-adjoint operators that do not commute with electrical charges cannot be physical. This constraint on physically permissible observables (and therefore states) is known as a *superselection rule*.

There are additional superselection rules in physics, though they vary greatly in nature, such as integer/semi-integer angular momenta and Bargmann's superselection rule on mass in non-relativistic physics.

An operator $\hat{\alpha}$ has a *pure spectrum* if every point in its spectra $\sigma(\hat{\alpha})$ is solely a limit point of the eigenvectors, and the continuous spectrum has no internal points; see Appendix A.2.1.

Definition 2.33 (Superselection rule) *Given a von Neumann algebra $\mathfrak{M} \subset \mathcal{B}(\mathcal{H})$, we say that there are* (abelian discrete) superselection rules *if there is a finite set $\mathfrak{O} = \{Q_1, \ldots, Q_n\}$ such that*

1. $\mathfrak{M} = \mathfrak{O}' = \{Q_1, \ldots, Q_n\}'$,
2. *the* superselection charges $Q_1, \ldots, Q_n$ *are self-adjoint operators with commuting PVMs and pure point spectra.*

For a more compact notation of the next result, denote the Cartesian product of the spectrum of Q_k by $\sigma(\mathfrak{O}) = \sigma_p(Q_1) \times \cdots \times \sigma_p(Q_n)$ with the element $\vec{q} = (q_1, \ldots, q_n)$.

Proposition 2.34 *Let $\mathfrak{M}$ be a von Neumann algebra with superselection rules $\mathfrak{O} = \{Q_1, \ldots, Q_n\}$. Then the following holds.*

1. *$\mathcal{H}$ admits a orthogonal decomposition into* superselection *(or* coherent*)* sectors $\mathcal{H}_{\vec{q}}$,

$$\mathcal{H} = \bigoplus_{\vec{q} \in \sigma(\mathfrak{O})} \mathcal{H}_{\vec{q}}, \tag{2.27}$$

where the superselection sectors $\mathcal{H}_{\vec{q}}$ are invariant and irreducible under the action of $\mathfrak{M}$.

2. *The algebra $\mathfrak{M}$ admits a direct decomposition*

$$\mathfrak{M} = \bigoplus_{\vec{q}\in\sigma(\mathfrak{O})} \mathfrak{M}_{\vec{q}} \tag{2.28}$$

where $\mathfrak{M}_{\vec{q}} = \{\hat{\alpha}|_{\mathcal{H}_{\vec{q}}} : \mathcal{H}_{\vec{q}} \to \mathcal{H}_{\vec{q}} \mid \hat{\alpha} \in \mathfrak{M}\} = \mathcal{B}(\mathcal{H})$.

3. *The representations of $\mathfrak{M}_{\vec{q}}$ on $\mathcal{H}_{\vec{q}}$ with distinct $\vec{q}$ are not unitarily equivalent.*

Superselection rules lead to a direct sum structure, partitioning the von Neumann algebra into superselection sectors where physical pure states reside. Within each superselection sector, we encounter the simpler scenario where every self-adjoint operator is an observable. In this context, the superselection charges merely act as fixed parameters.

It is worth noting that the requirement of a pure spectrum is essential for achieving a direct sum decomposition. Allowing for a continuous spectrum with a nonempty interior would result in a direct integral decomposition, making the physical meaning ambiguous [38, Remark 6.33].

Lastly, we highlight the significance of superselection in quantum nonlocality. For instance, [72] examines the superselection rule of particle numbers in a bipartite system. They construct separable states that cannot be prepared locally, as the states supposedly prepared locally are invalidated by the superselection rule.

2.4 Composite systems and independent subsystems

Composite systems and independent subsystems are vital in quantum information. Therefore, we dedicate this section exclusively to this topic and draw from [8, 33, 38] for our discussion.

In quantum theory, the standard approach to describe the composition of two systems represented by $\mathcal{H}_A$ and $\mathcal{H}_B$ involves the tensor product space $\mathcal{H}_A \otimes \mathcal{H}_B$. However, in general, $\mathcal{H}_A \otimes \mathcal{H}_B$ does not naturally form a Hilbert space since it is not complete. Despite this, one can verify that the form defined by $\langle \varphi_1 \otimes \psi_1 | \varphi_2 \otimes \psi_2 \rangle_{\mathcal{H}_A \otimes \mathcal{H}_B} = \langle \varphi_1 | \varphi_2 \rangle_A \langle \psi_1 | \psi_2 \rangle_B$ is an inner product on $\mathcal{H}_A \otimes \mathcal{H}_B$. Consequently, the completion $\mathcal{H}_A \overline{\otimes} \mathcal{H}_B$ with respect to this inner product becomes a Hilbert space. Moreover, this completion of Hilbert spaces is both *canonical* and *unique* [33, Appendix C.13].

On the other hand, when working with tensor products of C^*-algebras, there is an "*embarrassment of riches*" [33, Appendix C.13], meaning there are numerous norms available to complete a tensor product of C^*-algebras. As a

result, the notions of system composition and independence become much more intricate.

Conversely, when examining a global physical system, it is often valuable to understand its subsystems, which may be spatially or causally independent of one another. However, due to the more complex algebraic structures, the notion of independent subsystems becomes increasingly delicate. We will explore various notions of independence and their related results within the framework of von Neumann algebras.

2.4.1 Sptial and maximal tensor products of C^*-algebras

Definition 2.35 *Let $\mathfrak{A}$ and $\mathfrak{B}$ be C^*-algebras, the* algebraic tensor product $\mathfrak{A} \odot \mathfrak{B}$ *is a *-algebra with multiplication*

$$(\alpha_1 \otimes \beta_1)(\alpha_2 \otimes \beta_2) = \alpha_1\alpha_2 \otimes \beta_1\beta_2 \tag{2.29}$$

and involution $(\alpha \otimes \beta)^\dagger = \alpha^\dagger \otimes \beta^\dagger$. The vector space structure is from the usual tensor product.

If γ is a norm on $\mathfrak{A} \odot \mathfrak{B}$, then the completion with respect to γ is denoted as $\mathfrak{A} \otimes_\gamma \mathfrak{B}$.

Equivalently, algebraic tensor products can be defined through a universal property, and a standard exercise shows the equivalence [33, Appendix C.13].

Note that if π_A and π_B are representations of $\mathfrak{A}$ and $\mathfrak{B}$ on $\mathcal{H}_A$ and $\mathcal{H}_B$, then one can define a representation $\pi_A \otimes \pi_B$ of $\mathfrak{A} \odot \mathfrak{B}$ using the formula $(\pi_A \otimes \pi_B)(\alpha \otimes \beta) = \pi_A(\alpha) \otimes \pi_B(\beta)$. Moreover, if both π_A and π_B are faithful, then $\pi_A \otimes \pi_B$ is faithful [8, I.2.5.5].

We now introduce two important norms.

Definition 2.36 *Let $\mathfrak{A}$ and $\mathfrak{B}$ be C^*-algebras and consider an arbitrary element $\sum_{i=1}^n \alpha_i \otimes \beta_i \in \mathfrak{A} \odot \mathfrak{B}$.*

1. *The* spatial norm *(or* minimal norm*) is defined by*

$$\left\| \sum_{i=1}^n \alpha_i \otimes \beta_i \right\|_{\min} = \sup \left\| (\pi_A \otimes \pi_B)(\sum_{i=1}^n \alpha_i \otimes \beta_i) \right\|_{\mathcal{B}(\mathcal{H}_A \overline{\otimes} \mathcal{H}_B)}, \tag{2.30}$$

   where the supremum is taken over all representations $\pi_A : \mathfrak{A} \to \mathcal{B}(\mathcal{H}_A)$ of $\mathfrak{A}$ and $\pi_B\, \mathfrak{B} \to \mathcal{B}(\mathcal{H}_B)$ of $\mathfrak{B}$. The completion with respect to this norm is spatial tensor product *(or* minimum tensor product*), and is denoted by $\mathfrak{A} \otimes_{\min} \mathfrak{B}$.*

2. *The* maximal norm *is defined by*

$$\left\| \sum_{i=1}^n \alpha_i \otimes \beta_i \right\|_{\max} = \sup \left\| \pi(\sum_{i=1}^n \alpha_i \otimes \beta_i) \right\|_{\mathcal{B}(\mathcal{H})}, \tag{2.31}$$

where the supremum is taken over every representation $\pi : \mathfrak{A} \to \mathcal{B}(\mathcal{H})$. The completion with respect to this norm is maximal tensor product, *denoted by* $\mathfrak{A} \otimes_{\max} \mathfrak{B}$.

Clearly, a maximal tensor product space contains the respective spatial tensor product space, and the maximal norm is indeed larger than the spatial norm since the former supremum is a relaxation of the latter.

The spatial tensor product gets its name from being generated through tensor product representations of Hilbert spaces, which can be interpreted as spatial separation. In elementary quantum theory, the minimal completion seems more physical, given its connection with tensor products of Hilbert spaces. The fact that the spatial norm is the smallest of all possible norms on $\mathfrak{A} \odot \mathfrak{B}$ is highly nontrivial, as discussed in [8, II.9.5].

Conversely, the maximal tensor product's maximality is almost inherent due to the inequality $\|\alpha \otimes \beta\| \leq \|\alpha\|\|\beta\|$ [33, Lemma C.96]. Moreover, the maximal tensor product corresponds to the commutative quantum model. In other words, instead of requiring two independent parties to be separated by a tensor product, we only demand commutativity between their measurements. (It is worth noting that the commutative quantum model plays a significant role in Chapter 3 and Chapter 4.) This is due to the following universality result.

Theorem 2.37 *Let π be a nondegenerate representation of C^*-algebras $\mathfrak{A} \odot \mathfrak{B}$ on a Hilbert space $\mathcal{H}$. Then there exist unique nondegenerate representations π_A of $\mathfrak{A}$ and π_B of $\mathfrak{B}$ on $\mathcal{H}$, such that*

$$\pi(\alpha \otimes \beta) = \pi_A(\alpha)\pi_B(\beta) = \pi_B(\beta)\pi_A(\alpha) \tag{2.32}$$

for all $\alpha \in \mathfrak{A}$ and $\beta \in \mathfrak{B}$.

Remark 2.38 *Due to its importance in the physical interpretation of different norms, we summarise the discussion that precedes Theorem 2.37 in the case when $\mathfrak{A}$ and $\mathfrak{B}$ are both unital. Then, for any norm γ, there are natural embeddings $\mathfrak{A} \hookrightarrow \mathfrak{A} \otimes \mathbb{1}_{\mathfrak{B}} \subset \mathfrak{A} \otimes_\gamma \mathfrak{B}$ and $\mathfrak{B} \hookrightarrow \mathbb{1}_{\mathfrak{A}} \otimes \mathfrak{B} \subset \mathfrak{A} \otimes_\gamma \mathfrak{B}$.*

Therefore, $\mathfrak{A} \otimes_{\min} \mathfrak{B}$ can be described as the universal unital C^*-algebra acting on $\mathcal{H}_A \otimes \mathcal{H}_B$ that is generated by unital copies of $\mathfrak{A}$ and $\mathfrak{B}$, which act on $\mathcal{H}_A$ and $\mathcal{H}_B$, respectively. And $\mathfrak{A} \otimes_{\max} \mathfrak{B}$ is generated by unital copies of $\mathfrak{A}$ and $\mathfrak{B}$ that commute with each other.

We note that both spatial and maximal tensor products are associated [8, II.9.2.6]. But also $\mathcal{B}(\mathcal{H}) \otimes_{\min} \mathcal{B}(\mathcal{H}) \neq \mathcal{B}(\mathcal{H}) \otimes_{\max} \mathcal{B}(\mathcal{H})$ in general [29]. Only a very special (and important) class of C^*-algebras can have unique tensor product norms.

Definition 2.39 *A C^*-algebra $\mathfrak{A}$ is* nuclear *if for any C^*-algebra $\mathfrak{B}$, we have $\mathfrak{A} \otimes_{\min} \mathfrak{B} = \mathfrak{A} \otimes_{\max} \mathfrak{B}$. Consequently, the algebraic tensor product $\mathfrak{A} \odot \mathfrak{B}$ admits a unique norm since all norms γ coincide.*

Examples of nuclear C^*-algebra include any commutative C^*-algebras [33, Proposition C.100] and any finite-dimensional C^*-algebras [33, Proposition C.101].

2.4.2 States on tensor products of C^*-algebras

We now introduce the notion of states on algebraic tensor products of unital algebras and results from [33] that provide connections to different tensor products.

Definition 2.40 *Let $\mathfrak{A}$ and $\mathfrak{B}$ be unital C^*-algebras. The* state *ω on $\mathfrak{A} \odot \mathfrak{B}$ is a positive linear functional (positivity in the usual sense for elements in $\mathfrak{A} \odot \mathfrak{B}$) and $\omega(\mathbb{1}_{\mathfrak{A}} \otimes \mathbb{1}_{\mathfrak{B}}) = 1$.*

The state space is denoted by $\mathcal{S}(\mathfrak{A} \odot \mathfrak{B})$.

The following result suggests that the maximal product is more natural in describing states on tensor product spaces.

Proposition 2.41 *Let $\mathfrak{A}$ and $\mathfrak{B}$ be unital C^*-algebras. Then each state ω on $\mathfrak{A} \odot \mathfrak{B}$ is continuous with respect to the maximal norm $\|\cdot\|_{\max}$, and therefore can be extended to a state on the maximal tensor product $\mathfrak{A} \otimes_{\max} \mathfrak{B}$. Hence, we have the identification*

$$\mathcal{S}(\mathfrak{A} \odot \mathfrak{B}) = \mathcal{S}(\mathfrak{A} \otimes_{\max} \mathfrak{B}). \tag{2.33}$$

On the other hand, product states can be identified with the state of the spatial tensor products (therefore any norms γ).

Proposition 2.42 *Let ω_A and ω_B be states on $\mathfrak{A}$ and $\mathfrak{B}$, respectively. Then the product state $\omega_A \otimes \omega_B$, defined by*

$$\omega_A \otimes \omega_B(\sum_i \alpha_i \otimes \beta_i) = \sum_i \omega_A(\alpha_i)\omega_B(\beta_i), \tag{2.34}$$

is continuous with respect to the spatial norm $\|\cdot\|_{\min}$, and can therefore be extended to a state on $\mathfrak{A} \otimes_{\min} \mathfrak{B}$. Consequently, we have the analogous result for any norm γ.

2.4.3 Tensor products of von Neumann algebras

We quickly mention the tensor products of von Neumann algebras. For von Neumann algebras $\mathfrak{M}_1, \mathfrak{M}_2$ on Hilbert spaces $\mathcal{H}_A, \mathcal{H}_B$, the algebraic tensor product $\mathfrak{M}_A \odot \mathfrak{M}_B$ acts naturally on $\mathcal{H}_A \otimes \mathcal{H}_B$. Then as noted by [8, III.1.5.4] the spatial tensor product analogy is given by the weak closure

$$\mathfrak{M}_A \overline{\otimes} \mathfrak{M}_B = (\mathfrak{M}_A \odot \mathfrak{M}_B)''. \tag{2.35}$$

The *commutant theorem for tensor products* [8, III.4.5.8], which is a very profound result, shows that

$$(\mathfrak{M}_A \overline{\otimes} \mathfrak{M}_B)' = \mathfrak{M}_A' \odot \mathfrak{M}_B'. \tag{2.36}$$

There are also analogous notions of the maximal tensor product (the *binormal tensor product*) and of the nuclearity (semi-discreteness); see [8, IV.2].

2.4.4 Independent subsystems

In elementary quantum theory, two independent subsystems are modeled by the tensor product of two distinct Hilbert spaces, which, along with Born's rule, implies statistical independence. However, this is not generally true in the algebraic framework, where one begins with the entire system and then defines the subsystem within it. In the following discussion, we work with von Neumann algebras, focusing on the case of two subsystems for simplicity (the general case can be derived through inductive argument).

Definition 2.43 (Independent subsystems) *Suppose that there is a physical system with two subsystems.*

1. *There exists a von Neumann algebra $\mathfrak{M}$ on $\mathcal{H}$ describe the* total system *and the* subsystems *are described by von Neumann algebras $\mathfrak{M}_A, \mathfrak{M}_B \subset \mathfrak{M}$.*
2. *The subsystems are* compatible *with each other in the sense that $[\mathfrak{M}_A, \mathfrak{M}_B] = 0$.*
3. *There are different notions of how these subsystems are independent of each other:*
   (a) *(*W^*-independence*) For any two independent states on the different subsystems, there exists a total state encapsulates them. Mathematically, any normal states ρ on $\mathfrak{M}_A$ and σ on $\mathfrak{M}_B$ admit a common extension τ on $(\mathfrak{M}_A \cup \mathfrak{M}_B)''$ such that $\mathrm{Tr}(\tau\hat{\alpha}) = \mathrm{Tr}(\rho\hat{\alpha})$ and $\mathrm{Tr}(\tau\hat{\beta}) = \mathrm{Tr}(\sigma\hat{\beta})$ for any $\hat{\alpha} \in \mathfrak{M}_A$ and $\hat{\beta} \in \mathfrak{M}_B$.*
   (b) *(*Statistical independence*) In additionally to W^*-independence, τ further satisfies the factorisation condition $\mathrm{Tr}(\tau\hat{\alpha}\beta) = \mathrm{Tr}(\rho\hat{\alpha})\,\mathrm{Tr}(\sigma\hat{\beta})$[4]. In this case, τ is the* product state *of ρ and σ.*

In the above definition, we abuse the notation of normal states and their associated density operators. The notion of W^*-independence gives the following result.

Proposition 2.44 *Under the hypothesis of W^* independent subsystems (i.e., 1, 2, and 3.(a) of Definition 2.43), there is a natural unital $*$-algebra isomorphism between*

[4] A necessary condition for statistical independence is the *algebraic independence*, that is, if $\hat{\alpha}\hat{\beta} = 0$ then either $\hat{\alpha} = 0$ or $\hat{\beta} = 0$ [38, Chapter 6.4.1].

the $*$-algebra generated by finite linear combinations of finite products of elements in $\mathfrak{M}_A$ and $\mathfrak{M}_B$ to the algebraic tensor product $\mathfrak{M}_A \odot \mathfrak{M}_B$, given by φ the unique linear extension of $\hat{\alpha}\hat{\beta} \mapsto \hat{\alpha} \otimes \hat{\beta}$.

Moreover, if $\mathfrak{M}_A, \mathfrak{M}_B$, and $\mathfrak{M}$ are factors, the result can be strengthened: the weakly continuous extension Φ of φ is an isomorphism between $(\mathfrak{M}_A \cup \mathfrak{M}_B)''$ and $\mathfrak{M}_A \overline{\otimes} \mathfrak{M}_B$.

We note that W^*-independence does not force a tensor factorisation of the underlying Hilbert space $\mathcal{H}$, and hence is not statistically independent. This is different from the usual notion of independence in elementary quantum theory. But there is a special case discovered by von Neumann and Murray, which is fundamental to the discussion of Tsirelson's problem [58, 69].

Proposition 2.45 *Suppose that $\mathfrak{M}_A \subset \mathcal{B}(\mathcal{H})$ is a type I factor and $\mathfrak{M}_B = \mathfrak{M}_A'$. Then there exist Hilbert spaces $\mathcal{H}_A, \mathcal{H}_B$ and an isomorphism $u : \mathcal{H} \to \mathcal{H}_A \otimes \mathcal{H}_B$ such that $u\mathfrak{M}_A u^{-1} = \mathcal{B}(\mathcal{H}_A)\overline{\otimes}\mathbb{C}\mathbb{1}_B$ and $u\mathfrak{M}_B u^{-1} = \mathbb{C}\mathbb{1}_A \overline{\otimes} \mathcal{B}(\mathcal{H}_B)$. Consequently $\mathfrak{M}_A$ and $\mathfrak{M}_B$ are statistically independent of each other.*

If one replaces the type I condition in Proposition 2.45, then $\mathfrak{M}_A$ and $\mathfrak{M}_B$ fail to be statistically independent but only W^*-independent [38, Chapter 6.4.2]. However, type III factors naturally appear in more general frameworks such as quantum field theory, and hence we need a generalisation to model statistical independence. One generalisation is as follows.

Definition 2.46 *Two commuting von Neumann algebras $\mathfrak{M}_A$ and $\mathfrak{M}_B$ satisfy* split property *if there exists a type I factor $\mathfrak{R}$ such that $\mathfrak{M}_A \subset \mathfrak{R}$ and $\mathfrak{M}_B \subset \mathfrak{R}'$.*

Proposition 2.47 *If two commuting von Neumann algebras $\mathfrak{M}_A$ and $\mathfrak{M}_B$ satisfy the split property, then there exist Hilbert spaces $\mathcal{H}_i$ and faithful representations $\pi_i : \mathfrak{M}_i \to \mathcal{B}(\mathcal{H}_i)$, where $i = A, B$, such that $(\mathfrak{M}_A \cup \mathfrak{M}_B)''$ is spatially isomorphic to $\pi_A(\mathfrak{M}_A)\overline{\otimes}\pi_B(\mathfrak{M}_B)$. The isomorphism is the linear extension of the map $\hat{\alpha}\hat{\beta} \mapsto \pi_A(\hat{\alpha}) \otimes \pi_B(\hat{\beta})$ where $\hat{\alpha} \in \mathfrak{M}_A$ and $\hat{\beta} \in \mathfrak{M}_B$.*

The statistical independence is then implied by Proposition 2.47. Moreover, one can improve Proposition 2.47 to an "if and only if statement" by imposing normality on π_i [38, Proposition 6.50].

2.5 Algebraic formulation of quantum theory

We summarise the chapter by providing *algebraic quantum postulates*, modulo the symmetry and evolutionary aspect, in correspondence to the usual postulates A–C in Schrödinger picuture, in the beginning of the chapter.

A′ A physical system S is described by the self-adjoint elements of some (generally abstract) unital C^*-algebra $\mathfrak{A}_S$ (Definition 2.2), which are the *observables*, and a algebraic state ω_S on $\mathfrak{A}_S$ (Definition 2.9).

Recall that the notion of algebraic states is strictly more general than density operators; see Remark 2.12. In addition, the GNS construction Theorem 2.20 bridges the usual postulate by the unique (up to unitary transformation) cyclic representation $(\mathcal{H}_S, \pi_S, |\varphi_1^S\rangle)$.

B′ For any observable $O \in \mathfrak{A}_S$, $\omega(O)$ can be understood as the expectation value of O. Similarly, for any PVM $P : X \to \mathfrak{A}_S$ (as in Definition A.14), where X is the measure space and each Borel set $E \subset X$ is an event E, $\omega(P_E)$ can be understood as the probability of observing the event E.

See [38, Chapter 8.2.3] which justifies this probabilistic interpretation by addressing the potential problem.

C′ The composition of the systems is handled by *some sort of tensor products*. Which sort depends on the underlying algebras of observable (e.g. C^*-algebras or von Neumann algebras) and the types of state (e.g., normal or algebraic) under what kind of independence (e.g. W^* or statistical); see Section 2.4.

Due to the existence of many possible norms for tensor products of C^*-algebras, there are many possibilities in defining the composed systems depending on the specific problem that one wishes to address.

The complication of the Postulate C′ has direct consequence to Chapter 4, when we try to model independent quantum sources in network scenario.

Chapter 3

Basics of quantum nonlocal correlations

In this chapter, we provide an overview of quantum correlations and present the relevant methods used in their study. We begin by introducing the concept of nonlocal correlations within the framework of two-player nonlocal games, drawing on the work presented in [12]. Subsequently, we discuss the Navascués–Pironio–Acín (NPA) hierarchy [42], a powerful technique to analyse quantum correlations and characterise the set of feasible quantum correlations. This background information lays the foundation for a deeper understanding of the significance and implications of quantum correlations for the content in Chapter 4.

3.1 Nonlocal correlations

In this section, we introduce the concept of nonlocal correlations by considering a two-player nonlocal game, as described in [12].

A typical two-player nonlocal game (see Figure 3.1) involves spatially separated players, Alice and Bob, who receive inputs x and y from a referee. These inputs are chosen independently and uniformly at random from sets of possible inputs X and Y, respectively. Based on their inputs, Alice and Bob must produce outputs a and b from their respective output sets A and B. Prior to the game, Alice and Bob can establish a strategy, represented by the conditional probability $\vec{P} = p(ab|xy)$, but they are not allowed to communicate during the game. Their goal is to satisfy a specific condition determined by a predicate $V(ab|xy)$, which takes the value 1 if they win and 0 otherwise. This predicate depends on their outputs a and b, as well as their inputs x and y.

Nonlocal games serve as a powerful conceptual tool for studying quantum nonlocality, providing an intuitive understanding of nonlocal correlations.

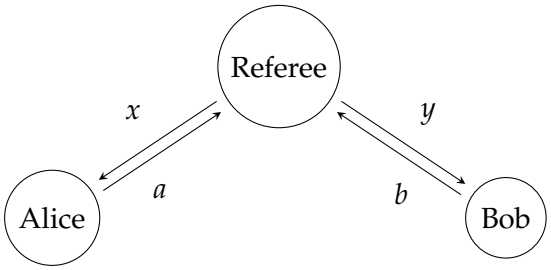


Figure 3.1: The diagram of a typical two-player nonlocal game, where Alice and Bob are spatially separated and receive inputs x and y from a referee, chosen independently and uniformly at random from a set of possible inputs, X and Y respectively. Based on their inputs, they must produce outputs a and b, respectively, from their respective output sets A and B. Before the game, Alice and Bob are allowed to determine a strategy, represented by the conditional probability $\vec{P} = \{p(ab|xy)\}$, but they cannot communicate during the game. The players win the game if their outputs satisfy a certain condition determined by a predicate $V(ab|xy)$, which takes the value 1 if they win and 0 otherwise.

Consequently, we will explore the operational and physical interpretations of non-signaling correlations, local correlations, and quantum correlations within the framework of two-player nonlocal games. This framework allows for straightforward generalisation to more complex scenarios.

Non-signaling correlations

A correlation $\vec{P} = p(ab|xy)$ is said to be *non-signaling* if the marginal probabilities of Alice's and Bob's outputs are independent of the other player's input. Formally, the non-signaling condition is given by:

$$\sum_{b\in B} p(ab|xy) = p(a|x) \quad \forall a \in A, x \in X, y \in Y, \tag{3.1}$$

$$\sum_{a\in A} p(ab|xy) = p(b|y) \quad \forall b \in B, x \in X, y \in Y. \tag{3.2}$$

The set of all non-signaling correlations is denoted by $\mathcal{NS}$. This condition ensures that information cannot be transmitted instantaneously between the two players, in accordance with the non-signaling principle of relativity.

Local correlations

A *local correlation* is one that can be modelled using a shared (hidden) random variable $\lambda \in \Lambda$ with a probability distribution q. The local model can be described as follows:

$$p(ab|xy) = \int_\Lambda P_\lambda(\lambda) p(a|x,\lambda) p(b|y,\lambda), \tag{3.3}$$

where $p(a|x,\lambda)$ and $p(b|y,\lambda)$ are conditional probability distributions for Alice and Bob, respectively. The set of all local correlations is

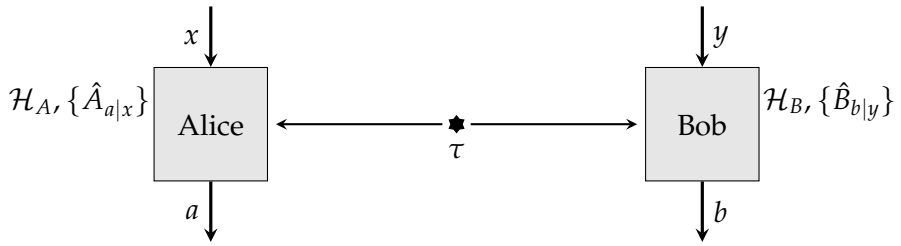


Figure 3.2: This figure illustrates the standard Bell scenario, where two parties, Alice and Bob, share a quantum state τ. Alice receives input x and produces output a, with a PVM $\hat{A}_{a|x}$ acting on her Hilbert space $\mathcal{H}_A$. Similarly, Bob receives input y and produces output b, with a PVM $\hat{B}_{b|y}$ acting on his Hilbert space $\mathcal{H}_B$.

denoted by $\mathcal{L}$. Local correlations have an interpretation as classical correlations that can be explained by a local hidden variable. Operationally, Alice will choose her answer a based on input x and randomness λ.

We note that the local assumption does not violate the no-communication rule of the game, since the additional Λ only provides information to Alice and Bob, but does not allow one to send information to the other. Also, any local correlation is non-signaling yet there exists non-signaling correlation that is not local (e.g. PR box), hence the strict inclusion $\mathcal{L} \subsetneq \mathcal{NS}$.

Quantum correlations

It is possible that Alice and Bob have access to some quantum devices and design their strategy around them. In this case, they share a global state $\tau \in \mathcal{B}(\mathcal{H}_A \otimes \mathcal{H}_B)$, and for each input x and y, they perform PVMs $\{\hat{A}_{a|x}\} \subset \mathcal{B}(\mathcal{H}_A)$ and $\{\hat{B}_{b|y}\} \subset \mathcal{B}(\mathcal{H}_B)$ on the state τ. Their outputs a and b depend on the measurement outcomes. This is known as the Bell scenario; see Figure 3.2.

Then by quantum theory, the *(tensor) quantum correlation* is given by:

$$p(ab|xy) = \mathrm{Tr}\Big(\tau(\hat{A}_{a|x} \otimes \hat{B}_{b|y})\Big), \tag{3.4}$$

The set of all quantum correlations is denoted by $\mathcal{Q}$.

Alternatively, one may relax the requirement that Alice and Bob's Hilbert space are spatially separated. That is, the share state τ is instead on a global Hilbert space $\mathcal{H}$, and the PVMs $\{\hat{A}_{a|x}\}, \{\hat{B}_{b|y}\}$ act on the same $\mathcal{H}$ as well. Instead of being separated by a tensor product, we impose only the commutation relation $[\hat{A}_{a|x}, \hat{B}_{b|y}] = 0$. The *commutator quantum correlation* is given by:

$$p(ab|xy) = \mathrm{Tr}\Big(\tau(\hat{A}_{a|x}\hat{B}_{b|y})\Big), \tag{3.5}$$

The set of all commutator quantum correlations is denoted by $\mathcal{Q}'$.

3.1.1 Tsirelson's problem

Clearly, $\mathcal{Q} \subset \mathcal{Q}'$ since $[\hat{A}_{a|x} \otimes \mathbb{1}_B, \mathbb{1}_A \otimes \hat{B}_{b|y}] = 0$. The converse, that is, whether the inclusion is strict, is known as the Tsirelson's problem [58, 69], turns out to be a highly nontrivial problem that is equivalent to the Kirchberg conjecture and the Connes' embedding conjecture [21]. Although it is proven by Tsirelson himself [58, 69] that if the Hilbert spaces $\mathcal{H}_A$, $\mathcal{H}_B$ are finite-dimensional then the two sets coincide, the general case is only recently *disproved* by [28] (indirectly, and a more intuitive counterexample is still absent).

The discussion of Tsirelson's problem is very interesting and highly relevant to the theory of operator algebras through its equivalence to the celebrated Connes' embedding conjecture. Unfortunately, we have to skip the discussion due to the scope of this thesis.

We nonetheless present the finite-dimensional Tsirelson's theorem.

Theorem 3.1 (Tsirelson) *Given a Hilbert space $\mathcal{H}$, let $\{\hat{A}_{a|x}\}$, $\{\hat{B}_{b|y}\}$ be two finite mutually commuting sets of positive operators in $\mathcal{B}(\mathcal{H})$, each generating a finite-dimensional von Neumann algebra $\mathcal{A}$ and $\mathcal{B}$. Let $\vec{Q} = \{q(ab|xy)\}$ be a probability distribution associated with $\{\hat{A}_{a|x}\}$, $\{\hat{B}_{b|y}\}$ and a global state τ.*

Then there exist Hilbert spaces $\mathcal{H}_A, \mathcal{H}_B$, such that $\{\hat{A}_{a|x}\}$ can be mapped into $\mathcal{B}(\mathcal{H}_A)$ and $\{\hat{B}_{b|y}\}$ can be mapped into $\mathcal{B}(\mathcal{H}_B)$ via injective *-homomorphisms. That is, extending these homomorphisms to the tensor product space $\mathcal{H}_A \otimes \mathcal{H}_B$, we have

$$\{\hat{A}_{a|x}\} \mapsto \{\hat{A}_{a|x} \otimes \mathbb{1}_B\}$$
$$\{\hat{B}_{b|y}\} \mapsto \{\mathbb{1}_A \otimes \hat{B}_{b|y}\}.$$

Moreover, there exists a state τ' on $\mathcal{H}_A \otimes \mathcal{H}_B$ recovering the probability distribution $\vec{Q}$ with the operators $\{\hat{A}_{a|x} \otimes \mathbb{1}_B\}$ and $\{\mathbb{1}_A \otimes \hat{B}_{b|y}\}$, namely,

$$q(ab|xy) = \mathrm{Tr}_\tau\left(\hat{A}_{a|x}\hat{B}_{b|y}\right) = \mathrm{Tr}_{\tau'}\left(\hat{A}_{a|x} \otimes \hat{B}_{b|y}\right)$$

The proof technique is known as the "doubling the centre", we refer to [58] and [57, Appendix H] for a proof. (Moreover, [57, Appendix H] asks for a generalisation to bilocal scenario which is later shown in [35], see Section 4.2.3.)

3.1.2 Geometry of the correlation sets

From the definitions above, it follows that $\mathcal{L} \subseteq \mathcal{Q} \subset \mathcal{Q}' \subseteq \mathcal{NS}$[1]. It can be shown that both the local correlations $\mathcal{L}$ and non-signaling correlations $\mathcal{NS}$

[1] One may also impose the finite-dimensionality in the definition of $\mathcal{Q}$, but it turns out both definitions have the same closure; see [58, 63].

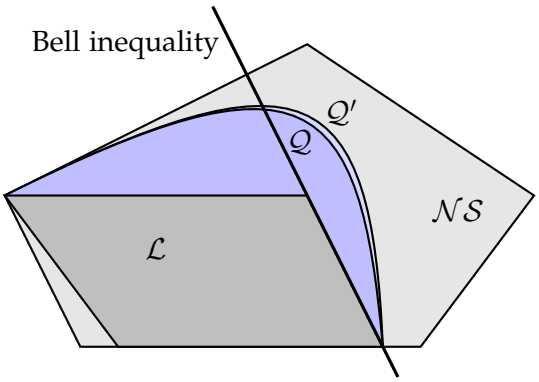


Figure 3.3: A sketch demonstrating the relations between the local $\mathcal{L}$ (dark gray), quantum $\mathcal{Q}$ (light blue), commutator quantum $\mathcal{Q}'$ (dark blue), and non-signaling $\mathcal{NS}$ (light gray) sets of correlations. Both $\mathcal{L}$ abd $\mathcal{NS}$ are polytopes, while $\mathcal{Q}$ and $\mathcal{Q}'$ are only convex in general. The sets $\mathcal{Q}$ and $\mathcal{Q}'$ do not coincide necessarily, but they do agree in finite dimension due to Tsirelson's Theorem 3.1. The hyperplane separating $\mathcal{L}$ and $\mathcal{Q}$ is a Bell inequality.

are in fact polytopes [12], i.e. can be defined as the convex combination of a *finite* number of extremal points. On the other hand, the quantum correlations $\mathcal{Q}$ and $\mathcal{Q}'$ are convex, but not polytopes.

The hyperplane separation theorem implies that, if a correlation $\vec{Q} \notin \mathcal{K}$, where $\mathcal{K} = \mathcal{L}, \mathcal{Q}, \mathcal{Q}', \mathcal{NS}$, then there exists a hyperplane separating the correlation $\vec{Q}$ and the set $\mathcal{K}$. Namely, there exists an equality of the form

$$\sum_{a,b,x,y} s_{ab|xy} p(ab|xy) \leq S_k \tag{3.6}$$

satisfied by all $\vec{P} \in \mathcal{K}$ but not $\vec{Q}$. Inequalities of this type are the well-known *Bell inequalities*, which indicate if a given correlation belongs to a specific set of correlations, and there can be many Bell inequalities depending on the specific problem. See Figure 3.3 for a pictorial illustration.

3.1.3 The CHSH game

Perhaps one of the most famous two-player nonlocal games is the Clauser-Horne-Shimony-Holt (CHSH) game [14], which we introduce to highlight the concept of nonlocal correlations.

In the CHSH game, we have $A = B = X = Y = \{0, 1\}$ and the winning condition is given by $V(a, b|x, y) = 1$ if $a \oplus b = xy$, where $\oplus$ denotes the bitwise exclusive OR operation (XOR), and 0 otherwise.

The CHSH inequality is an example of a Bell inequality that is derived from the CHSH game. It can be expressed in terms of the correlation function $\langle a_x, b_y \rangle = \sum_{a,b} (-1)^{a \oplus b} p(ab|xy)$. The famous CHSH inequality states that for

any local hidden variable model (i.e. any local correlation), the following inequality holds:

$$|S| = |\langle a_0 b_0\rangle + \langle a_0 b_1\rangle + \langle a_1 b_0\rangle - \langle a_1 b_1\rangle| \leq 2. \tag{3.7}$$

It can be shown that the winning probability of a given strategy $\vec{P}$ is related to the value S, that is,

$$p_{win} = \frac{1}{2}(1 + \frac{S}{4}). \tag{3.8}$$

It follows from the CHSH inequality that Alice and Bob can win at best 75% of the time using only the classical strategies (i.e., local correlations). The strategy that saturates the upper bound is the deterministic strategy.

For quantum correlations, it is known that the CHSH inequality can be violated up to a certain limit. The Tsirelson bound states that for any quantum strategy, the following inequality holds:

$$|S| = |\langle a_0 b_0\rangle + \langle a_0 b_1\rangle + \langle a_1 b_0\rangle - \langle a_1 b_1\rangle| \leq 2\sqrt{2}. \tag{3.9}$$

(We refer to [66] for a proof using the self-testing technique.) Therefore, the Tsirelson bound implies that with the access to quantum devices, Alice and Bob can do better than with only classical apparatus, with a winning probability of at most $\approx 86\%$!

The Tsirelson bound can be saturated using a maximally entangled state. For instance, consider the Bell state

$$|\Psi\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle). \tag{3.10}$$

Then, Alice measures the Pauli matrix Z when $x = 0$ and the Pauli matrix X when $x = 1$. For Bob, he measures $(Z + X)/\sqrt{2}$ if $y = 0$ and $(Z - X)/\sqrt{2}$ if $y = 1$. One can check that this is indeed the optimal quantum strategy.

Finally, we consider a strategy called *PR box*, that is, $\vec{P} = p(ab|xy) = 1/2\delta_{a\oplus b,xy}$. By construction, Alice and Bob win with certainty using this strategy. One can easily check that the PR box is non-signaling and $|S| = 4$, violating the quantum bound. Hence, such correlation cannot be of quantum origin.

3.2 Navascués-Pironio-Acín (NPA) hierarchy

Understanding sets of nonlocal correlations is crucial to investigate the nonlocal nature of quantum theory, as well as to certify the feasibility of a given probability distribution $\vec{P}$ with respect to a specified causal structure.

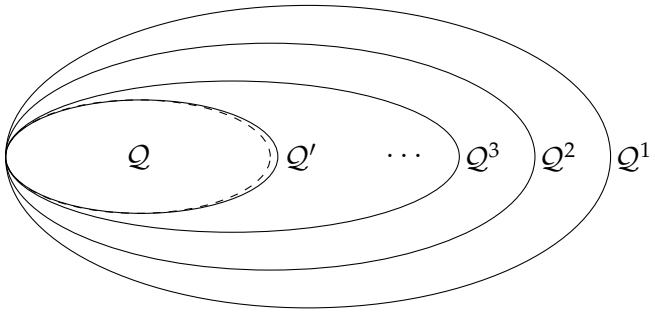


Figure 3.4: A pictorial representation of the NPA hierarchy as an outer approximation of the (tensor) quantum set $\mathcal{Q}$, represented by the dashed ellipse. Each ellipse $\mathcal{Q}^n$ represents the n-th level of relaxation/approximation provided by the NPA hierarchy. As the level increases, the approximation becomes more accurate and eventually converges towards the commutator quantum set $\mathcal{Q}'$, which is a relaxation/outer approximation of the quantum set $\mathcal{Q}$. Moreover by Tsirelson's Theorem 3.1, $\mathcal{Q}$ and $\mathcal{Q}'$ agrees in finite dimension, whence the hierarchy converges to the quantum set.

Identifying Bell inequalities is essential for characterising sets of correlations in quantum systems. For local correlations, this task is relatively easier since the set of correlations forms a polytope, which has a well-defined structure amenable to analysis. However, when it comes to quantum correlations, the situation becomes much more challenging, as the set of correlations is only convex yet no longer a polytope.

Inner approximations of quantum correlations can be obtained relatively easily by sampling over all Hilbert spaces and operators, which provides a lower bound for the Bell inequalities. Nevertheless, for outer approximations or upper bounds, the task is more intricate. One powerful technique for tackling this issue is the Navascués-Pironio-Acín (NPA) hierarchy [42], which provides a systematic approach to approximate the set of quantum correlations from the outside. Furthermore, the NPA hierarchy can be employed to determine whether a given probability distribution $\vec{P}$ is compatible with a specified causal structure. It is shown by [42] that a correlation $\vec{Q}$ is a commutator quantum correlation (in $\mathcal{Q}'$) if and only if it passes all NPA hierarchy tests. See Figure 3.4 for a sketch of the NPA hierarchy as an outer approximation of the feasible quantum set. We also refer to [13] for a comprehensive discussion from the viewpoint of theory of noncommuting polynomials optimisation.

3.2.1 Moment matrix and necessity of the hierarchy

Now, let us recall the standard bipartite Bell scenario as in Figure 3.2 and suppose that we have a quantum correlation $\vec{P} \in \mathcal{Q}$. Therefore, we have PVMs $\{\hat{A}_{a|x}\}$ and $\{\hat{B}_{b|y}\}$ and a global quantum density operator τ such that $p(ab|xy) = \mathrm{Tr}\Big(\tau \hat{A}_{a|x} \hat{B}_{b|y}\Big)$. (Note that here and below we abuse the notation by omitting $\otimes \mathbb{1}_B$ and $\mathbb{1}_A \otimes$).

We now fix the notation. For each $\hat{A}_{a|x}, \hat{B}_{b|y}$, we associate with a formal *letter* $A_{a|x}, B_{b|y}$ and impose the algebraic relations satisfied by the PVMs to the formal letters:

$$A_{a|x}^2 = A_{a|x},\ B_{b|y}^2 = B_{b|y} \tag{3.11}$$

$$[A_{a|x}, B_{b|y}] = 0 \tag{3.12}$$

$$A_{a|x}^\dagger = A_{a|x},\ B_{b|y}^\dagger = B_{b|y}. \tag{3.13}$$

The operator $\dagger$ is the involution on the letters that represents the complex conjugation of the measurement operators. A product of multiple letters (i.e., a monomial in letters) is a *word*, usually denoted by bolded Greek letter $\boldsymbol{\omega}, \boldsymbol{\nu}$. A word consisting only of $A_{a|x}$ (resp. $B_{b|y}$) is denoted by $\boldsymbol{\alpha}$ (resp. $\boldsymbol{\beta}$). And we write 1 for the *empty word*. Hence, any word admits a unique decomposition $\boldsymbol{\omega} = \boldsymbol{\alpha}\boldsymbol{\beta}$ where no letter is squared. Therefore, the equality of two words is checked in their minimal form, and the length of a word, denoted by $|\boldsymbol{\omega}|$, is the number of letters of $\boldsymbol{\omega}$ in its minimal form. For any formal word $\boldsymbol{\omega}$, we use the "hat" notation $\hat{\omega}$ to represent the corresponding product of PVMs.

Then for each $n \geq 2$, one can construct the moment matrix Γ^n of order n:

$$\begin{array}{c} \\ \mathbb{1} \\ (A_{a|x})^\dagger \\ (B_{b|y})^\dagger \\ (A_{a|x}A_{a'|x'})^\dagger \\ (A_{a|x}B_{b|y})^\dagger \\ \vdots \end{array} \begin{array}{c} \begin{array}{ccccccc} \mathbb{1} & A_{a|x} & & B_{b|y} & & A_{a|x}A_{a'|x'} & A_{a|x}B_{b|y} & \cdots \end{array} \\ \begin{bmatrix} 1 & & & & & & \mathrm{Tr}\Big(\tau \hat{A}_{a|x}\hat{B}_{b|y}\Big) & \\ & & \mathrm{Tr}\Big(\tau \hat{A}_{a|x}\hat{B}_{b|y}\Big) & & & & & \\ & & \mathrm{Tr}\Big(\tau \hat{B}_{b|y}\Big) & & & & & \\ & & \mathrm{Tr}\Big(\tau \hat{A}_{a'|x'}\hat{A}_{a|x}\hat{B}_{b|y}\Big) & & & & & \\ & & & & & & & \\ & & & & & & & \end{bmatrix} \end{array}, \tag{3.14}$$

where each entry is indexed by words in $A_{a|x}, B_{b|y}$ with length $\leq n$, and the tensor products between $\hat{A}_{a|x}, \hat{B}_{b|y}$ are omitted for simplicity. In other words, we construct the moment matrix via the formula

$$\Gamma^n_{\boldsymbol{\omega},\boldsymbol{\nu}} = \mathrm{Tr}\Big(\tau \hat{\omega}^\dagger \hat{\nu}\Big). \tag{3.15}$$

Observe that by construction, for all $n \geq 2$, the moment matrix satisfies the following properties:

(i) (normalised) $\Gamma^n_{1,1} = 1$;

(ii) (positive semi-definite) $\Gamma^n \succeq 0$;

(iii) (Hankel condition) $\Gamma^n_{\boldsymbol{\omega}',\boldsymbol{\nu}'} = \Gamma^n_{\boldsymbol{\omega},\boldsymbol{\nu}}$ whenever $\boldsymbol{\omega}'^\dagger \boldsymbol{\nu}' = \boldsymbol{\omega}^\dagger \boldsymbol{\nu}$.

(iv) (Compatibility) $\Gamma^n_{1,A_{a|x}B_{b|y}} = p(ab|xy)$.

We say that the correlation $\vec{P}$ passes the *n-th order NPA hierarchy test* if there exists a moment matrix Γ^n that satisfies the properties (i)–(iv).

Note that we have established the *necessity* of the NPA hierarchy, that is, if a given correlation $\vec{Q}$ is an element of $\mathcal{Q}$, then $\vec{Q}$ passes all levels of the NPA hierarchy tests (i.e. for all n there exists a moment matrix Γ^n). Equivalently, if at some n the correlation $\vec{Q}$ does not admit a moment matrix Γ^n, then $\vec{Q} \notin \mathcal{Q}$.

For each n, certifying the existence of a moment matrix Γ^n can be casted as a semi-definite programming (SDP), which can be solved by computers in polynomial time; see Section 3.2.4 for a brief introduction. Thus, we have found a practical method to check whether a given correlation $\vec{Q}$ is a (tensor or commutator) quantum correlation.

The exact same argument works for $\vec{Q} \in \mathcal{Q}'$ the commutator quantum correlations.

3.2.2 Sufficiency of the hierarchy

While the necessity is clear, is the NPA hierarchy *sufficient*? That is, if a correlation $\vec{Q}$ passes the NPA hierarchy test for all n, does it mean that $\vec{Q}$ is a quantum correlation? The answer is definitive, but rather than being a tensor quantum correlation, $\vec{Q}$ is a commutator quantum correlation in $\mathcal{Q}'$.

Theorem 3.2 (NPA hierarchy is sufficient) *Let $\vec{Q}$ be a bipartite probability distribution. If $\vec{Q}$ passes the NPA hierarchy test at all levels (that is, it admits a compatible moment matrix Γ^n for each $n \geq 2$), then $\vec{Q} \in \mathcal{Q}'$.*

With the observation we made above, it follows that $\vec{Q} \in \mathcal{Q}'$ if and only if $\vec{Q}$ passes all NPA hierarchy tests.

Proof We only provide a short sketch here; see [42] for the original proof. In fact, the proof is nothing more than a generalised GNS construction.

First, we wish to construct an infinite matrix Γ that acts as the algebraic state in GNS construction. It can be shown that $|\Gamma^n_{\omega,\nu}| \leq 1$ and realise that each Γ^n can be seen as an element in $l^\infty(\mathbb{N})$ (see Example A.5). Then the sequence (Γ^n) is contained in the unit ball $l^\infty(\mathbb{N})$ of with respect to the sup norm. We may then apply the Banach–Alaoglu Theorem A.24 to find a convergent subsequence of (Γ^{n_k}) with the weak-* limit Γ, which is what we want. The weak-* convergence in fact implies the pointwise convergence

$$\lim_k \Gamma^{n_k}_{\omega,\nu} = \Gamma_{\omega,\nu}. \tag{3.16}$$

Next, there are two ways to obtain the GNS construction. The original proof [42] applies the sequential Cholesky decomposition to realise the

moment matrix as a Gram matrix

$$\Gamma_{\omega,\nu} = \langle \varphi_\omega | \varphi_\nu \rangle, \tag{3.17}$$

with respect to an infinite family of vectors $\{|\varphi_\omega\rangle\}$ on all words. Then the Hilbert space can be defined as the closure of the span of $\{|\varphi_\omega\rangle\}$.

Alternatively, but *also equivalently*, one may note that the linear functional L evaluating the set of all words, defined as $L(\omega^\dagger \nu) = \Gamma_{\omega,\nu}$, is analogous to an algebraic state. Except that an algebraic state is defined over a C^*-algebra, while the set of all words is at best a *-algebra. Nevertheless, one can use the notion of archemedean modules to check that the set $N_L = \{\omega \mid L(\omega^\dagger \omega) = 0\}$ is a left-ideal; see [13, Theorem 1.27]. After this technical difficulty, the rest of the arguments in the proof of the GNS reconstruction Theorem 2.20 follow through, and we obtain a cyclic representation $(\mathcal{H}, \pi, |\varphi_1\rangle)$, where we denote by $\pi(\omega) = \hat{\omega}$ the operator acting by left multiplication on the equivalent classes of all words.

The representation π respects all the algebraic relations of ω, particularly the commutation relation between α and β. To see this, we check that

$$\langle \varphi_\omega | [\hat{A}_{a|x}, \hat{B}_{b|y}] | \varphi_\nu \rangle = \langle \varphi_\omega | \pi([A_{a|x}, B_{b|y}]) | \varphi_\nu \rangle = 0. \tag{3.18}$$

Then, since the set $\{|\varphi_\omega\rangle\}$ is dense in $\mathcal{H}$, the commutator

$$[\hat{A}_{a|x}, \hat{B}_{b|y}] = 0 \tag{3.19}$$

in the weak-* topology.

In particular, we have a density operator $\tau = |\varphi_1\rangle\langle\varphi_1|$ such that

$$\hat{\omega} \, |\varphi_\nu\rangle = \pi(\omega) \, |\varphi_\nu\rangle = |\varphi_{\omega\nu}\rangle \tag{3.20}$$

$$\mathrm{Tr}\Big(\tau \hat{\omega}^\dagger \hat{\nu}\Big) = \langle \varphi_1 | \hat{\omega}^\dagger \hat{\nu} | \varphi \rangle = \langle \varphi_\omega | \varphi_\nu \rangle = \Gamma_{\omega,\nu} \tag{3.21}$$

which implies $q(ab|xy) = \Gamma_{1, A_{a|x} B_{b|y}} = \mathrm{Tr}\Big(\tau \hat{A}_{a|x} \hat{B}_{b|y}\Big)$. Observe that we have constructed a Hilbert space $\mathcal{H}$, a density operator τ, and mutually commuting PVMs $\{\hat{A}_{a|x}\}$, $\{B_{b|y}\}$ that can recover the given probability correlation $\vec{Q}$ by Born's rule. That is, $\vec{Q} \in \mathcal{Q}'$. □

Finally, denote the $\mathcal{Q}^n$ to be the set of all correlations $\vec{Q}$ passes all NPA hierarchy at level $2, 3, \ldots, n$ Then Theorem 3.2 implies the inclusions $\mathcal{Q}^2 \subset \mathcal{Q}^3 \subset \ldots \mathcal{Q}'$ and $\cup_n \mathcal{Q}^n = \mathcal{Q}'$, which is exactly the pictorial sketch as in Figure 3.4.

3.2.3 Stopping criteria

Although Theorem 3.2 provides an excellent theoretical result, computers cannot handle infinite levels of calculation. In practice, we want to determine the stopping criteria for our NPA hierarchy, meaning the conditions under which the hierarchy is sufficient at a finite level n. Interestingly, a stopping criterion at a finite level corresponds to a finite-dimensional quantum system, where the concepts of tensor quantum correlations and commutator quantum correlations align according to Tsirelson's theorem 3.1.

We present a simplified (compared to the original paper) rank-loop definition presented in [57] that turns out to be our stopping criteria.

Definition 3.3 *Given $N \geq 2$, the moment matrix Γ^N satisfies the* rank-loop *condition if* $\text{rank}(\Gamma^{N-1}) = \text{rank}(\Gamma^{N-1})$.

Theorem 3.4 *A bipartite probability distribution $\vec{Q}$ admits a* finite-dimensional *commutator quantum realisation if and only if there exists an $N \geq 2$ such that Γ^N admits a rank-loop.*

It follows that if $\vec{Q}$ admits a moment matrix Γ^N with a rank-loop for some N, $\vec{Q}$ is also a tensor quantum correlation.

Proof We only provide a sketch, and refer to [42, Theorem 10] or [57, Theorem 3.3] for a more detailed proof.

The "$\Rightarrow$" direction is easy. With a quantum representation of dimension d we construct the moment matrix Γ^n for any n the usual way (Equation 3.15. Then $\text{rank}(\Gamma^n)$ is an increasing sequence (Γ^{n-1} is a submatrix Γ^n) and is upper bounded by d, which implies the rank-loop.

For the converse, suppose that we have Γ^N with rank-loop. Then we analogously use the Cholesky decomposition to write $\Gamma^N_{\boldsymbol{\omega},\boldsymbol{\nu}} = \langle \varphi_{\boldsymbol{\omega}} | \varphi_{\boldsymbol{\nu}} \rangle$ for a *finite* family of vectors $\{|\varphi_\omega\rangle\}_{|\boldsymbol{\omega}| \leq N}$. Then we define a *finite-dimensional* Hilbert space

$$\mathcal{H} = \text{Span}\,\{|\varphi_\omega\rangle\}_{|\boldsymbol{\omega}| \leq N} = \text{Span}\,\{|\varphi_\omega\rangle\}_{|\boldsymbol{\omega}| \leq N-1}\,, \tag{3.22}$$

where the second equality holds due to the rank-loop condition.

The only difference from having only a finite matrix Γ^N appears when one attempts to show $[\hat{A}_{a|x}, \hat{B}_{b|y}] = 0$. This is because the key equation $\langle \varphi_{\boldsymbol{\omega}} | [\hat{A}_{a|x}, \hat{B}_{b|y}] | \varphi_{\boldsymbol{\nu}} \rangle = 0$ does not make sense for $\boldsymbol{\omega}$ and $\boldsymbol{\nu}$ of length N, and in general, one cannot conclude $[\hat{A}_{a|x}, \hat{B}_{b|y}] = 0$ without checking the inner product over all $\text{Span}\,\{|\varphi_\omega\rangle\}_{|\boldsymbol{\omega}| \leq N}$. But thanks to the rank loop, not being able to check long words is no longer problematic as it is enough to check words of length $\leq N - 1$, for which the equation is well-defined.

It follows that the cyclic representation $(\mathcal{H}, \pi, |\varphi_1\rangle)$ is the desired finite-dimensional (commutator) quantum representation. Then Tsirelson's Theorem 3.1 shows the second claim. □

3.2.4 Semi-definite Programs

As mentioned earlier, the search for compatible moment matrices Γ^n associated with a given probability distribution can be cast as a *semi-definite program (SDP)*. We provide a concise introduction to the basics of SDP based on [42, Appendix A]. We refer to [70, 75] for more complete information.

SDP deals with the optimisation of a linear function subject to linear matrix inequality constraints. A general SDP problem can be written in the following *primal form*:

$$\begin{aligned} &\text{maximise } \langle C, X\rangle \\ &\text{subject to } \langle A_i, X\rangle = b_i, \quad i = 1, \ldots, m, \\ &\qquad\qquad X \succeq 0, \end{aligned} \tag{3.23}$$

where C and A_i are Hermitian matrices, X is the optimisation variable, and $\langle A, B\rangle = \mathrm{Tr}(AB)$ denotes the trace inner product. The notation $X \succeq 0$ indicates that X is a positive semi-definite matrix.

The *dual form* of the SDP problem can be written as:

$$\begin{aligned} &\text{minimise } b^T y \\ &\text{subject to } C - \sum_{i=1}^{m} y_i A_i \succeq 0, \end{aligned} \tag{3.24}$$

where y is the vector of dual variables.

The duality condition in SDP establishes a relationship between primal and dual problems. In many cases, the optimal values of the primal and dual problems are equal, indicating that the duality condition holds. For the NPA hierarchy, this condition is satisfied, allowing us to treat the NPA hierarchy as an SDP problem. The duality condition enables the use of efficient algorithms to find the moment matrices associated with the NPA hierarchy. Numerical packages such as SeDuMi [64] have been developed to solve these optimisation problems effectively.

Chapter 4

Algebraic approaches to quantum bilocal scenarios

In this chapter, we turn our attention to the application of the mathematical formalism introduced in Chapter 2 to study quantum nonlocality, with a particular focus on the quantum bilocal scenario. With the rising interest in network scenarios, where distant parties have access to multiple independent sources, characterising the quantum correlations feasible for a given network has become an important area of research for investigating quantum nonlocality. Unlike the standard Bell scenario (see Figure 4.1), where every party has access to a common quantum source, network scenarios, where distant parties have access to multiple independent sources, have garnered increasing interest in recent years. The bilocal scenario illustrated in Figure 4.2, in which

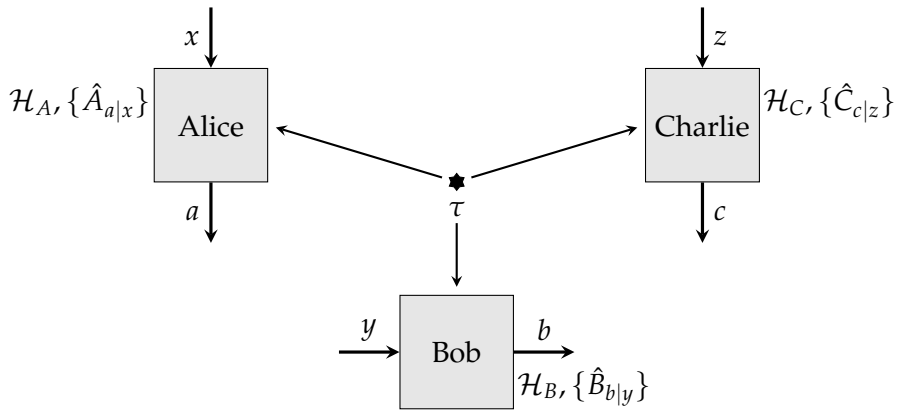


Figure 4.1: This figure illustrates the standard tripartite Bell scenario, where three parties, Alice, Bob, and Charlie, share a quantum state τ. Alice receives input x and produces output a, with a PVM $\hat{A}_{a|x}$ acting on her Hilbert space $\mathcal{H}_A$. Similarly, Bob receives input y and produces output b, with a PVM $\hat{B}_{b|y}$ acting on his Hilbert space $\mathcal{H}_B$. Charlie receives input z and produces output c, with a PVM $\hat{C}_{c|z}$ acting on his Hilbert space $\mathcal{H}_C$.

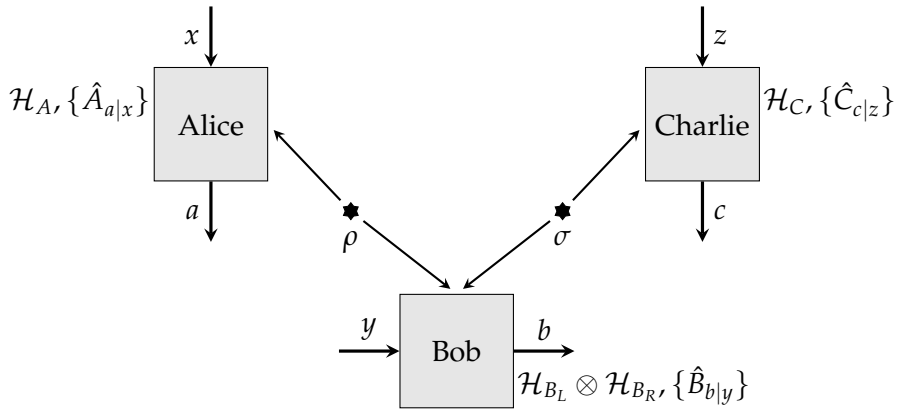


Figure 4.2: This figure illustrates the quantum bilocal scenario, where three parties, Alice, Bob, and Charlie, are involved in two independent quantum sources, ρ shared between Alice and Bob, and σ shared between Bob and Charlie. Denote the global state $\tau = \rho \otimes \sigma$. Alice receives input x and produces output a, with a PVM $\hat{A}_{a|x}$ acting on her Hilbert space $\mathcal{H}_A$. Bob has a tensor product Hilbert space $\mathcal{H}_{B_L} \otimes \mathcal{H}_{B_R}$, with a PVM $\hat{B}_{b|y}$ acting on it. Charlie receives input z and produces output c, with a PVM $\hat{C}_{c|z}$ acting on his Hilbert space $\mathcal{H}_C$. The shared quantum states ρ and σ allow for correlations between the outcomes of Alice and Bob, and Bob and Charlie, respectively. Alice and Charlie's measurements are uncorrelated, which is represented by the factorisation condition $\mathrm{Tr}\Big(\tau(\hat{A}_{a|x} \otimes \hat{C}_{c|z})\Big) = \mathrm{Tr}\Big(\rho\hat{A}_{a|x}\Big)\,\mathrm{Tr}\Big(\sigma\hat{C}_{c|z}\Big) = \mathrm{Tr}\Big(\tau\hat{A}_{a|x}\Big)\,\mathrm{Tr}\Big(\tau\hat{C}_{c|z}\Big)$.

three parties Alice, Bob, and Charlie are arranged in a line and Alice and Bob share one source while Bob and Charlie share another, represents the simplest network scenario. This setup also has practical applications, such as entanglement swapping. Thus, understanding the bilocal scenario is crucial for studying more general network scenarios.

In standard quantum theory, the bilocal quantum correlations in Figure 4.2 can be formalised as follows.

Definition 4.1 (Tensor bilocal quantum correlation) *The tripartite probability distribution $\vec{P}$ is said to be a* tensor bilocal quantum correlation *if there are*

(a) a composite Hilbert space $\mathcal{H}_A \otimes \mathcal{H}_{B_L} \otimes \mathcal{H}_{B_R} \otimes \mathcal{H}_C$,

(b) two pure density operators ρ on $\mathcal{H}_A \otimes \mathcal{H}_{B_L}$ and σ on $\mathcal{H}_{B_R} \otimes \mathcal{H}_C$,

(c) PVMs $\{\hat{A}_{a|x}\}$ act on $\mathcal{H}_A$, $\{\hat{B}_{b|y}\}$ act on $\mathcal{H}_{B_L} \otimes \mathcal{H}_{B_R}$, and $\{\hat{C}_{c|z}\}$ act on $\mathcal{H}_C$,

such that $p(abc|xyz) = \mathrm{Tr}\Big((\rho \otimes \sigma)(\hat{A}_{a|x} \otimes \hat{B}_{b|y} \otimes \hat{C}_{c|z})\Big)$.

To characterise the set of tensor bilocal quantum correlations in Definition 4.1, several methods have been proposed to address this issue, such as the scalar extension [52] (which is applicable to the bilocal scenario but lacks a proof of convergence), the inflation technique [74] (a more general method), and the SDP hierarchy of state polynomials in [32] that seems to approach

the optimisation problem from a primal perspective. In this chapter, we introduce [57] and [35], which provide methods to characterise the quantum correlations in the bilocal scenario. The algebraic perspective introduced in Chapter 2 proves to be a powerful tool for gaining a deeper understanding of these scenarios and elucidating the rich structure of quantum nonlocality in network settings.

For simplicity, in this chapter we omit the tensor product signs when the context is clear, e.g. $\hat{A}_{a|x}\hat{B}_{b|y}\hat{C}_{c|z}$ instead of $\hat{A}_{a|x} \otimes \hat{B}_{b|y} \otimes \hat{C}_{c|z}$ with the natural identification.

4.1 Bilocal quantum correlations with generalised NPA hierarchies

In this section we introduce selected results from [57].

4.1.1 Generalising NPA hierarchy for tripartite case

In order to characterise the set in Definition 4.1, one may consider the NPA hierarchy-type argument in Section 3.2. We extend the notation in Section 3.2.1 to the tripartite scenario: in addition to monomials $\boldsymbol{\alpha}, \boldsymbol{\beta}$ in $A_{a|x}, B_{b|y}$, we consider one more set of formal letters $C_{c|z}$ associated with Charlie's PVM $\hat{C}_{c|z}$ and denote monomials in $C_{c|z}$ by $\boldsymbol{\gamma}$. Analogously, we impose $C_{c|z}^2 = C_{c|z}$, $C_{c|z}^\dagger = C_{c|z}$, and commutation relations $[\boldsymbol{\alpha}, \boldsymbol{\beta}] = [\boldsymbol{\alpha}, \boldsymbol{\gamma}] = [\boldsymbol{\beta}, \boldsymbol{\gamma}] = 0$. Then any word $\boldsymbol{\omega}$ admits a unique decomposition $\boldsymbol{\omega} = \boldsymbol{\alpha}\boldsymbol{\beta}\boldsymbol{\gamma}$.

Analogously, for each $n \geq 3$, any tensor tripartite quantum correlation $\vec{P}$ gives rise to a n-th order moment matrix Γ^n

$$
\begin{array}{c}
\\
\mathbb{1} \\
(A_{a|x})^\dagger \\
(B_{b|y})^\dagger \\
(C_{c|z})^\dagger \\
(A_{a|x}A_{a'|x'})^\dagger \\
(A_{a|x}B_{b|y})^\dagger \\
\cdots
\end{array}
\begin{array}{c}
\begin{array}{ccccccc}
\mathbb{1} & A_{a|x} & B_{b|y} & C_{c|z} & A_{a|x}A_{a'|x'} & A_{a|x}B_{b|y} & \cdots
\end{array} \\
\left[\begin{array}{ccccccc}
1 & & & & & \mathrm{Tr}\left(\tau \hat{A}_{a|x}\hat{B}_{b|y}\right) & \\
 & \mathrm{Tr}\left(\tau \hat{A}_{a|x}\hat{B}_{b|y}\right) & & & & & \\
 & & & & & & \\
 & & & & & \mathrm{Tr}\left(\tau \hat{A}_{a|x}\hat{B}_{b|y}\hat{C}_{c|z}\right) & \\
 & \mathrm{Tr}\left(\tau \hat{A}_{a'|x'}\hat{A}_{a|x}\hat{B}_{b|y}\right) & & & & & \\
 & & & & & & \\
 & & & & & &
\end{array}\right]
\end{array},
\tag{4.1}
$$

which is labelled by formal monomials in $\{A_{a|x}, B_{b|y}, C_{c|z}\}$ of length at most n.

However, this direct generalisation of the moment matrix is too relaxed, in the sense that the resulting hierarchy converges only to the *commutator*

tripartite Bell scenario in Figure 4.1 rather than some relaxation of the bilocal quantum scenario in Figure 4.2. In fact, this claim holds for any multipartite Bell scenario. To prove the claim, one note that the proof of Theorem 3.2 can be straightforwardly generalised to the case of having more sets of formal letters.

4.1.2 Key concepts in factorisation bilocal NPA hierarchy

To obtain a tighter approximation to the set of tensor quantum bilocal correlations, the authors of [57] consider the factorisation constraints

$$\mathrm{Tr}(\tau\hat{\alpha}\hat{\gamma}) = \mathrm{Tr}(\tau\hat{\alpha})\,\mathrm{Tr}(\tau\hat{\gamma}), \tag{4.2}$$

which corresponds to the fact that Alice and Charlie's measurements are uncorrelated in a bilocal quantum network, and impose them to the standard NPA moment matrices.

Definition 4.2 (Factorisation bilocal NPA hierarchy) *Fix $n \in \mathbb{N}$, $n \geq 3$, let Γ^n be a square matrix indexed by all words $\boldsymbol{\omega}$ in formal letters $A_{a|x}, B_{b|y}, C_{c|z}$ of length $\leq n$. We say that Γ^n is a* factorisation bilocal moment matrix *of order n if*

(i) (normalised) $\Gamma^n_{1,1} = 1$;

(ii) (positive semi-definite) $\Gamma^n \succeq 0$;

(iii) (Hankel condition) it satisfies all the linear constraints

$$\Gamma^n_{\boldsymbol{\omega},\boldsymbol{\nu}} = \Gamma^n_{\boldsymbol{\omega}',\boldsymbol{\nu}'} \tag{4.3}$$

whenever $\boldsymbol{\omega}^\dagger\boldsymbol{\nu} = \boldsymbol{\omega}'^\dagger\boldsymbol{\nu}'$;

(iv) (factorisation) it satisfies the additional factorisation constraints

$$\Gamma^n_{\boldsymbol{\alpha},\boldsymbol{\gamma}} = \Gamma^n_{\boldsymbol{\alpha},1} \cdot \Gamma^n_{1,\boldsymbol{\gamma}}, \tag{4.4}$$

where $\boldsymbol{\alpha}$ is any monomial in $A_{a|x}$ and $\boldsymbol{\gamma}$ is any monomial in $C_{c|z}$.

Furthermore, given a tripartite probability distribution $\vec{Q}$, the factorisation moment matrix is compatible *to $\vec{Q}$ if*

(v) (compatibility) $\Gamma^n_{1,A_{a|x}B_{b|y}C_{c|z}} = q(abc|xyz)$.

For each $n \geq 3$, we say that $\vec{Q}$ passes the n-th order factorisation bilocal NPA hierarchy *if for there exists a factorisation bilocal moment matrix Γ^n compatible to $\vec{Q}$, that is, satisfying properties (i)–(v).*

It is clear that the factorisation bilocal NPA hierarchy provides an outer approximation to Definition 4.1 that is more restrictive than the standard

NPA hierarchy. However, the question remains as to what set does this hierarchy converge to.

To this end, the authors of [57] propose the following relaxation of tensor quantum bilocal correlations by considering only the algebraic relations among all mathematical objectives in Definition 4.1.

Definition 4.3 (Renou–Xu model) *A tripartite probability distribution $\vec{Q}$ is said to be a* Renou–Xu commutator bilocal quantum correlation *if there are*

(a) a Hilbert space $\mathcal{H}$,

(b) a pure density operator τ on $\mathcal{H}$,

(c) two commuting projectors ρ and σ (possibly infinite trace) on $\mathcal{H}$,

(d) PVMs $\{\hat{A}_{a|x}\}$, $\{\hat{B}_{b|y}\}$, and $\{\hat{C}_{c|z}\}$ on $\mathcal{H}$,

such that

(i) $\tau = \rho \cdot \sigma = \sigma \cdot \rho$,

(ii) $[\hat{A}_{a|x}, \hat{B}_{b|y}] = [\hat{B}_{b|y}, \hat{C}_{c|z}] = [\hat{C}_{c|z}, \hat{A}_{a|x}] = 0$,

(iii) $[\hat{A}_{a|x}, \sigma] = [\rho, \hat{C}_{c|z}] = 0$,

with $q(abc|xyz) = \mathrm{Tr}_\tau\Big(\hat{A}_{a|x}\hat{B}_{b|y}\hat{C}_{c|z}\Big)$ *for all* $\hat{A}_{a|x}, \hat{B}_{b|y}, \hat{C}_{c|z}$.

4.1.3 Convergence of factoristaion bilocal NPA hierarchy

Factorisation bilocal hierarchy indeed converges to the set of Renou–Xu commutator bilocal quantum correlations.

Theorem 4.4 (Factorisation bilocal NPA hierarchy is convergent) *Let $\vec{Q}$ be a tripartite probability distribution. Then $\vec{Q}$ passes the factorisation bilocal NPA hierarchy for all orders iff $\vec{Q}$ is a Renou–Xu commutator bilocal quantum correlation.*

Proof The proof is based on [57, Appendix B.1].

($\Leftarrow$) We first show the easy direction. Given any $n \geq 3$ and assume that we have a Renou–Xu commutator bilocal quantum correlation $\vec{Q}$, and suppose that the pure density operator $\tau = \rho \cdot \sigma = |\varphi_1\rangle\langle\varphi_1|$. Then, with the usual notation, we may define the moment matrix Γ^n by

$$\Gamma^n_{\omega,\nu} = \mathrm{Tr}\Big(\tau\hat{\omega}^\dagger\hat{v}\Big). \tag{4.5}$$

It remains to check that Γ satisfies properties (i)–(v) of Definition 4.2. Indeed, (i), (iii), and (v) clearly hold by construction. Positivity (ii) holds due to

the fact that Γ^n is a Gram matrix. We check factorisation constraints (iv) by computing

$$\begin{aligned}\mathrm{Tr}\Big(\tau\hat{\alpha}^\dagger\tau\hat{\gamma}\Big) &= \mathrm{Tr}\Big(\rho\sigma\hat{\alpha}^\dagger\rho\sigma\hat{\gamma}\Big)\\ &= \mathrm{Tr}\Big(\rho\hat{\alpha}^\dagger\sigma\rho\sigma\hat{\gamma}\Big)\\ &= \mathrm{Tr}\Big(\hat{\alpha}^\dagger\sigma\rho\sigma\hat{\gamma}\rho\Big)\\ &= \mathrm{Tr}\Big(\hat{\alpha}^\dagger\sigma\rho\sigma\rho\hat{\gamma}\Big)\\ &= \mathrm{Tr}\Big(\hat{\alpha}^\dagger\tau^2\hat{\gamma}\Big) = \mathrm{Tr}\Big(\tau\hat{\alpha}^\dagger\hat{\gamma}\Big),\end{aligned} \tag{4.6}$$

where commutativity and trace cyclicity are considered. It follows that

$$\begin{aligned}\Gamma^n_{\boldsymbol{\alpha},\boldsymbol{\gamma}} &= \mathrm{Tr}(\tau\hat{\alpha}\hat{\gamma}) = \mathrm{Tr}\Big(\tau\hat{\alpha}^\dagger\tau\hat{\gamma}\Big)\\ &= \langle\varphi_1|\hat{\alpha}^\dagger|\varphi_1\rangle\langle\varphi_1|\hat{\gamma}|\varphi_1\rangle = \Gamma^n_{\boldsymbol{\alpha},1}\Gamma^n_{1,\boldsymbol{\gamma}},\end{aligned} \tag{4.7}$$

as desired.

($\Rightarrow$) For the sufficiency of the hierarchy, we first construct a GNS representation $(\mathcal{H}, \pi, |\varphi_1\rangle = |v^{00}\rangle)$. This is completely analogous to the proof of Theorem 3.2 for the convergence of the standard NPA hierarchy, so we omit the details. In short, the representation π, with $\pi(\boldsymbol{\omega}) = \hat{\omega}$, respects all the commutation relations of $\boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\gamma}$. We also obtain the pure density operator $\tau = |\varphi_1\rangle\langle\varphi_1| = |v^{00}\rangle\langle v^{00}|$ such that

$$\hat{\omega}\,|\varphi_{\boldsymbol{\nu}}\rangle = \pi(\boldsymbol{\omega})\,|\varphi_{\boldsymbol{\nu}}\rangle = |\varphi_{\boldsymbol{\omega}\boldsymbol{\nu}}\rangle \tag{4.8}$$

$$\mathrm{Tr}\Big(\tau\hat{\omega}^\dagger\hat{\nu}\Big) = \langle\varphi_1|\hat{\omega}^\dagger\hat{\nu}|\varphi\rangle = \langle\varphi_{\boldsymbol{\omega}}|\varphi_{\boldsymbol{\nu}}\rangle = \Gamma_{\boldsymbol{\omega},\boldsymbol{\nu}}. \tag{4.9}$$

To finish the proof, it remains to construct the commuting projectors ρ, σ in Definition 4.3. Define $H_{AB_L} = \mathrm{Span}\,\{P(\{\hat{\alpha}\})\}_P$ as the subspace generated by polynomials in $\hat{\alpha}$, which is nonempty due to $|v^{00}\rangle = |\varphi_1\rangle \in H_{AB_L}$. Then, from $|v^{00}\rangle$, apply Gram–Schmidt procedure to obtain an orthogonal basis $\{P_i(\{\hat{\alpha}\})\,|v^{00}\rangle = |v^{i0}\rangle, i \in I\}$ for a countable set I. Analogously, we define $H_{B_RC} = \mathrm{Span}\,\{Q(\{\hat{\gamma}\})\}_Q$ with the orthogonal basis $\{Q_j(\{\hat{\gamma}\})\,|v^{00}\rangle = |v^{0j}\rangle, j \in J\}$ for a countable set J.

We claim that $\langle v^{i0}|v^{0j}\rangle = 0$ for all $i \in I \setminus \{0\}$ and $j \in J \setminus \{0\}$. Indeed, assume that $P_i(\{\hat{\alpha}\}) = \sum_{\hat{\alpha}} p^i_{\hat{\alpha}}\hat{\alpha}$ and $Q_j(\{\hat{\gamma}\}) = \sum_{\hat{\gamma}} q^j_{\hat{\gamma}}\hat{\gamma}$, for some real coefficients $p^i_{\hat{\alpha}}$ and $q^j_{\hat{\gamma}}$. We calculate, by orthogonality and linearity,

$$\begin{aligned}\langle v^{i0}|v^{0j}\rangle &= \langle v^{00}|P_i(\{\hat{\alpha}\})^\dagger \cdot Q_j(\{\hat{\gamma}\})|v^{00}\rangle\\ &= \sum_{\hat{\alpha},\hat{\gamma}} (p^i_{\hat{\alpha}})^\dagger q^j_{\hat{\gamma}} \cdot \langle v^{00}|\hat{\alpha}^\dagger\hat{\gamma}|v^{00}\rangle\end{aligned}$$

$$
\begin{aligned}
&= \sum_{\hat{\alpha},\hat{\gamma}} (p^i_{\hat{\alpha}})^\dagger q^j_{\hat{\gamma}} \cdot \Gamma^{\infty}_{\boldsymbol{\alpha},\gamma} \\
&= \sum_{\hat{\alpha},\hat{\gamma}} (p^i_{\hat{\alpha}})^\dagger q^j_{\hat{\gamma}} \cdot \Gamma^{\infty}_{\boldsymbol{\alpha},1} \cdot \Gamma^{\infty}_{1,\gamma} \\
&= \sum_{\hat{\alpha},\hat{\gamma}} (p^i_{\hat{\alpha}})^\dagger q^j_{\hat{\gamma}} \cdot \langle v^{00}|\hat{\alpha}^\dagger|v^{00}\rangle\langle v^{00}|\hat{\gamma}|v^{00}\rangle \\
&= \langle v^{00}|P_i(\{\hat{\alpha}\})^\dagger|v^{00}\rangle\langle v^{00}|Q_j(\{\hat{\gamma}\})|v^{00}\rangle \\
&= \langle v^{i0}|v^{00}\rangle\langle v^{00}|v^{0j}\rangle = 0.
\end{aligned}
$$

Now we define ρ, σ to be projectors on $H_{B_R C}$ and H_{AB_L}, respectively. Concretely, they are written as

$$\rho = \sum_j |v^{0j}\rangle\langle v^{0j}| \tag{4.10}$$

$$\sigma = \sum_i |v^{i0}\rangle\langle v^{i0}|. \tag{4.11}$$

Then $\rho \cdot \sigma = \sum_{ij} |v^{0j}\rangle\langle v^{0j}| \cdot |v^{i0}\rangle\langle v^{i0}| = |v^{00}\rangle\langle v^{00}| = \tau = \sigma \cdot \rho$ by the claim.

Finally, we show $[\rho, \hat{\gamma}]$. Since the support of $[\rho, \hat{\gamma}]$ is contained in $H_{B_R C}$ and for all $j, j' \in J$, we have

$$
\begin{aligned}
\langle v^{0j}|[\hat{\gamma}, \rho]|v^{0j'}\rangle &= \sum_k \langle v^{0j}|\hat{\gamma}|v^{0k}\rangle\langle v^{0k}|v^{0j'}\rangle - \sum_k \langle v^{0j}|v^{0k}\rangle\langle v^{0k}|\hat{\gamma}|v^{0j'}\rangle \\
&= \langle v^{0j}|\hat{\gamma}|v^{0j'}\rangle - \langle v^{0j}|\hat{\gamma}|v^{0j'}\rangle = 0,
\end{aligned} \tag{4.12}
$$

it follows that $[\rho, \hat{\gamma}] = 0$. Note that a similar argument shows that $[\hat{\alpha}, \sigma] = 0$, which completes the proof. □

A stopping criterion of the factorisation bilocal NPA hierarchy is also shown.

Theorem 4.5 (Stopping criteria) *Let $\vec{Q}$ be a tripartite probability distribution. Then $\vec{Q}$ admits a compatible factorisation bilocal moment matrix Γ^n with rank-loop iff $\vec{Q}$ admits a finite-dimensional Renou–Xu commutator bilocal quantum representation, where the rank-loop condition is a direct generalisation of Definition 3.3.*

Proof This is completely analogous to the stopping criterion of the standard NPA hierarchy in Theorem 3.4, hence we skip the proof; see [57, Appendix B.2] for details. □

4.1.4 Remarks on factorisation bilocal NPA hierarchy

We now discuss some open questions from [57].

Tsirelson's problem in bilocal scenario

We observe that the set of tensor bilocal quantum correlations in Definition 4.1 is contained within the set of Renou–Xu commutator bilocal

quantum correlations. However, the converse containment, as in the Bell scenario, has been proven to be false [28]. The authors of [57] leave open the question of whether a bilocal version of Tsirelson's theorem 3.1 exists, that is, if these two sets coincide under the condition of finite-dimensionality. This is later proven to be true by [35], as we will see in Section 4.2.3.

Theorem 4.4 offers an appealing theoretical result that outlines a hierarchy of increasingly tight outer approximations to tensor bilocal quantum correlations. Specifically, if the stopping criteria in Theorem 4.5 are met, then by bilocal Tsirelson's Theorem 4.12, the factorisation bilocal quantum scenario converges to the set of (finite-dimensional) tensor bilocal quantum correlations. However, this hierarchy's drawback is that it incorporates the nonlinear constraints in Equation 4.4, making it unsolvable by SDP.

Scalar extension hierarchy

In an effort to find a linear hierarchy equivalent to the factorisation bilocal NPA hierarchy, [57] formalised the concept of scalar extension, initially proposed in [52], using *scalar extension polynomials*. Scalar extension involves introducing a new set of scalar operators, denoted by $\hat{\kappa}_\alpha$ for each $\hat{\alpha}$, which commute with all other operators and are expected to be scalar operators $\mathrm{Tr}(\tau\hat{\alpha})\mathbb{1}_{\mathcal{H}}$. The factorisation constraint can then be linearised as the equation $\mathrm{Tr}(\tau\hat{\alpha}\hat{\gamma}) = \mathrm{Tr}(\tau\hat{\kappa}_\alpha\hat{\gamma}) = \mathrm{Tr}(\tau\,\mathrm{Tr}(\tau\hat{\alpha})\mathbb{1}_{\mathcal{H}}\hat{\gamma}) = \mathrm{Tr}(\tau\hat{\alpha})\,\mathrm{Tr}(\tau\hat{\gamma})$.

Using the corresponding commuting scalar letters κ_α, [57, Appendix C] introduces a linear hierarchy called *scalar extension bilocal hierarchy* with these new linearised factorisation constraints. However, the commutativity of κ_α does not guarantee that their GNS reconstructed operators $\hat{\kappa}_\alpha$ will be scalar multiples of the identity; see [57, Remark C.2]. Since factorisation constraints do not generally hold, the scalar extension bilocal hierarchy does not coincide with the factorisation bilocal NPA hierarchy.

Nonetheless, a group of mathematicians later demonstrated Positivstellensätze of *state polynomials*[32], which are a generalisation of scalar extension polynomials. The Positivstellensätze lead to a convergent SDP hierarchy that appears to imply the equivalence of the scalar extension NPA hierarchy with the factorisation NPA hierarchy, possibly with some extra conditions[1].

[1]Understand the implication of[32] in the context of [57] is still an ongoing work for the author of the thesis.

4.2 Algebraic insights into quantum bilocal network

In this section, we explore an alternative perspective in modelling the quantum bilocal scenario, referencing the content of [35] and building on the ideas presented in [57]. By adopting an operator algebraic approach and taking advantage of the powerful tools introduced in Chapter 2, we aim to answer some open questions posed in [57], such as the bilocal Tsirelson's problem. This change in perspective highlights the effectiveness of the algebraic approach in quantum information theory, and it justifies the effort we have invested in developing the mathematical formalism in Chapter 2. In the following, we delve into the details of this algebraic modelling of the quantum bilocal scenario and discuss the insights it provides into the underlying structure of quantum nonlocality in network settings.

4.2.1 Models for quantum bilocal scenario

We introduce several models for the quantum bilocal scenario in [35]. For easier reference, we first recall the tensor quantum bilocal correlations from Definition 4.1 and the Renou–Xu model 4.3.

Definition 4.6 (Tensor bilocal model) *The tripartite probability distribution $\vec{P}$ is said to be a* tensor bilocal quantum correlation *if there are*

(a) a composite Hilbert space $\mathcal{H}_A \otimes \mathcal{H}_{B_L} \otimes \mathcal{H}_{B_R} \otimes \mathcal{H}_C$,

(b) two pure density operators ρ on $\mathcal{H}_A \otimes \mathcal{H}$ and σ on $\mathcal{H}_{B_R} \otimes \mathcal{H}_C$,

(c) PVMs $\{\hat{A}_{a|x}\}$ over $\mathcal{H}_A$, $\{\hat{B}_{b|y}\}$ over $\mathcal{H}_{B_L} \otimes \mathcal{H}_{B_R}$, and $\{\hat{C}_{c|z}\}$ over $\mathcal{H}_C$,

such that $p(abc|xyz) = \mathrm{Tr}\Big((\rho \otimes \sigma)(\hat{A}_{a|x}\hat{B}_{b|y}\hat{C}_{c|z})\Big)$.

Definition 4.7 (Renou–Xu model) *A tripartite probability distribution $\vec{Q}$ is said to be a* Renou–Xu commutator bilocal quantum correlation *if there are*

(a) a Hilbert space $\mathcal{H}$,

(b) a pure density operator τ on $\mathcal{H}$,

(c) two commuting projectors ρ and σ (possibly infinite trace) on $\mathcal{H}$,

(d) PVMs $\{\hat{A}_{a|x}\}$, $\{\hat{B}_{b|y}\}$, and $\{\hat{C}_{c|z}\}$ on $\mathcal{H}$,

such that

(i) $\tau = \rho \cdot \sigma = \sigma \cdot \rho$,

(ii) $[\hat{A}_{a|x}, \hat{B}_{b|y}] = [\hat{B}_{b|y}, \hat{C}_{c|z}] = [\hat{C}_{c|z}, \hat{A}_{a|x}] = 0$,

(iii) $[\hat{A}_{a|x}, \sigma] = [\rho, \hat{C}_{c|z}] = 0$,

with $q(abc|xyz) = \mathrm{Tr}_\tau\Big(\hat{A}_{a|x}\hat{B}_{b|y}\hat{C}_{c|z}\Big)$ for all $\hat{A}_{a|x}, \hat{B}_{b|y}, \hat{C}_{c|z}$.

We note that the above two definitions are formulated with concrete Hilbert spaces and density operators in mind, that is, they both assume Schrödinger's picture and stick to postulates A–C in Chapter 2. Now we instead take Heisenberg's picture and use A'–C' to introduce another bilocal model[2] in [35].

Definition 4.8 (Algebraic commutator bilocal model) *The tripartite probability distribution $\vec{Q}$ is said to be an* algebraic commutator bilocal quantum correlation *if there are*

(a) a background C^*-algebra $\mathfrak{D}$,

(b) C^*-algebras $\mathfrak{A}, \mathfrak{B}, \mathfrak{C} \subset \mathfrak{D}$ that are multually commuting,

(c) the PVMs $\{\hat{A}_{a|x}\} \subset \mathfrak{A}$, $\{\hat{B}_{b|y}\} \subset \mathfrak{B}$, $\{\hat{C}_{c|z}\} \subset \mathfrak{C}$,

(d) an algebraic *state τ on $\mathfrak{D}$ satisfying factorisation constraints*

$$\tau(\hat{\alpha}\hat{\gamma}) = \tau(\hat{\alpha})\tau(\hat{\gamma}), \tag{4.13}$$

for any $\alpha \in \mathfrak{A}$, $\gamma \in \mathfrak{C}$,

such that $q(abc|xyz) = \tau(\hat{A}_{a|x}\hat{B}_{b|y}\hat{C}_{c|z})$.

Note that the global C^*-algebras $\mathfrak{D}$ can be an abstract algebra in general. Finally, we introduce a mixed model that bridges the algebraic formulation with the usual tensor formulation. Albeit contrasting to the tensor bilocal quantum model, the mixed model does not require spatial separation between Alice and Bob, Bob and Charlie, respectively.

Definition 4.9 (Mixed bilocal model) *The tripartite probability distribution $\vec{Q}$ is said to be a* mixed bilocal quantum correlation *if there are*

(a) Hilbert spaces $\mathcal{H}_{AB_L}$, $\mathcal{H}_{B_RC}$,

(b) C^*-algebras $\mathfrak{A}$, $\mathfrak{B}_L \subset \mathcal{B}(\mathcal{H}_{AB_L})$, $\mathfrak{B}_R$, $\mathfrak{C} \subset \mathcal{B}(\mathcal{H}_{B_RC})$ that are mutually commuting,

(c) PVMs $\{\hat{A}_{a|x}\} \subset \mathfrak{A}$, $\{\hat{B}_{b|y}\} \subset \mathfrak{B}_L \overline{\otimes} \mathfrak{B}_R$, $\{\hat{C}_{c|z}\} \subset \mathfrak{C}$,

(d) density operators ρ on $\mathcal{H}_{AB_L}$ and σ on $\mathcal{H}_{B_RC}$,

such that $p(abc|xyz) = \mathrm{Tr}\Big((\rho \otimes \sigma)(\hat{A}_{a|x} \otimes \hat{B}_{b|y} \otimes \hat{C}_{c|z})\Big)$.

[2] In this thesis, we omit [35, Definition 5], referred to as the "commuting observables model" in the original paper. Our focus is solely on the "algebraic commutator bilocal model", which the original paper terms the "reduced model" in [35, Corollary 6]. This notational difference should be taken into account when comparing the two sources.

4.2.2 Equivalence of bilocal models

One of the key results in [35] is the equivalence of the alternative models introduced in the previous section.

Theorem 4.10 (Equivalence of bilocal models) *The Renou–Xu model in Definition 4.7, the algebraic commutator bilocal model in Definition 4.8, and the mixed bilocal model in Definition 4.9 are equivalent to each other.*

We dedicate the remainder of this section to the proof of Theorem 4.10, which is a consequence of the following result.

Theorem 4.11 *Let $\mathfrak{A}, \mathfrak{B}, \mathfrak{C}$ be mutually commuting C^*-subalgebras of some global C^*-algebra $\mathfrak{D}$ and let τ be an algebraic state on $\mathfrak{D}$ that factorises, i.e.*

$$\tau(\hat{\alpha}\hat{\gamma}) = \tau(\hat{\alpha})\tau(\hat{\gamma}) \tag{4.14}$$

for any $\hat{\alpha} \in \mathfrak{A}$, $\hat{\gamma} \in \mathfrak{C}$.

Relative to the state τ, consider its GNS representations $(\mathcal{H}_{AB_L}, \pi_A, |\varphi_1^A\rangle)$ of $\mathfrak{A}$ and $(\mathcal{H}_{B_RC}, \pi_C, |\varphi_1^C\rangle)$ of $\mathfrak{C}$. Define $\mathfrak{B}_L$ as the commuter of $\pi_A(\mathfrak{A})$ in $\mathcal{H}_{AB_L}$ and $\mathfrak{B}_R$ as the commuter of $\pi_C(\mathfrak{C})$ in $\mathcal{H}_{B_RC}$.

Then there exists a completely positive unital map $\Lambda : \mathfrak{B} \to \mathfrak{B}_L \overline{\otimes} \mathfrak{B}_R$ such that

$$\tau(\hat{\alpha}\hat{\beta}\hat{\gamma}) = \mathrm{Tr}\Big((|\varphi_1^A\rangle\langle\varphi_1^A| \otimes |\varphi_1^C\rangle\langle\varphi_1^C|)\pi_A(\hat{\alpha})\Lambda(\hat{\beta})\pi_C(\hat{\gamma})\Big). \tag{4.15}$$

We remark that the choice of notation $\mathcal{H}_{AB_L}$ and $\mathcal{H}_{B_RC}$ in Theorem 4.11 is deliberate, since they in fact coincide with the subspaces $\mathcal{H}_{AB_L}$ and $\mathcal{H}_{B_RC}$ constructed in the proof of Theorem 4.4. This observation motivates the following proof in [35].

Proof (of Theorem 4.4) Considering the GNS representation $(\mathcal{H}_D, \pi_D, |\varphi_1^D\rangle)$ of $\mathfrak{D}$ relative to τ, we prove the theorem in three steps.

Step 1. We construct an isometric embedding $V : \mathcal{H}_{AB_L} \otimes \mathcal{H}_{B_RC} \to \mathcal{H}_D$ such that

$$V\left|\varphi_1^A\right\rangle \otimes \left|\varphi_1^C\right\rangle = \left|\varphi_1^D\right\rangle \tag{4.16}$$

$$V(\pi_A(\hat{\alpha}) \otimes \pi_C(\hat{\gamma}))V^\dagger = \pi_D(\hat{\alpha}\hat{\gamma}). \tag{4.17}$$

Indeed, with the factorisation constraint of τ, we calculate

$$\begin{aligned} &\left\langle\varphi_1^A\right| \otimes \left\langle\varphi_1^C\right| \pi_A(\hat{\alpha}) \otimes \pi_C(\hat{\gamma}) \left|\varphi_1^A\right\rangle \otimes \left|\varphi_1^C\right\rangle \\ &= \tau(\hat{\alpha})\tau(\hat{\gamma}) = \tau(\hat{\alpha}\hat{\gamma}) = \langle\varphi_1^D|\pi_D(\hat{\alpha}\hat{\gamma})|\varphi_1^D\rangle, \end{aligned} \tag{4.18}$$

which implies $\left|\varphi_1^D\right\rangle$ and $\left|\varphi_1^A\right\rangle \otimes \left|\varphi_1^A\right\rangle$ induce the same cyclic representations of $\mathfrak{AC}$. Then the existence of V is due to the uniqueness statement in Theorem 2.20.

We note that

$$V^\dagger V = \mathbb{1}_{\mathcal{H}_{AB_L}\otimes\mathcal{H}_{B_RC}},\ VV^\dagger = P \in \mathcal{B}(\mathcal{H}_D), \tag{4.19}$$

where P is a projector onto $\overline{\pi_D(\mathfrak{AC})\left|\varphi_1^D\right\rangle} \subset \mathcal{H}_D$.

Step 2. We show that

$$V^\dagger \pi_D(\mathfrak{B}) V \subset \mathfrak{B}_L \overline{\otimes} \mathfrak{B}_R. \tag{4.20}$$

To this end, we use *Step* 1 to compute, for any $\hat{\alpha} \in \mathfrak{A}$, $\hat{\beta} \in \mathfrak{B}$, $\hat{\gamma} \in \mathfrak{C}$,

$$\begin{aligned} &[V^\dagger \pi_D(\hat{\beta}) V, \pi_A(\hat{\alpha}) \otimes \pi_C(\hat{\gamma})] \\ &= [V^\dagger \pi_D(\hat{\beta}) V, V^\dagger V \pi_A(\hat{\alpha}) \otimes \pi_C(\hat{\gamma}) V^\dagger V] \\ &= V^\dagger [\pi_D(\hat{\beta}), V\pi_A(\hat{\alpha}) \otimes \pi_C(\hat{\gamma}) V^\dagger] \\ &= V^\dagger [\pi_D(\hat{\beta}), \pi_D(\hat{\alpha}\hat{\gamma})] V = 0, \end{aligned} \tag{4.21}$$

which implies that $V^\dagger \pi_D(\mathfrak{B}) V \subset (\pi_A(\mathfrak{A}) \odot \pi_C(\mathfrak{C}))'$. Recall Definition 2.35 that $\odot$ is the algebraic tensor product.

Then, by Proposition 2.24, Bicommutant Theorem 2.26, and the commutant theorem for tensor products in Section 2.4.3, we check that

$$\begin{aligned} (\pi_A(\mathfrak{A}) \odot \pi_C(\mathfrak{C}))' &= (\pi_A(\mathfrak{A}) \overline{\otimes} \pi_C(\mathfrak{C}))' \\ &= (\pi_A(\mathfrak{A})'' \overline{\otimes} \pi_C(\mathfrak{C})'')' \\ &= \pi_A(\mathfrak{A})''' \overline{\otimes} \pi_C(\mathfrak{C})''' \\ &= \pi_A(\mathfrak{A})' \overline{\otimes} \pi_C(\mathfrak{C})' = \mathfrak{B}_L \overline{\otimes} \mathfrak{B}_R. \end{aligned} \tag{4.22}$$

Thus, Equation 4.20 follows.

Step 3. We define the map $\Lambda : \mathfrak{B} \to \mathfrak{B}_L \overline{\otimes} \mathfrak{B}_R$ using the formula

$$\Lambda(\hat{\beta}) = V^\dagger \pi_D(\hat{\beta}) V, \tag{4.23}$$

which due to *Step* 2 has the correct range. Then, a straightforward calculation using the results in *Step* 1 shows that

$$\begin{aligned} &\left\langle \varphi_1^A \right| \left\langle \varphi_1^C \right| \pi_A(\hat{\alpha}) \Lambda(\hat{\beta}) \pi_C(\hat{\gamma}) \left| \varphi_1^A \right\rangle \left| \varphi_1^C \right\rangle \\ &= \left\langle \varphi_1^A \right| \left\langle \varphi_1^C \right| (V^\dagger V) \pi_A(\hat{\alpha}) (V^\dagger \pi_D(\hat{\beta}) V) \pi_C(\hat{\gamma}) (V^\dagger V) \left| \varphi_1^A \right\rangle \left| \varphi_1^C \right\rangle \\ &= \left\langle \varphi_1^D \right| (V\pi_A(\hat{\alpha})V^\dagger) \pi_D(\hat{\beta}) (V\pi_C(\hat{\gamma})V^\dagger) \left| \varphi_1^D \right\rangle \\ &= \left\langle \varphi_1^D \right| \pi_D(\hat{\alpha}) \pi_D(\hat{\beta}) \pi_D(\hat{\gamma}) \left| \varphi_1^D \right\rangle = \tau(\hat{\alpha}\hat{\beta}\hat{\gamma}), \end{aligned} \tag{4.24}$$

which completes the proof. □

Now we prove the main theorem.

Proof (of Theorem 4.10)

Step 1 Observe that the equivalence of the Renou–Xu model (Definition 4.7) and the algebraic commutator bilocal model (Definition 4.8) is a direct consequence of Theorem 4.4 of convergence of factorisation bilocal NPA hierarchy.

Indeed, given a tripartite probability distribution $\vec{Q}$ that admits algebraic commutator bilocal model, we use the algebraic state τ to define, for any $n \geq 3$, the factorisation bilocal moment matrix by

$$\Gamma^n_{\omega,\nu} = \tau(\hat{\omega}^\dagger \hat{v}). \tag{4.25}$$

Properties (i)–(iv) of Definition 4.2 hold since τ is an algebraic state, see Definition 2.9.

For the converse, suppose that $\vec{Q}$ is a Renou–Xu commutator bilocal quantum correlation. By Theorem 4.4 and its proof, there exists a "limit factorisation bilocal moment matrix" Γ compatible with $\vec{Q}$. Then using Equation 4.25 again we define an operator $\tau : \mathfrak{A}\mathfrak{B}\mathfrak{C} \to \mathbb{C}$. The defining properties of factorisation bilocal moment matrices lead to the conclusion that τ is indeed the desired algebraic state that factorises.

Step 2 The most difficult part of this proof is the fact that the algebraic commutator bilocal model implies the mixed bilocal model[3]. But we have done all the hard work in the proof of Theorem 4.11.

Indeed, given an algebraic commutator bilocal correlation $\vec{Q}$, we applying Theorem 4.11. In the notation of Theorem 4.11, we have found $\mathcal{H}_{AB_L}$, $\mathcal{H}_{B_RC}$, $\mathfrak{B}_L$, $\mathfrak{B}_R$ for the mixed model. Define $\pi_A(\mathfrak{A})$, $\pi_C(\mathfrak{C})$ as $\mathfrak{A}^{\text{mix}}$, $\mathfrak{C}^{\text{mix}}$, respectively. In addition, we identify the PVMs in the mixed model with $\pi_A(\hat{A}_{a|x})$, $\Lambda(\hat{B}_{b|y})$, $\pi_C(\hat{C}_{c|z})$. The independent density operators are $\rho = |\varphi_1^A\rangle\langle\varphi_1^A|$ and $\sigma = |\varphi_1^C\rangle\langle\varphi_1^C|$.

Step 3 Finally, the fact that the mixed model implies the algebraic commutator bilocal model is trivial. It also implies the Renou–Xu model by identifying the projector ρ with $|\varphi_1^A\rangle\langle\varphi_1^A| \otimes \mathbb{1}_{B_RC}$ and σ with $\mathbb{1}_{AB_L} \otimes |\varphi_1^C\rangle\langle\varphi_1^C|$, note the analogy with the proof of Theorem 4.4. □

[3] In fact, [57, Appendix H] encountered this problem, but the authors were unable to find a way to obtain a spatial separation (tensor product) through purely algebraic relations. It was not until the shift towards an algebraic perspective in [35] that such a separation became clear.

4.2.3 Bilocal Tsirelson's Problem

Finally, we come back to the bilocal Tsirelson's problem that is asked initially in [57, Appendix H], for which we briefly discuss in Section 4.1.4, and answer affirmatively with the following theorem.

Theorem 4.12 (Bilocal Tsirelson) *Let $\vec{Q}$ be a tripartite probability distribution and suppose that the C^*-algebras $\mathfrak{A}$, $\mathfrak{C}$ generated by the PVMs of Alice and Charlie are finite-dimensional (note that no assumption is made on Bob's PVM). Then the following statements are equivalent:*

(i) $\vec{Q}$ is compatible with the tensor bilocal quantum model;

(ii) $\vec{Q}$ is compatible with the Renou–Xu bilocal quantum model;

(iii) $\vec{Q}$ is compatible with the algebraic bilocal quantum model;

(iv) $\vec{Q}$ is compatible with the mixed bilocal quantum model;

(v) $\vec{Q}$ admits a factorisation bilocal moment matrix with rank-loop.

Proof Note that (ii)–(iv) are equivalent due to Theorem 4.10, and (ii) is equivalent to (v) according to Theorem 4.5. Clearly (i) implies (iv). Thus, we only need to show that (iv) implies (i).

Assume that $\vec{Q}$ admits a mixed bilocal quantum model with finite-dimensional $\mathfrak{A}$, $\mathfrak{C}$. As in the proof of Theorem 4.11, we conclude that $\mathcal{H}_{AB_L}$, $\mathcal{H}_{B_RC}$ are finite-dimensional. Then apply Tsirelson's Theorem 3.1 to $\mathfrak{A}, \mathfrak{B}_L$ on $\mathcal{H}_{AB_L}$ and $\mathfrak{B}_R, \mathfrak{C}$ on $\mathcal{H}_{B_RC}$, respectively. We obtain the Hilbert spaces $\mathcal{H}_A$, $\mathcal{H}_{B_L}$, $\mathcal{H}_{B_R}$, $\mathcal{H}_C$, and the injective *-homomorhpisms $\mathcal{H}_{AB_L} \to \mathcal{H}_A \otimes \mathcal{H}_{B_L}$ and $\mathcal{H}_{B_RC} \to \mathcal{H}_{B_R} \otimes \mathcal{H}_C$. It is then straightforward to check that the identification under the two *-homomorphisms leads to tensor bilocal quantum model. See [35, Lemma 13] for more details. □

4.3 Quantum inflation technique in bilocal scenario

In the second part of [35], beyond the equivalence statement of various models in the bilocal quantum scenario, it is demonstrated that two other SDP hierarchies, namely the polarisation and inflation hierarchies, converge to the bilocal scenario. While a comprehensive discussion of this topic requires much more background, which is beyond the scope of this thesis, we provide a high-level overview in this section.

The quantum inflation technique, proposed in [74], is utilised to characterise quantum network scenarios. The core concept of the quantum inflation technique involves introducing m copies of the associated Hilbert spaces and independent quantum sources for each $m \geq 2$. The inflated system exhibits numerous symmetries, which allow quantum de Finetti-style arguments to

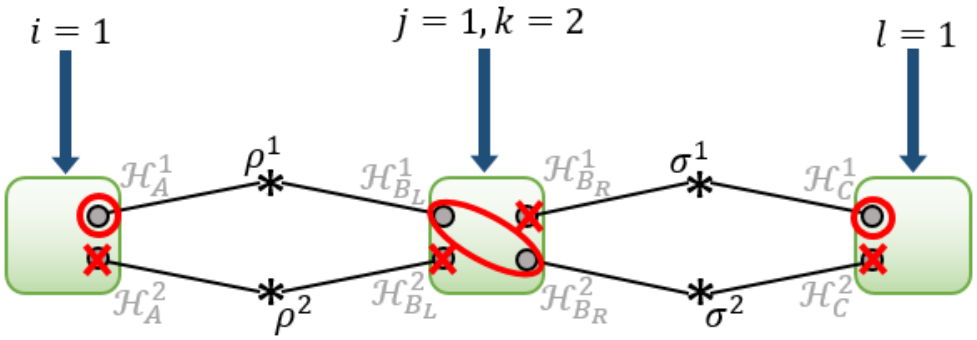


Figure 4.3: Adapted from [57, Figure 2], a visual representation of the inflation technique in the bilocal scenario (for simplicity, inputs x, y, z and outputs a, b, c are not shown). Suppose we have a distribution $\vec{P}$ obtained in the bilocal scenario; we can then imagine a new setup where the sources are duplicated, and each party receives extra inputs i; j, k; and l. The parties perform the measurements from the original scenario on the copies of the states associated with these additional inputs. In this example, Alice measures $A_{a|x}$ on her share of state ρ^1 using the new operator $A^1_{a|x}$. Bob measures operator $B^{12}_{b|y}$ on his shares of states ρ^1 and σ^2, and Charlie measures $C^1_{c|z}$ on his share of state σ^1. Notice that in this case, Charlie's measurement is independent of the others, which should result in a factorisation in the measurement outcome. Since the states and PVMs are duplicates, the behavior should be invariant under the $S_2 \times S_2$ permutation group. Lastly, if all parties measure all diagonal operators, i.e., Alice measures $A^1_{a|x}$ and $A^2_{a|x}$, Bob measures $B^{11}_{b|y}$ and $B^{22}_{b|y}$, and Charlie measures $C^1_{c|z}$ and $C^2_{c|z}$, we expect the probability to factorise along the diagonal and is consistent with the pre-inflated probability distribution.

linearise all possible factorisation constraints. As an example, [34] presents a convergence argument that requires the introduction of an additional Schmidt rank. Inflation technique appears to provide tighter approximations in more general network scenario. For instance, it can be used to detect the infeasibility of the shared random bit distribution in the triangle quantum scenario, while other methods such as scalar extension hierarchy is trivial in triangle network [57, Appendix E.2]

In the bilocal scenario, the inflation technique is thoroughly formalised in [57, Section 4.1]. Suppose that we have a distribution $\vec{P}$ obtained in the bilocal scenario. We can then consider a new setup in which the sources are duplicated, and each party receives extra inputs: i for Alice, j and k for Bob, and l for Charlie. The parties then perform the measurements from the original scenario on the copies of the states associated with these additional inputs.

For example, we consider Figure 4.3, adapted from [57, Figure 2], showcasing the inflated bilocal network at level $m = 2$. In this case, Alice measures $A_{a|x}$ on her share of state ρ^1 using the new operator $A^1_{a|x}$. Bob measures with operator $B^{12}_{b|y}$ on his shares of states ρ^1 and σ^2, while Charlie measures $C^1_{c|z}$ on his share of state σ^1. It is important to note that in this case, Charlie's

measurement is independent of the others, resulting in a factorised behaviour. Also, since acting on different copies, PVMs with nonoverlapping indices should commute with each other.

As the states and PVMs are duplicates in the inflation technique, the measurement outcomes exhibit a certain symmetry under permutation. This symmetry arises because when measuring the same state or PVM, regardless of the copy, the measurement outcomes should be identical. For instance, when Alice and Bob measure states ρ^1, they should obtain the same outcomes when they instead measure state ρ^2 together, which corresponds to a S_2 permutation on index i for Alice and j for Bob. Similarly, the measurement outcomes should be invariant under the permutation of the input index k for Bob and l for Charlie. Therefore, in the case where there are two copies, this results in an $S_2 \times S_2$ permutation group symmetry. The first S_2 corresponds to the permutation of indices for Alice and Bob, while the second S_2 corresponds to the permutation of indices for Bob and Charlie. (In a more general setting with m copies, the group symmetry extends to an $S_m \times S_m$ permutation group.)

In addition, when all parties measure all diagonal operators (i.e., Alice measures $A^1_{a|x}$ and $A^2_{a_*|x_*}$, Bob measures $B^{11}_{b|y}$ and $B^{22}_{b_*|y_*}$, and Charlie measures $C^1_{c|z}$ and $C^2_{c_*|z_*}$), we expect the probability to factorise as

$$\begin{aligned} & p(a^1 a_*^2 b^{11} b_*^{22} c^1 c_*^2 | x^1 x_*^2 y^{11} y_*^{22} c^1 c_*^2) \\ &= p(a^1 b^{11} c^1 | x^1 y^{11} c^1) p(a_*^2 b_*^{22} c_*^2 | x_*^2 y_*^{22} c_*^2) \\ &= p(abc|xyz) p(a_* b_* c_* | x_* y_* z_*), \end{aligned} \tag{4.26}$$

which is compatible with the pre-inflated probability distribution.

With the additional operators $A^i_{a|x}, B^{jk}_{b|y}, C^l_{c|z}$, a new type of moment matrix can be defined by incorporating all symmetric constraints and new commutation relations. This results in an *inflation-NPA hierarchy* that is SDP.

Building on the results from [34], it is shown in [35] that the inflation NPA hierarchy converges to commutator bilocal quantum sets in the bilocal scenario without requiring Schmidt rank. Furthermore, the convergence of a subcase of the inflation hierarchy, referred to as the polarization hierarchy, is also demonstrated. In the polarization hierarchy, instead of considering two inputs j, k for Bob, only one input j is considered, along with a smaller symmetric group S_m for each inflation level $m \geq 2$. That is, both SDP hiearchies—inflation and polarisation—completely characterise the commutator relaxation of the tensor bilocal quantum set.

Bibliography

[1] A. Acín, N. Brunner, N. Gisin, S. Massar, S. Pironio, and V. Scarani, "Device-independent security of quantum cryptography against collective attacks", *Physical Review Letters*, vol. 98, no. 23, p. 230 501, 2007.

[2] A. Acín, N. Gisin, and L. Masanes, "From bell's theorem to secure quantum key distribution", *Physical review letters*, vol. 97, no. 12, p. 120 405, 2006.

[3] H. Araki, "C^*-algebra approach in quantum field theory", *Physica Scripta*, vol. 24, no. 5, p. 881, 1981.

[4] R. Arnon-Friedman, R. Renner, and T. Vidick, "Simple and tight device-independent security proofs", *SIAM Journal on Computing*, vol. 48, no. 1, pp. 181–225, 2019.

[5] P. Aschieri, M. Dimitrijević, F. Meyer, and J. Wess, "Noncommutative geometry and gravity", *Classical and Quantum Gravity*, vol. 23, no. 6, p. 1883, 2006.

[6] J. Barrett, L. Hardy, and A. Kent, "No signaling and quantum key distribution", *Physical review letters*, vol. 95, no. 1, p. 010 503, 2005.

[7] J. S. Bell, "On the einstein ppodolsky rosen paradox", *Physics Physique Fizika*, vol. 1, no. 3, p. 195, 1964.

[8] B. Blackadar, *Operator algebras: theory of C*-algebras and von Neumann algebras*. Springer Science & Business Media, 2006, vol. 122.

[9] C. Branciard, N. Gisin, and S. Pironio, "Characterizing the nonlocal correlations created via entanglement swapping", *Physical review letters*, vol. 104, no. 17, p. 170 401, 2010.

[10] C. Branciard, D. Rosset, N. Gisin, and S. Pironio, "Bilocal versus nonbilocal correlations in entanglement-swapping experiments", *Physical Review A*, vol. 85, no. 3, p. 032 119, 2012.

[11] O. Bratteli and D. W. Robinson, *Operator algebras and quantum statistical mechanics: Volume 1: C*-and W*-Algebras. Symmetry Groups. Decomposition of States*. Springer Science & Business Media, 2012.

[12] N. Brunner, D. Cavalcanti, S. Pironio, V. Scarani, and S. Wehner, "Bell nonlocality", *Rev. Mod. Phys.*, vol. 86, pp. 419–478, 2 Apr. 2014. DOI: 10.1103/RevModPhys.86.419. [Online]. Available: https://link.aps.org/doi/10.1103/RevModPhys.86.419.

[13] S. Burgdorf, I. Klep, J. Povh, *et al.*, *Optimization of polynomials in non-commuting variables*. Springer, 2016, vol. 2.

[14] J. F. Clauser, M. A. Horne, A. Shimony, and R. A. Holt, "Proposed experiment to test local hidden-variable theories", *Phys. Rev. Lett.*, vol. 23, pp. 880–884, 15 Oct. 1969. DOI: 10.1103/PhysRevLett.23.880. [Online]. Available: https://link.aps.org/doi/10.1103/PhysRevLett.23.880.

[15] X. Coiteux-Roy, E. Wolfe, and M.-O. Renou, "Any physical theory of nature must be boundlessly multipartite nonlocal", *Physical Review A*, vol. 104, no. 5, p. 052 207, 2021.

[16] X. Coiteux-Roy, E. Wolfe, and M.-O. Renou, "No bipartite-nonlocal causal theory can explain nature's correlations", *Physical review letters*, vol. 127, no. 20, p. 200 401, 2021.

[17] R. Colbeck and R. Renner, "Free randomness can be amplified", *Nature Physics*, vol. 8, no. 6, pp. 450–453, 2012.

[18] A. Connes, J. Cuntz, M. A. Rieffel, and G. Yu, "Noncommutative geometry", *Oberwolfach Reports*, vol. 10, no. 3, pp. 2553–2629, 2014.

[19] A. Connes and M. Marcolli, *Noncommutative geometry, quantum fields and motives*. American Mathematical Soc., 2019, vol. 55.

[20] A. Einstein, B. Podolsky, and N. Rosen, "Can quantum-mechanical description of physical reality be considered complete?", *Physical review*, vol. 47, no. 10, p. 777, 1935.

[21] T. Fritz, "Tsirelson's problem and kirchberg's conjecture", *Reviews in Mathematical Physics*, vol. 24, no. 05, p. 1 250 012, 2012.

[22] T. Fritz, "Beyond bell's theorem: Correlation scenarios", *New Journal of Physics*, vol. 14, no. 10, p. 103 001, 2012.

[23] I. Gelfand and M. Neumark, "On the imbedding of normed rings into the ring of operators in hilbert space", *Matematicheskii Sbornik*, vol. 12, no. 2, pp. 197–217, 1943.

[24] R. Haag, *Local quantum physics: Fields, particles, algebras*. Springer Science & Business Media, 2012.

[25] H. Halvorson and M. Müger, "Algebraic quantum field theory", *arXiv preprint math-ph/0602036*, 2006.

[26] W. Heisenberg, *Ü̈ber Quantentheoretische umdeutung Kinematischer und mechanischer Beziehungen*. Springer, 1985.

[27] J. Henson, R. Lal, and M. F. Pusey, "Theory-independent limits on correlations from generalized bayesian networks", *New Journal of Physics*, vol. 16, no. 11, p. 113 043, 2014.

[28] Z. Ji, A. Natarajan, T. Vidick, J. Wright, and H. Yuen, "Mip*=re", *Communications of the ACM*, vol. 64, no. 11, pp. 131–138, 2021.

[29] M. Junge and G. Pisier, "Bilinear forms on exact operator spaces and $B(H) \otimes B(H)$", *Geometric & Functional Analysis GAFA*, vol. 5, no. 2, pp. 329–363, 1995.

[30] R. V. Kadison and J. R. Ringrose, *Fundamentals of the theory of operator algebras. Volume II: Advanced theory*. Academic press New York, 1986.

[31] I. Klep, V. Magron, and J. Volčič, "Optimization over trace polynomials", in *Annales Henri Poincaré*, Springer, vol. 23, 2022, pp. 67–100.

[32] I. Klep, V. Magron, J. Volčič, and J. Wang, "State polynomials: Positivity, optimization and nonlinear bell inequalities", *arXiv preprint arXiv:2301.12513*, 2023.

[33] K. Landsman, *Foundations of quantum theory: From classical concepts to operator algebras*. Springer Nature, 2017.

[34] L. T. Ligthart, M. Gachechiladze, and D. Gross, "A convergent inflation hierarchy for quantum causal structures", *arXiv preprint arXiv:2110.14659*, 2021.

[35] L. T. Ligthart and D. Gross, "The inflation hierarchy and the polarization hierarchy are complete for the quantum bilocal scenario", *arXiv preprint arXiv:2212.11299*, 2022.

[36] D. Mayers and A. Yao, "Quantum cryptography with imperfect apparatus", in *Proceedings 39th Annual Symposium on Foundations of Computer Science (Cat. No. 98CB36280)*, IEEE, 1998, pp. 503–509.

[37] J. Moffat, "Noncommutative quantum gravity", *Physics Letters B*, vol. 491, no. 3-4, pp. 345–352, 2000.

[38] V. Moretti, *Fundamental Mathematical Structures of Quantum Theory*. Springer, 2019.

[39] F. J. Murray and J. von Neumann, "On rings of operators", *Annals of Mathematics*, pp. 116–229, 1936.

[40] F. J. Murray and J. von Neumann, "On rings of operators. ii", *Transactions of the American Mathematical Society*, vol. 41, no. 2, pp. 208–248, 1937.

[41] F. J. Murray and J. von Neumann, "On rings of operators. iv", *Annals of Mathematics*, pp. 716–808, 1943.

[42] M. Navascués, S. Pironio, and A. Acín, "A convergent hierarchy of semidefinite programs characterizing the set of quantum correlations", *New Journal of Physics*, vol. 10, no. 7, p. 073 013, 2008.

[43] J. von Neumann, *Mathematische Grundlagen der Quantenmechanik*. Springer-Verlag, 2013, vol. 38.

[44] J. von Neumann, "On infinite direct products", *Compositio mathematica*, vol. 6, pp. 1–77, 1939.

[45] J. von Neumann, "On rings of operators. iii", *Annals of Mathematics*, pp. 94–161, 1940.

[46] J. von Neumann, "On rings of operators. reduction theory", *Annals of Mathematics*, pp. 401–485, 1949.

[47] J. von Neumann, "Über funktionen von funktionaloperatoren", *Annals of Mathematics*, pp. 191–226, 1931.

[48] J. von Neumann, "Wahrscheinlichkeitstheoretischer aufbau der quantenmechanik", *Nachrichten von der Gesellschaft der Wissenschaften zu Göttingen, Mathematisch-Physikalische Klasse*, vol. 1927, pp. 245–272, 1927.

[49] J. von Neumann, "Zur algebra der funktionaloperationen und theorie der normalen operatoren", *Mathematische Annalen*, vol. 102, no. 1, pp. 370–427, 1930.

[50] M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information: 10th Anniversary Edition*. Cambridge University Press, 2010. DOI: 10.1017/CBO9780511976667.

[51] S. Pironio, A. Acín, S. Massar, A. B. de La Giroday, D. N. Matsukevich, P. Maunz, S. Olmschenk, D. Hayes, L. Luo, T. A. Manning, *et al.*, "Random numbers certified by bell's theorem", *Nature*, vol. 464, no. 7291, pp. 1021–1024, 2010.

[52] A. Pozas-Kerstjens, R. Rabelo, Ł. Rudnicki, R. Chaves, D. Cavalcanti, M. Navascués, and A. Acín, "Bounding the sets of classical and quantum correlations in networks", *Phys. Rev. Lett.*, vol. 123, p. 140 503, 14 Oct. 2019. DOI: 10.1103/PhysRevLett.123.140503. [Online]. Available: https://link.aps.org/doi/10.1103/PhysRevLett.123.140503.

[53] M. O. Renou, J. Kaniewski, and N. Brunner, "Self-testing entangled measurements in quantum networks", *Physical review letters*, vol. 121, no. 25, p. 250 507, 2018.

[54] M.-O. Renou, E. Bäumer, S. Boreiri, N. Brunner, N. Gisin, and S. Beigi, "Genuine quantum nonlocality in the triangle network", *Physical review letters*, vol. 123, no. 14, p. 140 401, 2019.

[55] M.-O. Renou and S. Beigi, "Nonlocality for generic networks", *Physical Review Letters*, vol. 128, no. 6, p. 060 401, 2022.

[56] M.-O. Renou, D. Trillo, M. Weilenmann, T. P. Le, A. Tavakoli, N. Gisin, A. Acín, and M. Navascués, "Quantum theory based on real numbers can be experimentally falsified", *Nature*, vol. 600, no. 7890, pp. 625–629, 2021.

[57] M.-O. Renou and X. Xu, "Two convergent npa-like hierarchies for the quantum bilocal scenario", *arXiv preprint arXiv:2210.09065*, 2022.

[58] V. B. Scholz and R. F. Werner, "Tsirelson's Problem", *arXiv e-prints*, arXiv:0812.4305, arXiv:0812.4305, Dec. 2008. arXiv: 0812.4305 [math-ph].

[59] E. Schrödinger, "Quantisierung als eigenwertproblem", *Annalen der physik*, vol. 385, no. 13, pp. 437–490, 1926.

[60] I. Segal, "Irreducible representations of operator algebras", *Bulletin of the American Mathematical Society*, vol. 53, no. 2, pp. 73–88, 1947.

[61] I. E. Segal, "Postulates for general quantum mechanics", *Annals of Mathematics*, pp. 930–948, 1947.

[62] N. Seiberg and E. Witten, "String theory and noncommutative geometry", *Journal of High Energy Physics*, vol. 1999, no. 09, p. 032, 1999.

[63] W. Slofstra, "The set of quantum correlations is not closed", in *Forum of Mathematics, Pi*, Cambridge University Press, vol. 7, 2019, e1.

[64] J. F. Sturm, "Using sedumi 1.02, a matlab toolbox for optimization over symmetric cones", *Optimization Methods and Software*, vol. 11, no. 1-4, pp. 625–653, 1999. DOI: 10.1080/10556789908805766. eprint: https://doi.org/10.1080/10556789908805766. [Online]. Available: https://doi.org/10.1080/10556789908805766.

[65] W. van Suijlekom, *Noncommutative Geometry and Particle Physics*, ser. Mathematical Physics Studies. Springer Netherlands, 2014, ISBN: 9789401791625. [Online]. Available: https://books.google.ch/books?id=6PcsBAAAQBAJ.

[66] I. Šupić and J. Bowles, "Self-testing of quantum systems: A review", *Quantum*, vol. 4, p. 337, 2020.

[67] I. Šupić, J. Bowles, M.-O. Renou, A. Acín, and M. J. Hoban, "Quantum networks self-test all entangled states", *Nature Physics*, pp. 1–6, 2023.

[68] A. Tavakoli, A. Pozas-Kerstjens, M.-X. Luo, and M.-O. Renou, "Bell nonlocality in networks", *Reports on Progress in Physics*, vol. 85, no. 5, p. 056 001, 2022.

[69] B. Tsirelson, *Bell inequalities and operator algebras*, 2006. [Online]. Available: https://www.tau.ac.il/~tsirel/Research/bellopalg/main.html.

[70] L. Vandenberghe and S. Boyd, "Semidefinite programming", *SIAM review*, vol. 38, no. 1, pp. 49–95, 1996.

[71] U. Vazirani and T. Vidick, "Fully device independent quantum key distribution", *Communications of the ACM*, vol. 62, no. 4, pp. 133–133, 2019.

[72] F. Verstraete and J. I. Cirac, "Quantum nonlocality in the presence of superselection rules and data hiding protocols", *Physical review letters*, vol. 91, no. 1, p. 010 404, 2003.

[73] M. Weilenmann and R. Colbeck, "Self-testing of physical theories, or, is quantum theory optimal with respect to some information-processing task?", *Physical Review Letters*, vol. 125, no. 6, p. 060 406, 2020.

[74] E. Wolfe, A. Pozas-Kerstjens, M. Grinberg, D. Rosset, A. Acín, and M. Navascués, "Quantum inflation: A general approach to quantum causal compatibility", *Phys. Rev. X*, vol. 11, p. 021 043, 2 May 2021. DOI: 10.1103/PhysRevX.11.021043. [Online]. Available: https://link.aps.org/doi/10.1103/PhysRevX.11.021043.

[75] H. Wolkowicz, R. Saigal, and L. Vandenberghe, *Handbook of semidefinite programming: theory, algorithms, and applications*. Springer Science & Business Media, 2012, vol. 27.

Appendix A

Selected functional analysis concepts for quantum mechanics

Functional analysis during its development has deep roots in mathematical physics, and in return has been fundamental to the development of physics, for example quantum mechanics.

Based on [8, 30, 33, 38], this appendix provides a brief review of the basics on infinite-dimensional Banach spaces, spectral theorem for bounded self-adjoint operators, roundup of Hilbert space operator topologies, and some important functional analysis results that are essential to Quantum Mechanics and the theory of operator algebra in the main text. Due to the focus of this thesis, in this appendix the proofs are generally omitted.

A.1 Banach spaces and Banach algebras

The central mathematical object in Quantum Mechanics, Hilbert spaces, is a subclass of its more general counterpart, Banach spaces. Analogously, Banach algebras include the subclass C^*-algebras, which is the main object we treat in operator algebras for Quantum Mechanics.

A.1.1 Metrics, norms, inner products, and completeness

We start with the well-known definition of metrics, norms, and inner products.

Definition A.1 *Let V be a vector space and denote $v, w, u \in V$ as arbitrary elements with $c \in \mathbb{C}$.*

1. *A* metric *on V is a function $d : V \times V \to \mathbb{R}+$ such that we have:*

   (a) $d(v, w) \leq d(v, u) + d(u, w)$ *(*triangular inequality*);*

   (b) $d(v,w) = d(w,v)$ (symmetry);

   (c) $d(v,w) = 0$ *iff* $v = w$ (positive definiteness).

2. *A* norm *on V is a function $\|\cdot\| : V \to \mathbb{R}^+$ such that*

   (a) $\|v+w\| \leq \|v\| + \|w\|$ (triangle inequality);

   (b) $\|cv\| = |c|\|v\|$ (homogeneity);

   (c) $\|v\| = 0$ *iff* $v = 0$ (positive definiteness).

3. *A* sesquilinear form *on V is a map $\langle\cdot|\cdot\rangle : V \times V \to \mathbb{C}$, $(v,w) \mapsto \langle v|w\rangle$, such that it is anti-linear in its first argument and linear in its second argument. Moreover:*

   (a) *a* hermitian *form on V is a sesquilinear form such that $\langle v|w\rangle = \langle w|b\rangle^\dagger$;*

   (b) *a* pre-inner product *is a positive hermitian form, that is, $\langle v|v\rangle \geq 0$;*

   (c) *an* inner product *is a positive definite hermitian form, that is, $\langle v|v\rangle = 0$ iff $v = 0$.*

As this thesis focusses on physical aspects, we use † to refer to the action of conjugate transposition, which coincides with complex conjugation in $\mathbb{C}$. Similarly, we adopt the Dirac bra-ket notation for sesquilinear forms, identifying $f \in V$ with $|f\rangle$, therefore, for any map $\alpha : V \to V$ we denote $\langle v|\alpha w\rangle = \langle v|\alpha|w\rangle$.

We state the following well-known *Cauchy–Schwarz inequality* satisfied by any pre-inner products

$$|\langle v|w\rangle|^2 \leq \langle v|v\rangle\langle w|w\rangle. \tag{A.1}$$

The proof of the inequality is elementary. It can be straightforwardly checked that any inner product induces a norm via

$$\|v\| = \sqrt{\langle v|v\rangle}, \tag{A.2}$$

and any norm induces a metric by

$$d(v,w) = \|v - w\|. \tag{A.3}$$

The following definition classifies norms.

Definition A.2 *Two norms $\|\cdot\|$ and $\|\cdot\|'$ are said to be* equivalent *if there exist constants $m, M > 0$ such that for all $v \in V$,*

$$m\|v\|' \leq \|v\| \leq M\|v\|.$$

For the purpose of this appendix, we shall work with normed vector spaces $(V, \|\cdot\|)$, so let us define the notion of convergence and completeness for normed vector spaces. The definitions for metric spaces are easily generalised with the equations above.

Definition A.3 *Consider the sequence (v_n) in a vector space with the norm $(V, \|\cdot\|)$.*

1. *We say that $v_n \to v$ for some $v \in V$, or v_n converges to v in norm, when $\lim_{n\to\infty}\|v_n - v\| = 0$.*
2. *We say that (v_n) is a* Cauchy sequence *if $\|v_n - v_m\| \to 0$ whenever $n, m \to \infty$.*
3. *We say that V is* complete *if every Cauchy sequence in V converges (to some element in V).*

We note that two equivalent norms of some vector space induce the same topology, and the notions of convergence and completeness are the same.

A.1.2 Basic notions of Banach spaces

Now we are ready for the following definition.

Definition A.4 *A* Banach space *is a vector space equipped with a norm that is also complete. A* Hilbert space *is a vector space equipped with inner product that is completed with respect to the induced norm; equivalently, a Hilbert space is a Banach space equipped with inner product-induced norm.*

Example A.5 *Finite-dimensional cases are well-examined in linear algebra, and we present some important infinite-dimensional examples.*

- *One example of Banach spaces is*

$$l^p(\mathbb{N}) = \left\{ f : \mathbb{N} \to \mathbb{C} \mid \|f\|_p = \left(\sum_{x\in\mathbb{N}} |f(x)|^p \right)^{1/p} < \infty \right\} \tag{A.4}$$

  for $1 \leq p < \infty$ and

$$l^\infty(\mathbb{N}) = \left\{ f : \mathbb{N} \to \mathbb{C} \mid \|f\|_\infty = \sup_{x\in\mathbb{N}}\{|f(x)|\} < \infty \right\}. \tag{A.5}$$

  The requirement that $p \geq 1$ is due to Minkowski inequality, otherwise $\|\cdot\|_p$ fails to satisfy triangular inequality. We also remark that $l^2(\mathbb{N})$ is in fact a Hilbert space with the usual inner product that is ubiquitous in Quantum Mechanics.

- *Recall the notion of measure spaces (X, Σ, μ), where X is a set, $\Sigma \subset \mathcal{P}(X)$ is a σ-algebra*[1]*, and $\mu : \Sigma \to [0, \infty]$ is a measure*[2]*. Consider the norm for measurable functions*[3,4] *$f : X \to \mathbb{C}$*

$$\|f\|_p = \left(\int_X d\mu |f|^p \right)^{1/p} \tag{A.6}$$

for $1 \leq p < \infty$ and

$$\|f\|_\infty^{\text{ess}} = \{ t \in [0, \infty] \mid \mu(\{ x \in X \mid |f(x)| > t \}) = 0 \} . \tag{A.7}$$

Then the following sets

$$L^p(X) = \left\{ f : X \to \mathbb{C} \mid \|f\|_p < \infty \right\} \tag{A.8}$$

$$L^\infty(X) = \{ f : X \to \mathbb{C} \mid \|f\|_\infty^{\text{ess}} < \infty \} , \tag{A.9}$$

where $1 \leq p < \infty$, with pointwise operation and the respective norms defined above, are Banach spaces. Similarly to the discrete case, $p = 2$ gives the usual inner product on the space of functions and is a Hilbert space commonly appearing in physics.

- *Let $C(X)$ denote the set of all continuous functions $f : X \to \mathbb{C}$ for some locally compact space X. Then $C_0(X)$, the subset of $C(X)$ with functions vanish at infinity, and $C_b(X)$, the subset of $C(X)$ with bounded functions, are Banach spaces with the sup-norm $\|\cdot\|_\infty$.*

As in the last example, let us consider the morphisms between Banach spaces.

Definition A.6 *Let V, W be Banach spaces, a* morphism *$\alpha : V \to W$ is a* bounded linear map*, that is,*

$$\|\alpha v\|_W \leq C \|v\|_V$$

for all $v \in V$ and some constant $C > 0$. The set of all morphisms $\alpha : V \to W$ is denoted by $\mathcal{B}(V, W)$. In particular, a morphism $\alpha \in \mathcal{B}(V, V) = \mathcal{B}(V)$ is said to be an operator *on V, and an element $\varphi \in \mathcal{B}(V, \mathbb{C})$ is said to be a* functional *on V.*

For any $\alpha \in \mathcal{B}(V, W)$, define the norm[5]

$$\|\alpha\| = \sup_{v \in V, \|v\|_V \leq 1} \{ \|\alpha v\|_W \}, \tag{A.10}$$

[1] A σ-algebra of subsets of X, Σ, contains the whole set X, and is closed under set completement and arbitrary unions,

[2] That is, $\mu(\cup_n A_n) = \sum_n \mu(A_n)$ for a countable disjoint union of $A_n \in \Sigma$. In particular, (X, Σ, μ) is said to be a probability measure if $\mu(X) = 1$.

[3] Maps such that the preimage of any measurable set is again measurable.

[4] Recall also the integral of a positive measurable function $f : X \to \mathbb{C}$ can be defined as the supremum of suitable simple functions, and any measurable functions can be decomposed into sum of positive functions.

[5] When for operators, i.e. V=W, this is equivalent to the usual operator norm.

which we can directly check it is a norm as the notation suggested. Note that the equation in Definition A.6 is equivalent to the condition that $\|\alpha\| < \infty$, we also remark that it is straightforward to check the equivalence to continuity with respect to the norm-induced topology. Using a standard analysis argument [33, Theorem B.33], we can check that $\mathcal{B}(V, W)$ is complete with respect to the norm $\|\cdot\|$, and with the natural pointwise linear structure it follows that $\mathcal{B}(V, W)$ is also a Banach space.

We finally introduce the notion of isometric isomorphism, which is widely used in quantum mechanics. Note that the following definition can actually be generalised to normed spaces.

Definition A.7 *Let V, W be Banach spaces. An* isometry $\hat{u} : V \to W$ *is a linear map satisfying, for all $v \in V$,*

$$\|\hat{u}v\|_W = \|v\|_V.$$

An isometric isomorphism $\hat{u} : V \to W$ *is a surjective isometry.*

One can clearly see that the isometric condition implies both boundedness and injectivity, and hence any isometric isomorphism $\hat{u} : V \to W$ is a bijective morphism in $\mathcal{B}(V, W)$. The (non-trivial) facts that the set-theoretical inverse $\hat{u}^{-1} : W \to V$ belongs to $\mathcal{B}(W, V)$ and is isometric require a corollary of the famous open mapping theorem (see Theorem A.18 and Corollary A.19).

A.1.3 Banach algebras and specturm

For the space of bounded operators $\mathcal{B}(V)$, in addition to being a Banach space, there is a natural algebraic structure induced by composition. This leads to the following definition.

Definition A.8 *A vector space $\mathfrak{A}$ over $\mathbb{C}$ is an* algebra *if it is equipped with an associative and distributive multiplication law such that for each pair $\alpha, \beta \in \mathfrak{A}$ we have the product $\alpha\beta \in \mathfrak{A}$.*

The algebra $\mathfrak{A}$ is unital *if it has a unit $\mathbb{1}$ such that $\mathbb{1}\alpha = \alpha$ for any $\alpha \in \mathfrak{A}$*

A unital algebra $\mathfrak{A}$ is a Banach algebra, *if it has a norm such that*

$$\|\alpha\beta\| \leq \|\alpha\|\|\beta\| \tag{A.11}$$

for all $\alpha, \beta \in \mathfrak{A}$, and it is a Banach space with respect to this norm.

Note that the requirement of being unital is not essential, and a standard procedure can be applied to adjoin an identity to the algebra [11, Proposition 2.1.5]. For our purposes, it is enough to consider the unital case.

Therefore, the space of bounded operators $\mathcal{B}(V)$ is an example of a Banach algebra Additionally, the $C_0(X)$ and $C_b(X)$ discussed in the previous section

are also Banach algebras, where the algebraic structure comes from pointwise multiplication.

We now seek a generalisation of the concept of eigenvalues to Banach algebras. Indeed, if $a \in \mathcal{B}(V)$ for some finite-dimensional vector space, then λ is an eigenvalue iff $a - \lambda \mathbb{1}_V$ is not invertible, leading to the following definition.

Definition A.9 *The* specturm *of $a \in \mathfrak{A}$ is defined as the set*

$$\sigma(a) = \{\lambda \in \mathbb{C} \mid a - \lambda \mathbb{1} \text{ not invertible in } \mathfrak{A}\}.$$

The complement $\rho(a) = \mathbb{C} \setminus \sigma(a)$ is called resolvent.

Note that while the eigenvalues of a are always in the spectrum $\sigma(a)$, the converse is not always true, as shown in the following example.

Example A.10 *Given $L^2([0,1])$ with respect to the Lebesgue measure, we consider the associated position operator $\hat{x} \in \mathcal{B}(L^2([0,1]))$ in QM such that $(\hat{x}f)(x) = xf(x)$.*

Suppose that $f \in L^2([0,1])$ is an eigenfunction of $\hat{x}$, then the eigenequation $(\hat{x}f) = \lambda f$ implies that $f = 0$ except for $x = \lambda$. But then f is 0 almost every, hence $f = 0$ in $L^2([0,1])$. We comment that many physics textbooks claim that δ_x, "misleadingly" called Dirac-delta functions, are the eigenfunctions for $\hat{x}$. However, note that $\delta_x \notin L^2([0,1])$ but instead is of a more general class, namely distribution.

Nevertheless, we claim that the spectrum $\sigma(\hat{x}) = [0,1]$. To this end, we first assume $\lambda \in (0,1)$, the situation in which $\lambda = 0, 1$ can be checked with a natural modification of the following argument. Define a sequence $f_n = \sqrt{n}\chi_{[\lambda - 1/2n, \lambda + 1/2n]}$ in $L^2([0,1])$, where χ_A denotes the characteristic function of the subset A, assuming that the first n_0 of the sequence is sufficiently large. We note that with the $\|f_n\|_2 = 1$ and $\|(\hat{x} - \lambda\mathbb{1})f_n\|_2 = \sqrt{3}/6n$. These f_n can be thinked of as a sequence of "approximate unit eigenvector" converging to the Dirac-delta "function" δ_λ.

Next, for contradiction, suppose α is an inverse of $\hat{x} - \lambda\mathbb{1}$, then

$$1 = \|\mathbb{1} f_n\|_2 = \|\alpha(\hat{x} - \lambda\mathbb{1})f_n\|_2 \leq \|\alpha\|_2 \|(\hat{x} - \lambda\mathbb{1})f_n\|_2 = \|\alpha\|_2 \sqrt{3}/6n, \quad \text{(A.12)}$$

whence $\|\alpha\|_2 \geq 2\sqrt{3}n$ for any n and hence α is unbounded.

Finally, assume $\lambda \notin [0,1]$, then $g(x) = (x - \lambda)^{-1}$ is a continuous function in $[0,1]$. Then $\alpha_g(f) = gf$ is a bounded operator, and the inverse to $\hat{x} - \lambda\mathbb{1}$.

This example actually demonstrates the difference between two different types of spectrum, namely *point spectrum* (the usual eigenvalues) vs. *continuous spectrum* (the example above). We remark that while the proper eigenvectors to points in the continuous spectrum do not exist, it is a mathematical fact that they can be approximated arbitrarily well, and the sequence f_n constructed above is an example. We introduce the last definition of this section.

Definition A.11 *For $\alpha \in \mathfrak{A}$, the* spectral radius *of α is $r(\alpha) = \sup_{\lambda \in \sigma(\alpha)}\{|\lambda|\}$.*

A.2 Spectral Theorem for bounded self-adjoint operators

We recall that in the finite-dimension case when $\mathcal{H} = \mathbb{C}^n$, $\mathcal{B}(\mathcal{H}) = \mathrm{Mat}_n(\mathbb{C})$, the spectral Theorem of linear algebra states that for any self-adjoint matrix $A \in \mathrm{Mat}_n(\mathbb{C})$, i.e. $A^\dagger = A$, we have

$$A = \sum_i \lambda_i |v_i\rangle\langle v_i|, \tag{A.13}$$

where for each i, λ_i is an eigenvalue with eigenvector v_i and $|v_i\rangle\langle v_i|$ is the orthogonal projector on v_i. This theorem is immensely powerful in mathematics, and it is fundamental to quantum theory's postulation for observable and their corresponding measurements.

In this section we seek a generalisation in infinite-dimensional Hilbert space setting. For simplicity, we consider the Spectral Theorem for bounded self-adjoint operators in $\mathcal{B}(\mathcal{H})$ of some Hilbert space $\mathcal{H}$.

A.2.1 Spectrum of Hilbert space operators

Let us resume the discussion of point spectrum vs. continuous spectrum in Appendix A.1.3 and focus on the case of Hilbert space operators. In fact, there is one more type of spectrum called the *residue spectrum*.

Definition A.12 *Let α be an operator (possibly unbounded) in the Hilbert space $\mathcal{H}$. The resolvent of α is*

$$\rho(\alpha) = \{\lambda \in \mathbb{C} \mid (\alpha - \lambda\mathbb{1}) \text{ is injective}, \overline{\mathrm{range}(\alpha - \lambda\mathbb{1})} = \mathcal{H}, \\ (\alpha - \lambda\mathbb{1})^{-1} \text{ is bounded}\},$$

where $\bar{\cdot}$ indicates the topological closure and range *indicates the range of the corresponding operator.*

The spectrum $\sigma(\alpha) = \mathbb{C} \setminus \rho(\alpha)$ consists of three pairwise disjoint parts:

1. *The* point spectrum $\sigma_p(\alpha)$*, where $\alpha - \lambda\mathbb{1}$ is not injective;*
2. *the* continuous spectrum $\sigma_c(\alpha)$*, where $\alpha - \lambda\mathbb{1}$ is injective, $\overline{\mathrm{range}(\alpha - \lambda\mathbb{1})} = \mathcal{H}$, but $(\alpha - \lambda\mathbb{1})^{-1}$ is not bounded;*
3. *the* residue spectrum $\sigma_r(\alpha)$*, where $\alpha - \lambda\mathbb{1}$ is injective but $\overline{\mathrm{range}(\alpha - \lambda\mathbb{1})} \neq \mathcal{H}$.*

Recall that an operator $\alpha \in \mathcal{B}(\mathcal{H})$ is:

- *self-adjoint* if $\langle \alpha v | w \rangle = \langle v | \alpha \rangle$ ($\alpha^\dagger = \alpha$);
- *unitary* if $\alpha^\dagger \alpha = \alpha \alpha^\dagger = \mathbb{1}_{\mathcal{H}}$;
- *normal* if $\alpha^\dagger \alpha = \alpha \alpha^\dagger$.

We note that self-adjointness implies normality and so is unitary condition. In addition, any operator admits a self-adjoint decomposition, $\alpha = \hat{\beta} + i\hat{\gamma}$, where $\hat{\beta} = (\alpha + \alpha^\dagger)/2$ and $\hat{\gamma} = (\alpha - \alpha^\dagger)/2i$.

We state the properties of spectrum for these special types of operator.

Theorem A.13 *Consider $\alpha \in \mathcal{B}(\mathcal{H})$ for some Hilbert space $\mathcal{H}$. If α is normal (in particular self-adjoint or unitary), then*

(a) $\sigma_r(\alpha) = \sigma_r(\alpha^\dagger) = \emptyset$.

(b) $\sigma_p(\alpha) = \sigma_p(\alpha^\dagger)^\dagger$; *i.e.* $\alpha v = \lambda v$ *iff* $\alpha^\dagger v = \lambda^\dagger v$ *for* $v \neq 0$.

(c) $\sigma_c(\alpha) = \sigma_c(\alpha^\dagger)^\dagger$

(d) the eigenvectors corresponding to the distinct eigenvalues are orthogonal.

Moreover, $\sigma(\alpha) \subset \mathbb{R}$ if α is self-adjoint and $\sigma(\alpha) \subset \{z \in \mathbb{C} \mid |z|\}$ if α is unitary.

In other words, a normal operator behaves as what we expect for operators in finite dimension, other than the existence of continuous spectral. The properties of spectrum of self-adjoint operators are important for the following discussion.

A.2.2 Projector-valued measures (PVMs) and integration

We now define the general notion of projector-valued measures, which are very ubiquitous quantum theory. Recall that an operator $P \in \mathcal{B}(\mathcal{H})$ is an *orthogonal projector* when $P^2 = P$ and $P^\dagger = P$.

Denote $\mathscr{B}(\sigma(\alpha))$ the Borel σ-algebra[6] of the spectrum of α. We wish to construct a map $P : \mathcal{B}(\sigma(\alpha)) \to \mathcal{B}(\mathcal{H})$ that allows us to find a projector onto the subspace $V_E \subset \mathcal{H}$ of "generalised eigenvectors" of α associated with a given subset $E \subset \sigma(\alpha)$. This motivates the following definition.

Definition A.14 *Let $\mathcal{H}$ be a Hilbert space and $\Sigma(X)$ be a σ-algebra on some set X. A* projector-valued measure (PVM) *is a map $P : \Sigma(X) \to \mathcal{B}(\mathcal{H})$ that satisfies the following properties:*

(a) $P(E) \coloneqq P_E$ *is an orthogonal projector for all* $E \in \Sigma(X)$;

(b) $P(\emptyset) = 0$ *and* $P(X) = P_X = \mathbb{1}_{\mathcal{H}}$;

(c) $P_{E \cap F} = P_E P_F$;

[6] That is, the smallest σ-algebra that contains all open sets in $\sigma(\alpha)$

(d) for (E_i) a countable pairwise disjoint family in $\Sigma(X)$,

$$P_{\cup_i E_i}\varphi = \sum_i P_{E_i}\varphi$$

for all $\varphi \in \mathcal{H}$.

Note that we recover the familiar notion of PVMs in quantum information theory if we take X to be a finite set.

We now motivate the definition of integratal with respect to PVMs. Fix a PVM $P : \Sigma(X) \to \mathcal{B}(\mathcal{H})$, note that for any $\varphi, \psi \in \mathcal{H}$, the map $\mu^P_{\varphi,\psi} : \Sigma(X) \to \mathbb{C}$, $\mu^P_{\varphi,\psi}(E) \coloneqq \langle\varphi|P_E|\psi\rangle$, is a complex measure. Moreover, $\mu^P_{\varphi,\varphi}$ is a real measure.

Suppose that $s = \sum_k s_k \chi_{E_k}$ is a simple function[7], where $s_k \in \mathbb{C}$ and E_k are pairwise disjoint. If we define

$$\int_X s(\lambda)dP(\lambda) = \int_X \sum_k s_k \chi_{E_k} dP = \sum_k s_k P_{E_k}, \tag{A.14}$$

it can then be shown

$$\int_X s d\mu^P_{\varphi,\psi} = \sum_k s_k \mu^P_{\varphi,\psi}(E_k) = \langle\varphi| \sum_k s_k P_{E_k}|\psi\rangle = \langle\varphi| \int_X s(\lambda)dP(\lambda)|\psi\rangle. \tag{A.15}$$

By generalising simple functions to general measurable functions, Equation A.15 leads to the integration machinery required for Spectral Theorem.

Theorem A.15 (PVM integral operator) *Let $P : \Sigma(X) \to \mathcal{B}(\mathcal{H})$ be a PVM and let $f : X \to \mathbb{C}$ be a bounded[8] measurable function. Recall the complex measure defined by $\mu^P_{\varphi,\psi} : \Sigma(X) \to \mathbb{C}$, $\mu^P_{\varphi,\psi}(E) \coloneqq \langle\varphi|P_E|\psi\rangle$.*

Then there exists a unique *bounded operator of $\mathcal{H}$, denoted as*

$$\int_X f(\lambda)dP(\lambda) \in \mathcal{B}(\mathcal{H}).$$

The operator satisfies the following:

(a)

$$\langle\varphi| \int_X f(\lambda)dP(\lambda)|\psi\rangle = \int_X f(\lambda)d\mu^P_{\varphi,\psi}(\lambda). \tag{A.16}$$

(b)

$$\left\| \int_X f(\lambda)dP(\lambda) \right\| \leq \|f\|_\infty. \tag{A.17}$$

[7] A simple function is a finite linear combination of characteristic functions.

[8] We may substitute boundedness to essential boundedness relative to the induced measure of P, this generalisation is omitted for our purposes.

(c) For the characteristic function χ_E we have

$$\int_X \chi_E dP = P_E \tag{A.18}$$

for all $E \in \Sigma(X)$, and in particular $\int_X 1 dP = \mathbb{1}_{\mathcal{H}}$.

(d) Its adjoint operator is given by

$$(\int_X f(\lambda) dP(\lambda))^\dagger = \int_X f(\lambda)^\dagger dP(\lambda). \tag{A.19}$$

Furthermore, the map $f \mapsto \int_X f(\lambda) dP(\lambda)$ satisfies the following properties:

(i) (Continuity) Let f and f_n be bounded measurable functions and assume that $\|f\|_\infty, \|f_n\|_\infty \leq M < \infty$ for all $n \in \mathbb{N}$. If $f_n \to f$ pointwise as $n \to \infty$, then

$$\left(\int_X f_n dP\right) \varphi \to \left(\int_X f dP\right) \varphi \tag{A.20}$$

as $n \to \infty$, for any $\varphi \in \mathcal{H}$.

(ii) (Ring homomorphism) For f, g measurable bounded and $a, b \in \mathbb{C}$, we have

$$\int_X (af + bg) dP = a \int_X f dP + b \int_X g dP, \tag{A.21}$$

$$\int_X f dP \cdot \int_X f dP = \int_X f \cdot g dP. \tag{A.22}$$

Observe that (b) and (ii) of Theorem A.15 imply that, for a simple function $\sum_k s_k \chi_{E_k}$, we have

$$\int_X \sum_k s_k \chi_{E_k} dP = \sum_k s_k P_{E_k}, \tag{A.23}$$

agreeing with the motivational discussion precedes the theorem.

A.2.3 Spectral Theorem

With the technical tools introduced, we finally can state the celebrated Spectral Theorem for bounded self-adjoint operators on Hilbert spaces.

Theorem A.16 (Spectral Theorem, bounded self-adjoint operators) *For any self-adjoint $\alpha \in \mathcal{B}(\mathcal{H})$ on a Hilbert space $\mathcal{H}$, there exists a unique PVM $P^{(\alpha)} : \mathscr{B}(\sigma(\alpha)) \to \mathcal{B}\mathcal{H}$ such that*

$$\alpha = \int_{\sigma(\alpha)} \lambda dP^{(\alpha)}(\lambda). \tag{A.24}$$

Moreover, we have criteria for distinguishing the types of spectrum:

(a) $\lambda \in \sigma_p(\alpha)$ *iff* $P^{(\alpha)}(\{\lambda\}) \neq 0$. $P^{(\alpha)}(\{\lambda\})$ *is the orthogonal projector onto the* λ*-eigenspace. (This situation occurs particularly when* λ *is an isolated point of* $\sigma(\alpha)$*.)*

(b) $\lambda \in \sigma_c(\alpha)$ *iff* $P^{(\alpha)}(\{\lambda\}) = 0$ *but* $P^{(\alpha)}(E) \neq 0$ *for any open set* $E \subset \mathbb{R}$ *containing* λ*.*

The general version for unbounded self-adjoint operators is similar, albeit one needs to be careful with the domain

We conclude the section with an even nicer version of spectral theorem for compact self-adjoint operator which is very useful to Quantum Mechanics. Recall that a *compact operator* on normed spaces $\alpha : V \to W$ if for any bounded sequence (v_n) in V, the sequence (αv_n) contains a convergent subsequence. Any compact operator is automatically bounded and hence continuous.

Theorem A.17 (Spectral Theorem, compact self-adjoint operators) *Let* $\alpha \in \mathcal{B}(\mathcal{H})$ *be a compact self-adjoint operator. Then there exists an orthonormal eigenbasis* $(v_x)_{x\in I}$ *of* α*, with the associated eigenvalues* $(\lambda_x)_{x\in I}$*. Moreover, for all* $\varphi \in \mathcal{H}$*, it holds that*

$$\alpha\varphi = \left(\sum_{x\in I} \lambda_x |v_x\rangle\langle v_x|\right)\varphi = \sum_{x\in I} \lambda_x \langle v_x|\varphi\rangle v_x. \tag{A.25}$$

One straightforward application of Theorem A.15 and Spectral Theorem is to justify the following common manipulation of the time evolution operator of a given Hamiltonian

$$e^{-it\hat{H}/\hbar}\varphi_k = e^{-it\lambda_k/\hbar}\varphi_k, \tag{A.26}$$

where λ_k is an eigenvalue with eigenvector φ_k, assuming that (φ_k) forms an orthonormal basis.

We note that quantum operators in general are not compact. Even more unfortunately, boundedness is not guaranteed for quantum operators. To cover the unbounded case, a technique called *functional calculus* is required [38, Chapter 3.2–3.3].

A.3 Hilbert space operator topologies

The study of functional analysis requires extensive consideration on the topological structures of infinite-dimensional vector spaces. In particular, it is crucial to understand the interplay between algebraic structures and topological structures in the study of operator algebras and their application in Quantum Mechanics. In this section, we introduce some of the more

relevant topologies for the space of bounded linear operators $\mathcal{B}(\mathcal{H})$ given a Hilbert space $\mathcal{H}$, among others that we omit.

We first recall that the operator norm on $\mathcal{B}(\mathcal{H})$ is given by

$$\|\alpha\| = \sup_{\varphi\in\mathcal{H},\|\varphi\|=1} \{\|\alpha\varphi\|\} \tag{A.27}$$

for any $\alpha \in \mathcal{B}(\mathcal{H})$.

- The *uniform operator topology* on $\mathcal{B}(\mathcal{H})$, also known as the norm topology, is induced by the operator norm on $\mathcal{B}(\mathcal{H})$. A sequence of operators (α_n) in $\mathcal{B}(\mathcal{H})$ converges to α *uniformly* if $\|\alpha_n - \alpha\| \to 0$ as $n \to \infty$.
- The *strong operator topology* on $\mathcal{B}(\mathcal{H})$ is induced by a family of seminorms p_φ for each $\varphi \in \mathcal{H}$, where $p_\varphi(\alpha) = \|\alpha\varphi\|$. Equivalently, it is the coarsest topology such that the evaluation map $\alpha \mapsto \alpha\varphi$ is continuous for any fixed $\varphi \in \mathcal{H}$.

  A sequence (α_n) in $\mathcal{B}(\mathcal{H})$ converges to α *strongly* if $\|(\alpha_n - \alpha)\varphi\| \to 0$ as $n \to \infty$ for any $\varphi \in \mathcal{H}$.
- The *weak operator topology* on $\mathcal{B}(\mathcal{H})$ is induced by a family of seminorms $p_{\varphi,\psi}$ for each $\varphi, \psi \in \mathcal{H}$, where $p_{\varphi,\psi}(\alpha) = \langle\varphi|\alpha\psi\rangle = \langle\varphi|\alpha|\psi\rangle$. Equivalently, it is the weakest topology so that the functionals $p_{\varphi,\psi}$ are continuous for all $\varphi, \psi \in \mathcal{H}$.

  A sequence α_n in $\mathcal{B}(\mathcal{H})$ converges to α *weakly* if $|\langle\varphi|(\alpha_n - \alpha)|\psi\rangle| \to 0$ as $n \to \infty$ for any $\varphi, \psi \in \mathcal{H}$.
- We consider the topological dual of $\mathcal{B}(\mathcal{H})$.

  $$\mathcal{B}(\mathcal{H})^* = \{f : \mathcal{B}(\mathcal{H}) \to \mathbb{C} \mid f \text{ is linear and continuous}\} \tag{A.28}$$

  The *weak-* topology* on $\mathcal{B}(\mathcal{H})^*$ is the weakest topology such that all functionals in $\mathcal{B}(\mathcal{H})^*$ are continuous.

  A sequence of functionals f_n in $\mathcal{B}(\mathcal{H})^*$ converges to f *weakly* if $f_n(\alpha) \to f(\alpha)$ as $n \to \infty$ for any $\alpha \in \mathcal{B}(\mathcal{H})$.

  We note the definition of weak-* topology can be easily generalised to the dual V^* of any normed vector space V.

Note that uniform convergence implies strong convergence, and strong convergence implies weak convergence [38, Chapter 3.5].

A.4 Important theorems in functional analysis

Let us now turn our attention to Hilbert spaces. We present the three pillars in functional analysis, namely, the Open Mapping Theorem, the Hahn–Banach

Theorem, and the Banach–Steinhaus Theorem. Although they do not have very explicit associations with our main text, they are fundamental to many results and notions in functional analysis.

Then we preceed to the discussion of the Banach–Alaoglu theorem, the Krein–Milman theorem which are highly relevant to noncommutative polynomial optimisation and NPA hierarchy [13, 31, 42, 57].

A.4.1 Three cornerstones in functional analysis

Let us start with the Open Mapping Theorem, we say that a map $\alpha \in \mathcal{B}(V, W)$ between two Banach spaces is *open* if it maps open sets to open sets.

Theorem A.18 (Open Mapping) *If V, W are Banach spaces and $\alpha \in \mathcal{B}(V, W)$ is surjective, then α is open.*

A corollary is important for the notion of isometric isomorphism.

Corollary A.19 *The set-theoretical inverse α^{-1} of a bijective morphism $\alpha \in \mathcal{B}(V, W)$ is linear and bounded, i.e. $\alpha^{-1} \in \mathcal{B}(W, V)$.*

Another consequence is the well-known *Closed Graph Theorem*, which provides the criterion for when a linear map is bounded. Define the graph of a linear map $\alpha : V \to W$ as $\mathrm{graph}(\alpha) = \{(v, \alpha v) \mid v \in V\} \subset V \oplus W$.

Theorem A.20 *For $\alpha : V \to W$, the graph $\mathrm{graph}(\alpha)$ is closed (relative to the induced norm of $V \oplus W$) iff α is bounded, i.e. $\alpha \in \mathcal{B}(V, W)$.*

In general, a *topological vector space* is endowed with a Hausdorff topology such that vector addition and scalar multiplication are continuous. The second fundamental result is the Hahn–Banach Theorem, which deals with the extensions of functionals on real topological vector spaces.

We say that a map $p : V \to \mathbb{R}$ is a *sublinear functional* if

$$p(v + w) \leq p(v) + p(w);\ p(tv) = tp(v),$$

for all $v, w \in V$ and $t \in \mathbb{R}$. Simple examples of sublinear functionals include norms and seminorms (norms without positive definiteness).

Theorem A.21 (Hanh–Banach) *Consider V a real topological vector space with a sublinear functional p, and let $W \subset V$ be a linear subspace. Suppose there is a linear map $\varphi_W : W \to \mathbb{R}$ dominated by p on W (i.e. $\varphi_W(w) \leq p(w)$ for all $w \in W$).*

Then φ_W admits a linear extension $\varphi : V \to \mathbb{R}$ which is also dominated by p on V.

We remark that the proof for infinite-dimensional case of Hahn–Banach theorem requires Zorn's Lemma, and the theorem can be generalised to the complex number.

Corollary A.22 *Let V be a normed vector space with subspace W. Then each linear functional $\varphi_W \in W^*$ admits an extension $\varphi \in V^*$.*

An important application of the Hahn–Banach Theorem to Krein–Milman Theorem, which we shall discuss later.

We finally turn to the Banach–Steinhaus Theorem, also known as *Uniform Boundedness Principle*, which asserts that pointwise boundedness implies norm boundedness. This result is crucial since this fact is taken as granted by physicists in their works.

Theorem A.23 (Banach–Steinhaus) *Let V, W be Banach spaces, and let X be an arbitrary index set to the family (α_x) of operators in $\mathcal{B}(V, W)$. If the family (α_x) is pointwise bounded*

$$\sup_{x \in X} \{\|\alpha_x v\|\} \leq \infty$$

for any $v \in V$, then it is also uniformly bounded

$$\sup_{x \in X} \{\|\alpha_x\|\} \leq \infty.$$

A.4.2 Banach–Alaoglu and Krein–Milman Theorems

Recall the weak-* topology of any given normed space (Appendix A.3).

Theorem A.24 (Banach–Alaoglu) *Let V be a normed vector space, then any closed d-ball*

$$B_d^* = \{\varphi \in V^* \mid \|\varphi\| \leq d\}$$

in the dual V^* is compact in the weak-* topology. More generally, let $U \subset V$ be a neighbourhood of 0, then the set

$$B_U^* = \{\varphi \in V^* \mid |\varphi(u)| \; \forall u \in U\}$$

is weak-* compact.

Another observation with Banach–Alaoglu Theorem leads to the conclusion that the state spaces in quamtum theory are weak-* compact.

We now turn to another important result for our main text, which requires some elementary definitions on convexity, which is also pivotal for quantum theory.

Definition A.25 *Consider a vector space (real or complex) V.*

1. *A subset $K \subset V$ is* convex *if one has $tv + (1-t)w \in K$ whenever $v, w \in K$ and $t \in [0, 1]$; the sum of the form $tv + (1-t)w$ for is called a* convex combination of v and w.

2. *The* boundary $\mathrm{ex}(K)$ *of the convex set K contains the* extreme points *of K, that is, if any $x \in \mathrm{ex}(K)$ is a non-trivial convex combination (i.e. $t \neq 0, 1$) of some $v, w \in K$, then we must have $x = v = w$.*

3. *The* convex hull *of a subset $X \subset V$ is the smallest convex set containing X, denoted by $\mathrm{co}(X)$. Equivalently, $\mathrm{co}(X) = \{tx + (1-t)y \mid x, y \in X, t \in [0,1]\}$.*

We state the Krein–Milman Theorem, whose proof relies on the Hahn–Banach Theorem.

Theorem A.26 (Krein–Milman) *Suppose V is a normed vector space and $K \subset V$ is an nonempty compact convex subset. Then $\mathrm{ex}(K)$ is nonempty, and all points in K lie in the topological closure of the convex hull of its extreme points. That is,*

$$K = \overline{\mathrm{co}(\mathrm{ex}(K))}.$$

Note that we can applied Krein–Milman Theorem to the dual V^* as well. Combining this observation and Banach–Alaoglu Theorem, the nonemptiness assertion indicates the abundance of pure states in quantum theory, while the rest of the statement shows the connection between mixed states and pure states.

Eidgenössische Technische Hochschule Zürich
Swiss Federal Institute of Technology Zurich

Declaration of originality

The signed declaration of originality is a component of every semester paper, Bachelor's thesis, Master's thesis and any other degree paper undertaken during the course of studies, including the respective electronic versions.

Lecturers may also require a declaration of originality for other written papers compiled for their courses.

I hereby confirm that I am the sole author of the written work here enclosed and that I have compiled it in my own words. Parts excepted are corrections of form and content by the supervisor.

Title of work (in block letters):

Quantum Nonlocality in Bilocal Networks: An Operator Algebraic Perspective

Authored by (in block letters):

For papers written by groups the names of all authors are required.

Name(s):	**First name(s):**
Xu	Xiangling

With my signature I confirm that

- I have committed none of the forms of plagiarism described in the 'Citation etiquette' information sheet.
- I have documented all methods, data and processes truthfully.
- I have not manipulated any data.
- I have mentioned all persons who were significant facilitators of the work.

I am aware that the work may be screened electronically for plagiarism.

Place, date	**Signature(s)**
Zürich, 02.05.2023	Xiangling Xu 许湘龄

For papers written by groups the names of all authors are required. Their signatures collectively guarantee the entire content of the written paper.